Cambridge International AS Level Physics (9702)

316 Extreme-Difficulty Practice Questions

Detailed Answers & Worked Solutions

AS syllabus 2025-2027 | Target series: Oct / Nov 2026 | Topics 1–11

Full method, working and mark-by-mark reasoning for every question

Take g = 9.81 m s^{-2} and c = 3.00 × 10^{8} m s^{-1} unless stated otherwise.

SECTION A — Structured (theory) answers [Q1–Q194]

1 Physical quantities and units

Q1. The viscous drag on a small sphere of radius *r* moving slowly at speed *v* through a fluid of viscosity *η* is $F = 6\pi\eta rv$. **[6]**

1. Determine the SI base units of *η*.
2. A student claims the terminal speed of a sphere falling in the fluid is $v = 2r^2(\rho_s - \rho_f)g / 9\eta$, where the *ρ* are densities. By checking homogeneity, state whether this equation is dimensionally consistent.

Answer. **(a)** Rearranging, $\eta = F/(6\pi rv)$. Base units = (kg m s^{-2}) / (m · m s^{-1}) = **kg m^{-1} s^{-1}**. (The 6π is a pure number.)
(b) RHS base units: numerator $r^2\rho g$ = m^2 · (kg m^{-3}) · (m s^{-2}) = kg m^0 s^{-2} = kg s^{-2}. Dividing by η (kg m^{-1} s^{-1}): (kg s^{-2})/(kg m^{-1} s^{-1}) = m s^{-1}. This equals the units of speed, so the equation **is homogeneous**. (Note: homogeneity does not prove the numerical factor 2/9 is correct.)

Q2. Estimate, showing your reasoning and stating any assumptions, the average useful power developed by an adult who runs up a flight of stairs of typical height in a typical time. Give your answer to one significant figure and comment on why more figures would be unjustified. **[5]**

Answer. Reasonable estimates: mass $m \approx$ 70 kg; vertical height of one storey ≈ 3 m; time to run up ≈ 5 s.

Useful work against gravity $W = mg\Delta h$ = 70 × 9.81 × 3 ≈ 2060 J.

Power $P = W/t$ = 2060 / 5 ≈ 410 W → ≈ **400 W** (4 × 10^2 W).
Only 1 s.f. is justified because each input quantity is itself only a rough estimate (uncertain by tens of per cent), so extra figures would imply a false precision.

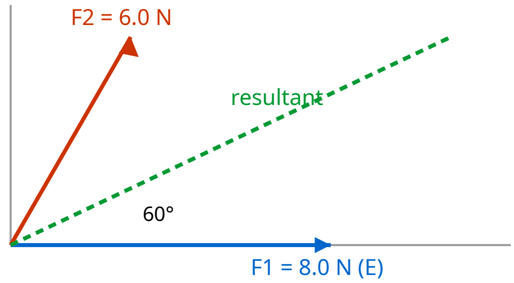
Q3. A resistance is found from $R = V/I$. In an experiment $V = (6.0 \pm 0.1)$ V and $I = (0.40 \pm 0.02)$ A. **[6]**

1. Calculate *R* and its absolute uncertainty.
2. The power is $P = V^2/R$. Determine the percentage uncertainty in *P*.

Answer. **(a)** $R = 6.0/0.40 = 15\ \Omega$. Percentage uncertainties: V: 0.1/6.0 = 1.67%; I: 0.02/0.40 = 5.0%. For a quotient, add: ΔR/R = 6.67%. ΔR = 0.0667 × 15 = 1.0 Ω. So **R = (15 ± 1) Ω.**

(b) $P = V^2/R \Rightarrow$ %unc = 2(%V) + (%R) = 2(1.67%) + 6.67% = 3.33% + 6.67% = **10%**. (Equivalently $P = V^2/R$ uses R already containing the I uncertainty; adding independently as P = VI would give 2× nothing — here use the stated formula.)

Q4. Two forces act at a point: F_1 = 8.0 N due east and F_2 = 6.0 N in a direction 60° north of east. By resolving into components, determine the magnitude and direction of the resultant force. **[5]**



> **Answer.** Components (east = x, north = y):
> x: 8.0 + 6.0 cos60° = 8.0 + 3.0 = 11.0 N.
> y: 0 + 6.0 sin60° = 5.20 N.
> Magnitude = $\sqrt{11.0^2 + 5.20^2} = \sqrt{121 + 27.0} = \sqrt{148}$ = **12.2 N**.
> Direction = $\tan^{-1}$(5.20/11.0) = **25.3° north of east**.

Q5. An aircraft has a velocity relative to the air of 250 m s^{-1} due north. A wind blows from the west (i.e. towards the east) at 40 m s^{-1}. **[7]**

1. Find the velocity of the aircraft relative to the ground (magnitude and direction).
2. State the direction in which the pilot must instead head, relative to north, so that the aircraft actually travels due north over the ground, and find the resulting ground speed.

> **Answer.** **(a)** Ground velocity = air velocity + wind velocity. North component 250, east component 40. Magnitude = $\sqrt{250^2 + 40^2} = \sqrt{62500 + 1600} = \sqrt{64100}$ = **253 m s^{-1}**. Direction = $\tan^{-1}$(40/250) = 9.1° **east of north**.
> **(b)** To cancel the eastward drift, the aircraft's own eastward component must be 40 m s^{-1} towards the west. So it heads at angle θ west of north with 250 sinθ = 40 ⇒ θ = $\sin^{-1}$(40/250) = **9.2° west of north**. Ground speed (northward) = 250 cosθ = 250 × 0.987 = **247 m s^{-1}**.

Q6. Distinguish between **random** and **systematic** errors. A micrometer with a zero error of +0.02 mm (it reads 0.02 mm when fully closed) is used to measure the diameter of a wire; five readings are 0.52, 0.53, 0.51, 0.54, 0.52 mm. **[7]**

1. State the type of error caused by the zero error and its effect on the mean.
2. Determine the best estimate of the true diameter and its absolute uncertainty.

> **Answer.** Random errors cause readings to scatter unpredictably about the true value (reduced by averaging); systematic errors shift all readings by the same amount in the same direction (not reduced by averaging).
> **(a)** The zero error is a **systematic error**; every reading is 0.02 mm too large, so the mean is 0.02 mm too high.
> **(b)** Mean of readings = (0.52+0.53+0.51+0.54+0.52)/5 = 0.524 mm. Correct for zero error: 0.524 - 0.02 = **0.504 mm**. Random uncertainty $\approx$ half the range = (0.54-0.51)/2 = 0.015 mm. So diameter = **(0.50 ± 0.02) mm** (uncertainty rounded up to 1 s.f., value to matching decimal place).

Q7. The Young modulus *E* of a wire is found from $E = 4FL / (\pi d^2 e)$. Measured percentage uncertainties are: *F* 2%, *L* 0.5%, *d* 1.5%, extension *e* 4%. Determine the percentage uncertainty in *E* and explain which single measurement it would be most worthwhile to improve. **[5]**

> **Answer.** For a product/quotient, add percentage uncertainties, multiplying each by its power. *d* appears squared, so it contributes 2 × 1.5% = 3%.
>
> %E = %F + %L + 2(%d) + %e = 2 + 0.5 + 3 + 4 = **9.5%** ($\approx$ 10%).
> The largest single contribution is the extension *e* (4%). Since *e* is a small quantity (difference of two scale readings), improving how it is measured — e.g. using a vernier/travelling microscope or a longer wire to give a larger *e* — would most reduce the overall uncertainty.

Q8. State what is meant by a **scalar** and a **vector** quantity. For each of the following, state whether it is a scalar or a vector: (i) work done, (ii) momentum, (iii) electric current, (iv) gravitational field strength, (v) power. Give a reason for classifying electric current as you do. **[6]**

> **Answer.** A scalar has magnitude only; a vector has magnitude **and** direction (and adds by the triangle/parallelogram rule).
> (i) work — **scalar**; (ii) momentum — **vector**; (iii) current — **scalar**; (iv) gravitational field strength — **vector**; (v) power — **scalar**.
> Although current has a direction of flow, currents do not combine by vector addition (they add algebraically at a junction, as in Kirchhoff's first law), so current is treated as a scalar.

Q9. A ball is thrown at a wall and rebounds. Just before impact its velocity is 12 m s^{-1} directed 30° below the horizontal; just after, it is 9.0 m s^{-1} directed 40° above the horizontal. Treating the horizontal (towards the wall) as positive x and upward as positive y, find the magnitude and direction of the **change in velocity** Δv. **[7]**

> **Answer.** Take x towards the wall, y up. Before: v_x = +12 cos30° = +10.39, v_y = -12 sin30° = -6.00 m s^{-1}. After (moving away from wall, so x negative): v_x = -9.0 cos40° = -6.89, v_y = +9.0 sin40° = +5.79 m s^{-1}. Δv = v_{after} - v_{before}: Δv_x = -6.89 - 10.39 = -17.28; Δv_y = 5.79 - (-6.00) = +11.79 m s^{-1}.
>
> Magnitude = $\sqrt{17.28^2 + 11.79^2} = \sqrt{298.6 + 139.0} = \sqrt{437.6}$ = **20.9 m s^{-1}**. Direction = $\tan^{-1}$(11.79/17.28) = 34.3° above the horizontal, pointing **away from the wall and upward** (i.e. 34° above horizontal in the -x, +y quadrant).

Q10. The pressure at depth in a liquid is given by $p = h\rho g$. **[5]**

1. Show that this equation is homogeneous.
2. Hence deduce the base units of pressure and confirm they are equivalent to the pascal expressed in terms of the newton.

> **Answer. (a)** RHS units: $h\rho g$ = m · kg m^{-3} · m s^{-2} = kg m^{-1} s^{-2}. LHS pressure = force/area = (kg m s^{-2})/m² = kg m^{-1} s^{-2}. Both sides identical ⇒ homogeneous.
> **(b)** Base units of pressure = **kg m^{-1} s^{-2}**. 1 Pa = 1 N m^{-2} = (kg m s^{-2}) m^{-2} = kg m^{-1} s^{-2} — the same, confirming consistency.

Q11. A student measures the period *T* of a pendulum by timing 20 oscillations with a stopwatch of resolution 0.01 s, obtaining a total time of 18.4 s. The reaction time in starting/stopping introduces an uncertainty of about ±0.3 s in the total time. **[6]**

1. Explain why timing 20 oscillations rather than 1 is good practice.
2. Determine *T* and its percentage uncertainty.

> **Answer. (a)** The reaction-time uncertainty (≈ 0.3 s) is fixed per measurement, so timing many oscillations makes it a much smaller fraction of the total time; dividing by 20 also divides the absolute uncertainty in T by 20. This reduces the percentage (random) uncertainty. (The stopwatch resolution is negligible compared with reaction time.)
> **(b)** T = 18.4/20 = **0.920 s**. The dominant uncertainty is ±0.3 s in the total: %unc = 0.3/18.4 = **1.6%** (this is the same for the total time and for T, since dividing by an exact 20 does not change the percentage). Hence T = (0.920 ± 0.015) s.

Q12. Estimate the order of magnitude of the number of air molecules in a typical classroom. State your assumptions clearly. (Take the number of molecules per mole as 6×10^{23} and the molar volume of a gas as about 0.024 m^3 mol^{-1} at room conditions.) **[5]**

> **Answer.** Assume a room 8 m × 6 m × 3 m $\approx$ 150 m^3.
>
> Moles of air = volume / molar volume = 150 / 0.024 $\approx$ 6.3×10^3 mol.
>
> Number of molecules = $6.3 \times 10^3 \times 6 \times 10^{23} \approx 3.8 \times 10^{27}$.
>
> **Order of magnitude $\approx$ 10^{27} molecules.** (Any room volume 50–300 m^3 gives the same order.)

Q13. Three coplanar forces act on a point object which is in equilibrium: a known force of 10.0 N acting horizontally, and two unknown forces *P* and *Q*. *P* acts at 90° to the 10.0 N force (vertically) and *Q* acts at 150° to the 10.0 N force (measured anticlockwise). Using the fact that the vector sum is zero, find *P* and *Q*. **[7]**

> **Answer.** For equilibrium the components in each direction sum to zero. Let the 10.0 N force be along +x. Q makes 150° with +x, so Q_x = Q cos150° = -0.866Q, Q_y = Q sin150° = +0.500Q. P is along +y: P_x = 0, P_y = P.
>
> x: 10.0 + (-0.866Q) = 0 $\Rightarrow$ Q = 10.0/0.866 = **11.5 N**.
>
> y: P + 0.500Q = 0 $\Rightarrow$ P = -0.500 × 11.5 = -5.77 N, i.e. **P = 5.8 N directed vertically downward** (opposite to the assumed +y). A closed vector triangle confirms equilibrium.

Q14. The drag force on a car at speed *v* is modelled by $F = bv^2$. **[5]**

1. Determine the SI base units of the constant *b*.
2. The power needed to overcome this drag is $P = Fv$. Show, using units, that $P = bv^3$ is homogeneous.

> **Answer.** **(a)** b = F/v^2 = (kg m s^{-2})/(m s^{-1})² = (kg m s^{-2})/(m² s^{-2}) = **kg m^{-1}**.
> **(b)** bv^3 = kg m^{-1} × (m s^{-1})³ = kg m^{-1} × m³ s^{-3} = kg m² s^{-3}. Power = energy/time = (kg m² s^{-2})/s = kg m² s^{-3}. Identical $\Rightarrow$ homogeneous.

Q15. A quantity is calculated from $g = 4\pi^2 L/T^2$ in a pendulum experiment. *L* = (1.000 ± 0.002) m and the period is measured as *T* = (2.008 ± 0.005) s. **[7]**

1. Calculate *g*.
2. Determine the absolute uncertainty in *g* and state the result to an appropriate number of significant figures.

> **Answer.** **(a)** g = $4\pi^2$ × 1.000 / $(2.008)^2$ = 39.478 / 4.032 = **9.79 m s^{-2}**.
> **(b)** %L = 0.002/1.000 = 0.20%; %T = 0.005/2.008 = 0.249%, and T is squared so contributes 2 × 0.249% = 0.498%. Total %g = 0.20% + 0.50% = 0.70%. Δg = 0.0070 × 9.79 = 0.068 $\approx$ 0.07 m s^{-2}. Result: **g = (9.79 ± 0.07) m s^{-2}**.

Q16. The period T of oscillation of a liquid drop is thought to depend on its radius r, its density ρ and the surface tension γ (which has base units kg s^{-2}). It is proposed that $T = k\, r^a \rho^b \gamma^c$, where k is a dimensionless constant. **[7]**

1. Use the method of dimensions (base units) to find the values of a, b and c.
2. State one thing this method cannot tell you about the equation.

Answer. **(a)** Base units: T = s; r = m; ρ = kg m^{-3}; γ = kg s^{-2}. So s = m^a(kg m^{-3})^b(kg s^{-2})^c = kg^{b+c} m^{a-3b} s^{-2c}.

Match powers: kg: b + c = 0; s: -2c = 1 ⇒ c = -½, so b = +½; m: a - 3b = 0 ⇒ a = 3b = 3/2. So **a = 3/2, b = 1/2, c = -1/2** (i.e. $T = k\sqrt{\rho r^3/\gamma}$).
(b) It cannot give the value of the dimensionless constant *k* (nor confirm the equation is physically correct) — only that it is dimensionally consistent.

Q17. A solid cylinder is used to find the density of a metal. Measurements: mass m = (48.6 ± 0.1) g; diameter d = (12.4 ± 0.1) mm; length L = (35.0 ± 0.5) mm. **[8]**

1. Calculate the density in kg m^{-3}.
2. Determine the percentage uncertainty in the density and hence the absolute uncertainty.

Answer. Volume $V = \pi(d/2)^2L = \pi(6.2\times10^{-3})^2(35.0\times10^{-3}) = \pi \times 3.844\times10^{-5} \times 0.0350 = 4.226\times10^{-6}$ m^3.
(a) $\rho = m/V = 48.6\times10^{-3}/4.226\times10^{-6}$ = **1.15×10^4 kg m^{-3}**.

(b) %m = 0.1/48.6 = 0.21%; %d = 0.1/12.4 = 0.81% (d is squared ⇒ 1.61%); %L = 0.5/35.0 = 1.43%. Total %ρ = 0.21 + 1.61 + 1.43 = **3.3%**. $\Delta\rho = 0.033 \times 1.15\times10^4 = 0.38\times10^3$, so ρ = **(11.5 ± 0.4)×$10^3$ kg m^{-3}**.

Q18. A swimmer who can swim at 1.2 m s^{-1} in still water wishes to cross a river 60 m wide that flows at 0.90 m s^{-1}. **[8]**

1. If the swimmer heads straight across (perpendicular to the bank), find the resultant velocity and how far downstream they land.
2. Find the direction in which the swimmer should instead head to land directly opposite the start, and the time this crossing takes.

Answer. **(a)** Perpendicular component 1.2 m s^{-1}, downstream 0.90 m s^{-1}. Resultant = $\sqrt{1.2^2 + 0.90^2}$ = $\sqrt{1.44 + 0.81}$ = $\sqrt{2.25}$ = **1.5 m s^{-1}** at $\tan^{-1}$(0.90/1.2) = 36.9° to the bank normal. Time across = 60/1.2 = 50 s; drift = 0.90 × 50 = **45 m downstream**.
(b) To land opposite, the upstream component of the swim velocity must cancel the current: 1.2 sinθ = 0.90 ⇒ θ = $\sin^{-1}$(0.75) = **48.6° upstream** of the normal. Effective across-speed = 1.2 cos48.6° = 0.794 m s^{-1}; time = 60/0.794 = **76 s**.

Q19. Explain the difference between **precision** and **accuracy**. A student takes four readings of a resistance using two methods. Method P gives 47.1, 47.2, 47.0, 47.1 Ω; method Q gives 44.0, 49.5, 46.8, 45.2 Ω. The accepted value is 45.4 Ω. **[7]**

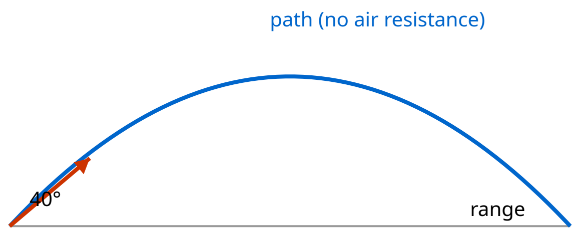
1. State which method is more precise and which is more accurate, with reasons.
2. Identify the likely type of error in the less accurate method.

Answer. Precision: how closely repeated readings agree with each other (small random scatter). Accuracy: how close a reading (or the mean) is to the true value.
(a) Method P is **more precise** (readings tightly clustered, range 0.2 Ω) but **less accurate** (mean 47.1 Ω, far from 45.4). Method Q is less precise (large scatter, range 5.5 Ω) but its mean (46.4 Ω) is **closer to** the true value than P's... in fact P's mean is 47.1 (1.7 off) and Q's is 46.4 (1.0 off), so Q is more accurate.
(b) Method P shows a consistent offset from the true value with little scatter — characteristic of a **systematic error** (e.g. a zero/calibration error in the meter).

2 Kinematics

Q20. A ball is projected from the ground with speed 25 m s^{-1} at 40° above the horizontal. Air resistance is negligible. **[8]**

1. Calculate the maximum height reached.
2. Calculate the horizontal range.
3. Determine the speed and direction of the ball 2.5 s after projection.



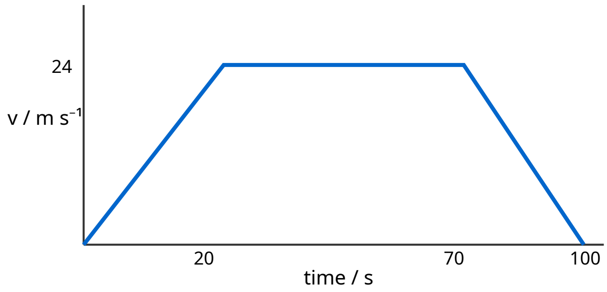
Answer. Resolve: u_x = 25 cos40° = 19.15 m s^{-1}; u_y = 25 sin40° = 16.07 m s^{-1}.

(a) At max height $v_y = 0$: $v_y^2 = u_y^2 - 2gh \Rightarrow h = 16.07^2/(2\times9.81) = 258/19.62 =$ **13.2 m**.
(b) Time of flight: $t = 2u_y/g = 2\times16.07/9.81 = 3.28$ s. Range = $u_x t = 19.15 \times 3.28 =$ **62.8 m**.
(c) At t = 2.5 s: v_x = 19.15 (constant); $v_y = 16.07 - 9.81\times2.5 = 16.07 - 24.53 = -8.46$ m s^{-1} (downward).

Speed = $\sqrt{19.15^2 + 8.46^2} = \sqrt{366.7 + 71.6} = \sqrt{438.3} =$ **20.9 m s^{-1}**, directed $\tan^{-1}(8.46/19.15)$ = **23.8° below the horizontal**.

Q21. The graph shows how the velocity of a train varies with time over a 100 s journey between two stations. Using the graph: **[8]**

1. describe the motion in each of the three phases;
2. determine the total distance travelled between the stations;
3. calculate the magnitude of the (constant) deceleration in the final phase.



> **Answer.** **(a)** 0–20 s: uniform acceleration from rest to 24 m s^{-1}; 20–70 s: constant velocity 24 m s^{-1}; 70–100 s: uniform deceleration to rest.
> **(b)** Distance = area under graph. Phase 1 (triangle) = ½×20×24 = 240 m; Phase 2 (rectangle) = 50×24 = 1200 m; Phase 3 (triangle) = ½×30×24 = 360 m. Total = 240 + 1200 + 360 = **1800 m**.
> **(c)** Deceleration = gradient magnitude = 24/(100-70) = 24/30 = **0.80 m s^{-2}**.

Q22. A stone is dropped from rest at the top of a cliff. During the last second before it hits the sea, it falls a distance equal to one quarter of the total height of the cliff. Ignoring air resistance, determine the height of the cliff. (Take g = 9.81 m s^{-2}.) **[7]**

> **Answer.** Let total fall time be T and height $H = ½gT^2$. Distance fallen in first (T-1) s: $½g(T-1)^2$. The last second's fall = $H - ½g(T-1)^2 = ¼H$.
>
> So $½g(T-1)^2 = ¾H = ¾(½gT^2)$. Cancel ½g: $(T-1)^2 = ¾T^2 \Rightarrow T-1 = (\sqrt{3}/2)T = 0.8660T \Rightarrow 0.1340T = 1 \Rightarrow T =$ 7.46 s.
> $H = ½ \times 9.81 \times 7.46^2 = ½ \times 9.81 \times 55.7 =$ **273 m** (2.7×10^2 m).

Q23. Describe an experiment, using a falling object and appropriate apparatus, to determine the acceleration of free fall g. Include: the measurements taken, how they are used, one significant source of error, and how it is reduced. **[7]**

> **Answer.** A steel ball is released from an electromagnet; switching off the magnet simultaneously starts an electronic timer. The ball falls a measured height h (metre rule) and strikes a trapdoor switch, stopping the timer, giving time t.
> Using $h = ½gt^2$, a graph of h (y-axis) against t^2 (x-axis) is a straight line through the origin of gradient ½g, so g = 2 × gradient. Repeating for several heights and plotting reduces random error and averages out timing scatter.
> A significant systematic error is the delay between cutting the current and the ball actually releasing (residual magnetism), which makes t too large. This is reduced by using the graphical gradient method (a constant time offset shifts the line but changes its intercept, not its gradient), so g is unaffected.

Q24. Two cars, A and B, travel along the same straight road. At t = 0 both are level. A moves at a constant 30 m s^{-1}. B starts from rest at t = 0 with constant acceleration 4.0 m s^{-2}. **[8]**

1. Find the time at which B catches up with A.
2. Find the distance travelled by then.
3. Find B's speed at that instant and comment on why it exceeds A's speed.

Answer. **(a)** Equal displacement: $30t = ½(4.0)t^2 = 2t^2 \Rightarrow 2t^2 - 30t = 0 \Rightarrow t(2t - 30) = 0 \Rightarrow t =$ **15 s** (rejecting t = 0).
(b) Distance = 30 × 15 = **450 m**.
(c) $v_B = 0 + 4.0 \times 15 =$ **60 m s^{-1}**. B is faster because, although they cover the same distance in the same time (so equal *average* speeds of 30 m s^{-1}), B started slower and finished faster; its speed at the catch-up point is twice the average, i.e. 60 m s^{-1}.

Q25. A displacement–time graph for an object moving in a straight line is a smooth curve. State how you would obtain, from such a graph, (i) the instantaneous velocity at a point and (ii) the instantaneous acceleration. Explain why the second is harder to obtain accurately. **[6]**

Answer. **(i)** The instantaneous velocity is the gradient of the tangent to the s–t curve at that point (draw the tangent, take $\Delta s/\Delta t$ over a large span of the tangent line).
(ii) Acceleration is the rate of change of velocity, so first obtain velocities at several points (tangent gradients), plot a v–t graph, and take the gradient of *that* graph.
It is harder/less accurate because it requires differentiating twice: each tangent drawn by eye carries appreciable uncertainty, and taking the gradient of a graph that is itself built from uncertain gradients compounds the error.

Q26. A projectile is launched horizontally from the top of a tower of height 45 m with speed 18 m s^{-1}. **[7]**

1. Calculate the time to reach the ground.
2. Calculate the horizontal distance from the base of the tower where it lands.
3. Determine the angle the velocity makes with the horizontal on landing.

Answer. **(a)** Vertical: $45 = ½(9.81)t^2 \Rightarrow t^2 = 9.174 \Rightarrow t =$ **3.03 s**.
(b) Horizontal distance = 18 × 3.03 = **54.5 m**.
(c) $v_y = gt = 9.81 \times 3.03 = 29.7$ m s^{-1}; $v_x = 18$. Angle below horizontal = $\tan^{-1}(29.7/18) =$ **58.8°**.

Q27. Explain, in terms of the independence of horizontal and vertical motion, why a bullet fired horizontally and a bullet dropped from the same height at the same instant reach the ground at the same time (ignoring air resistance and Earth's curvature). State what would change if air resistance were significant. **[6]**

Answer. Vertical and horizontal motions are independent. Both bullets start with zero vertical velocity and experience the same downward acceleration g; the horizontal velocity of the fired bullet has no vertical component and does not affect the vertical motion. Hence both take the same time $t = \sqrt{2h/g}$ to fall the same height h and land together.
With significant air resistance, the drag force acts opposite to the actual velocity, which for the fast horizontally-moving bullet has an upward vertical component; this reduces its vertical acceleration, so the fired bullet would take slightly **longer** to land, and the two would no longer land simultaneously.

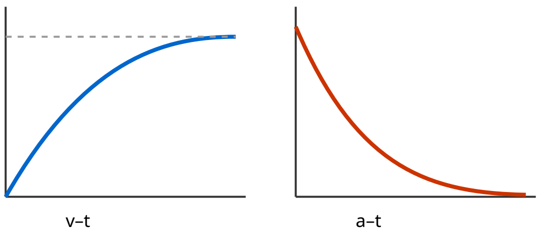
Q28. A car accelerates uniformly from 8.0 m s^{-1} to 20 m s^{-1} while travelling 168 m. **[8]**

1. Calculate the acceleration.
2. Calculate the time taken.
3. Calculate the distance travelled during the final 2.0 s of this period.

Answer. **(a)** $v^2 = u^2 + 2as$: $20^2 = 8.0^2 + 2a(168) \Rightarrow 400 = 64 + 336a \Rightarrow a = 336/336 =$ **1.00 m s^{-2}**.

(b) $v = u + at$: $20 = 8.0 + 1.00t \Rightarrow t =$ **12 s**.
(c) Final 2.0 s is from t = 10 s to 12 s. At t = 10 s, $v = 8.0 + 1.00 \times 10 = 18$ m s^{-1}. Distance = $18(2.0) + ½(1.00)(2.0)^2 = 36 + 2.0 =$ **38 m**.

Q29. The velocity–time graph shown consists of a curve whose gradient decreases with time, levelling off to a horizontal asymptote. State what kind of motion this represents, sketch the corresponding acceleration–time graph, and identify a real physical situation the graph could describe. **[6]**



Answer. The velocity increases but at a decreasing rate, approaching a constant (maximum) value — the object approaches a **terminal velocity**. The acceleration (gradient of v–t) starts large and positive and decreases towards zero: an a–t graph that is a decaying curve from a high value at t = 0 down to a = 0.
Physical situation: an object (e.g. a skydiver or a ball bearing) falling through a fluid under gravity with a drag force that increases with speed, until drag equals weight and the resultant force (hence acceleration) becomes zero.

Q30. A ball is thrown vertically upward from a height of 1.5 m above the ground with an initial speed of 12 m s^{-1}. Taking upward as positive and g = 9.81 m s^{-2}: **[8]**

1. find the total time before it strikes the ground;
2. find its speed just before impact;
3. sketch (with values) the velocity–time graph for the whole motion.

> **Answer.** Take the ground as displacement s = -1.5 m relative to the launch point.
>
> **(a)** $s = ut + ½at^2$: $-1.5 = 12t - 4.905t^2$. Rearranged: $4.905t^2 - 12t - 1.5 = 0$. $t = [12 \pm \sqrt{144 + 29.43}]/(9.81)$ = [12 ± 13.17]/9.81. Take positive root: t = 25.17/9.81 = **2.57 s**.
> **(b)** v = u + at = 12 - 9.81×2.57 = -13.2 m s^{-1}, i.e. speed **13.2 m s^{-1}** downward. (Check: $v^2 = 12^2 + 2(9.81)(1.5)$ = 144 + 29.4 = 173.4, v = 13.2.)
> **(c)** A straight line of constant negative gradient (-9.81 m s^{-2}) starting at +12 m s^{-1}, crossing v = 0 at t = 1.22 s (top of flight), reaching -13.2 m s^{-1} at t = 2.57 s.

Q31. An object moves so that its displacement is given by successive equal time intervals of 1.0 s: the distances travelled in the 1st, 2nd, 3rd and 4th seconds are 5, 15, 25 and 35 m respectively. **[7]**

1. Show that the motion is uniformly accelerated and find the acceleration.
2. Find the initial velocity.

> **Answer.** For uniform acceleration from initial velocity u, the distance in the n-th second is $s_n = u + ½a(2n-1)$. The differences between consecutive seconds are constant: 15-5 = 10, 25-15 = 10, 35-25 = 10 m. A constant difference confirms uniform acceleration.
> **(a)** The common difference equals a × $(1.0\ s)^2$... more precisely $s_{n+1} - s_n = a(\Delta t)^2$ with Δt = 1 s? Using $s_n = u + ½a(2n-1)$: difference = a. So **a = 10 m s^{-2}**.
>
> **(b)** 1st second: $s_1 = u + ½a = u + 5 = 5 \Rightarrow$ **u = 0** (the object starts from rest).

Q32. A cyclist travels the first half of a straight journey at 6.0 m s^{-1} and the second half (equal distance) at 12 m s^{-1}. Later the cyclist repeats the journey travelling at 6.0 m s^{-1} for the first half of the *time* and 12 m s^{-1} for the second half of the time. **[8]**

1. Find the average speed for the equal-distance journey.
2. Find the average speed for the equal-time journey.
3. Explain which is larger and why.

> **Answer.** **(a)** For equal distances d each: total distance 2d, total time = d/6 + d/12 = d(2+1)/12 = 3d/12 = d/4. Average = 2d/(d/4) = **8.0 m s^{-1}** (the harmonic mean).
> **(b)** For equal times T each: distances 6T and 12T, total 18T over 2T. Average = 18T/2T = **9.0 m s^{-1}** (the arithmetic mean).
> **(c)** The equal-time average (9.0) is larger. In the equal-distance case more time is spent at the slower speed (it takes longer to cover the slow half), which weights the average towards the lower speed;
>
> the arithmetic mean is always $\geq$ the harmonic mean.

Q33. During a tennis serve the ball is struck at a height of 2.4 m and travels horizontally (initially) at 40 m s^{-1} towards the net 12 m away. The net is 0.91 m high. Ignoring air resistance and any vertical launch component, determine whether the ball clears the net, and by what margin. **[7]**

Answer. Horizontal: time to reach the net t = 12/40 = 0.30 s.
Vertical drop in this time: $\Delta y = \frac{1}{2}gt^2 = \frac{1}{2}(9.81)(0.30)^2 = \frac{1}{2}(9.81)(0.090) = 0.441$ m.
Height of ball at net = 2.4 - 0.441 = 1.96 m.

Since 1.96 m > 0.91 m, the ball **clears the net**, by 1.96 - 0.91 = **1.05 m** ($\approx$ 1.0 m).

Q34. A lift (elevator) starts from rest, accelerates upward at 1.2 m s^{-2} for 3.0 s, then moves at constant velocity for 5.0 s, then decelerates uniformly to rest in a further 2.0 s. **[8]**

1. Sketch the velocity–time graph.
2. Find the total distance travelled.
3. Find the average velocity for the whole journey.

Answer. Peak velocity v = 0 + 1.2×3.0 = 3.6 m s^{-1}.

(a) Straight rise 0 $\rightarrow$ 3.6 m s^{-1} over 0–3 s; flat at 3.6 over 3–8 s; straight fall 3.6 $\rightarrow$ 0 over 8–10 s.
(b) Areas: accel ½(3.0)(3.6) = 5.4 m; constant (5.0)(3.6) = 18.0 m; decel ½(2.0)(3.6) = 3.6 m. Total = **27 m**.
(c) Total time = 10 s, so average velocity = 27/10 = **2.7 m s^{-1}** upward.

Q35. A particle is projected up a smooth plane inclined at 25° to the horizontal with an initial speed of 9.0 m s^{-1} along the plane. The component of gravitational acceleration along the plane is g sin25°. **[7]**

1. Find the distance travelled up the plane before it momentarily stops.
2. Find the time taken to return to its starting point.

Answer. Deceleration along plane a = g sin25° = 9.81 × 0.4226 = 4.15 m s^{-2}.

(a) $v^2 = u^2 - 2as$, v = 0: $s = u^2/(2a) = 9.0^2/(2\times4.15)$ = 81/8.30 = **9.76 m** ($\approx$ 9.8 m).
(b) Time up: t_{up} = u/a = 9.0/4.15 = 2.17 s. By symmetry (smooth plane), time down equals time up,

so total time = 2 × 2.17 = **4.34 s** ($\approx$ 4.3 s).

Q36. A stone is thrown from the top of a cliff 55 m above the sea with a speed of 20 m s^{-1} at 35° **above** the horizontal. Air resistance is negligible. **[9]**

1. Calculate the maximum height above the cliff top reached by the stone.
2. Calculate the total time of flight until it hits the sea.
3. Calculate the horizontal distance from the base of the cliff at which it lands.

> **Answer.** u_x = 20 cos35° = 16.4 m s^{-1}; u_y = 20 sin35° = 11.5 m s^{-1} (up).
> **(a)** $h_{max} = u_y^2/(2g)$ = $11.5^2/(2×9.81)$ = 131.6/19.62 = **6.7 m** above the cliff top.
>
> **(b)** Taking up positive, sea at -55 m: -55 = 11.5t - 4.905t^2 ⇒ 4.905t^2 - 11.5t - 55 = 0. t = [11.5 ± $\sqrt{132.3 + 1079}$]/9.81 = [11.5 ± 34.8]/9.81. Positive root: t = 46.3/9.81 = **4.72 s**.
> **(c)** Range = $u_x t$ = 16.4 × 4.72 = **77 m**.

Q37. The graph of velocity against time for a bouncing ball (taking upward as positive) shows a series of straight sloping lines of equal gradient, with sudden reversals of velocity that decrease in magnitude at each bounce. **[7]**

1. Explain why every sloping section of the graph has the same gradient.
2. Explain what the sudden vertical jumps represent and why they get shorter each time.
3. State how the displacement–time graph would differ in appearance.

> **Answer. (a)** While in the air the only force is gravity, so the acceleration is g downward at all times; the gradient of a v–t graph is the acceleration, so every free-flight section has the same gradient (-9.81 m s^{-2}).
> **(b)** Each vertical jump is the rapid reversal of velocity during the brief contact with the ground (downward velocity to upward velocity). They get shorter because the ball loses kinetic energy at each bounce (collision is inelastic), so it leaves the ground with a smaller speed each time.
> **(c)** The displacement–time graph would be a series of parabolic arcs (not straight lines), each lower than the last, touching the time axis at each bounce.

Q38. A car and a lorry are travelling in the same direction along a straight road. At t = 0 the car, moving at 8.0 m s^{-1}, is 15 m behind the lorry which moves at a constant 12 m s^{-1}. At t = 0 the car begins to accelerate uniformly at 2.5 m s^{-2}. **[8]**

1. Show that the car draws level with the lorry, and find the time taken.
2. Find the car's speed at that moment.

> **Answer.** Let the car's position start at 0 and the lorry's at +15 m.
> Car: x_c = 8.0t + ½(2.5)t^2 = 8.0t + 1.25t^2. Lorry: x_l = 15 + 12t.
>
> **(a)** Level when $x_c = x_l$: 8.0t + 1.25t^2 = 15 + 12t ⇒ 1.25t^2 - 4.0t - 15 = 0. t = [4.0 ± $\sqrt{16 + 75}$]/2.5 = [4.0 ± 9.54]/2.5. Positive root: t = 13.54/2.5 = **5.4 s** (a real positive solution exists, so they meet).
> **(b)** v = 8.0 + 2.5 × 5.42 = 8.0 + 13.5 = **21.6 m s^{-1}**.

Q39. Two students investigate the acceleration of free fall by dropping a ball through a set of light gates and by video analysis. Their processed data give displacement *s* against time *t* for a ball released from rest. **[7]**

1. State what graph they should plot to obtain a straight line, and how *g* is found from its gradient.
2. Their straight line does not pass through the origin but has a small positive intercept on the *s*-axis. Explain what this systematic feature indicates and why the gradient still gives a valid value of *g*.

Answer. **(a)** From $s = ut + ½gt^2$ with $u = 0$, $s = ½gt^2$. Plot *s* (y-axis) against t^2 (x-axis): a straight line through the origin of gradient ½g, so **g = 2 × gradient**.
(b) A positive intercept on the s-axis means the ball had already fallen a small distance (or timing started slightly late / the ball had a small initial speed) — a systematic offset. Because this adds a constant to every s value, it shifts the whole line up but does not change its slope; since g is obtained from the **gradient**, the offset does not affect the value of g.

3 Dynamics

Q40. State Newton's second law in terms of momentum. A jet of water from a hose leaves a nozzle of cross-sectional area 4.0×10^{-4} m² at a speed of 25 m s^{-1} and strikes a vertical wall at right angles, running down the wall without rebounding. (Density of water = 1.0×10^{3} kg m^{-3}.) **[7]**

1. Calculate the mass of water hitting the wall per second.
2. Calculate the force exerted on the wall.

> **Answer.** Newton's second law: the resultant force on a body equals the rate of change of its momentum, $F = \Delta p/\Delta t$ (in the direction of the force).
> **(a)** Mass per second = $\rho Av = 1.0\times10^{3} \times 4.0\times10^{-4} \times 25$ = **10 kg s^{-1}**.
> **(b)** The water's horizontal velocity changes from 25 m s^{-1} to 0 (it does not rebound). Rate of change of momentum = (mass/s) × Δv = 10 × 25 = **250 N**. By Newton's third law the wall pushes back on the water with 250 N, so the water exerts 250 N on the wall.

Q41. A trolley of mass 0.80 kg moving at 3.0 m s^{-1} collides head-on with a stationary trolley of mass 1.2 kg. After the collision they move off separately; the 0.80 kg trolley continues in the same direction at 0.60 m s^{-1}. **[8]**

1. Find the velocity of the 1.2 kg trolley after the collision.
2. Determine whether the collision is elastic, showing your working.

> **Answer. (a)** Conservation of momentum: 0.80(3.0) = 0.80(0.60) + 1.2v. 2.40 = 0.48 + 1.2v $\Rightarrow$ v = 1.92/1.2 = **1.6 m s^{-1}** (same direction).
> **(b)** KE before = ½(0.80)(3.0)² = 3.60 J. KE after = ½(0.80)(0.60)² + ½(1.2)(1.6)² = 0.144 + 1.536 = 1.68 J.
>
> KE is not conserved (3.60 J $\rightarrow$ 1.68 J), so the collision is **inelastic**. (Alternatively: relative speed of approach 3.0 vs relative speed of separation 1.6-0.6 = 1.0 — not equal, so inelastic.)

Q42. A ball of mass 0.15 kg strikes a wall horizontally at 8.0 m s^{-1} and rebounds horizontally at 6.0 m s^{-1}. The contact time is 0.020 s. **[7]**

1. Calculate the magnitude of the change in momentum of the ball.
2. Calculate the average force exerted by the wall on the ball.
3. State the force exerted by the ball on the wall and justify it.

> **Answer.** Take the initial direction as positive.
> **(a)** $\Delta p = m(v - u)$ = 0.15(-6.0 - 8.0) = 0.15(-14.0) = -2.10 kg m s^{-1}; magnitude **2.1 kg m s^{-1}**.
> **(b)** $F = \Delta p/\Delta t$ = 2.10/0.020 = **105 N** (directed away from the wall).
> **(c)** By Newton's third law the ball exerts an equal and opposite force of **105 N** on the wall, directed into the wall. The two forces act on different bodies, are equal in magnitude, opposite in direction and of the same type (contact).

Q43. Explain, using Newton's laws, why a person of weight 700 N standing in a lift experiences a different reading on bathroom scales placed under their feet when the lift (a) accelerates upward at 2.0 m s^{-2} and (b) accelerates downward at 2.0 m s^{-2}. Calculate the scale readings. **[8]**

> **Answer.** The scales read the normal contact force N they exert on the person (Newton's third law). Mass m = 700/9.81 = 71.4 kg. Resultant force = ma (Newton's second law), with up positive.
>
> **(a)** N - mg = ma ⇒ N = m(g + a) = 71.4(9.81 + 2.0) = 71.4 × 11.81 = **843 N**. The reading increases because an upward resultant force is needed, so N must exceed the weight.
>
> **(b)** N - mg = m(-a) ⇒ N = m(g - a) = 71.4(9.81 - 2.0) = 71.4 × 7.81 = **558 N**. The reading decreases because the resultant force is downward, so N is less than the weight.

Q44. A rocket works by ejecting exhaust gas. In a test, a rocket ejects gas at a rate of 45 kg s^{-1} with a speed of 1200 m s^{-1} relative to the rocket. **[7]**

1. Calculate the thrust produced.
2. If the rocket has an initial total mass of 6.0×10^3 kg, find its initial acceleration at lift-off (vertically upward).

> **Answer.** **(a)** Thrust = rate of change of momentum of the exhaust = (dm/dt)v = 45 × 1200 = 54000 N = **5.4×10^4 N**.
> **(b)** Weight = mg = 6.0×10^3 × 9.81 = 5.886×10^4 N. Resultant upward force = thrust - weight = 5.4×10^4 - 5.886×10^4 = -4.86×10^3 N. This is negative, so the rocket **cannot lift off** initially (thrust < weight); acceleration would only become positive once enough fuel has burned to reduce the weight below the thrust.

Q45. Two ice skaters, of masses 60 kg and 45 kg, stand at rest facing each other and push apart. The 60 kg skater moves off at 1.5 m s^{-1}. **[7]**

1. Find the speed of the 45 kg skater.
2. Find the total kinetic energy generated and state where it came from.
3. State the total momentum of the system after the push and explain your answer.

> **Answer.** **(a)** Momentum conserved (initially zero): 0 = 60(1.5) + 45(-v) ⇒ v = 90/45 = **2.0 m s^{-1}** in the opposite direction.
> **(b)** KE = ½(60)$(1.5)^2$ + ½(45)$(2.0)^2$ = 67.5 + 90 = **157.5 J**. This came from chemical (stored) energy in the skaters' muscles, converted to kinetic energy.
> **(c)** The total momentum after the push is **zero**, the same as before. There is no external resultant horizontal force (the ice is frictionless), so by conservation of momentum the vector sum of the two equal-and-opposite momenta remains zero.

Q46. A ball A of mass m moving at speed u makes a one-dimensional elastic collision with a stationary ball B of mass 3m. Using conservation of momentum and the elastic-collision condition, determine the velocities of A and B after the collision in terms of u, and comment on the direction of A's motion. **[8]**

Answer. Let final velocities be v_A and v_B.

Momentum: $mu = mv_A + 3mv_B \Rightarrow u = v_A + 3v_B$.

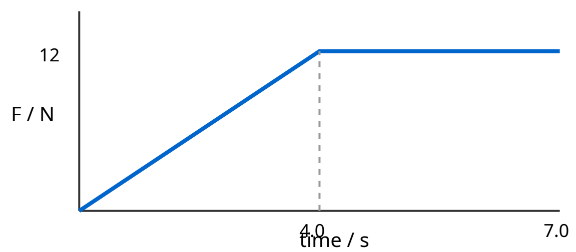
Elastic $\Rightarrow$ relative speed of approach = relative speed of separation: $u = v_B - v_A$.
Subtract: from the two equations, adding v_A: substitute $v_A = v_B - u$ into the first: $u = (v_B - u) + 3v_B = 4v_B - u \Rightarrow v_B = u/2$. Then $v_A = u/2 - u =$ **$-u/2$**.
So B moves off at **u/2** forward, and A **rebounds backward at u/2**. Since B is more massive than A, A bounces back — consistent with an elastic collision with a heavier target.

Q47. Define linear momentum and state the principle of conservation of momentum, including the condition under which it applies. Explain how this principle is consistent with Newton's third law for two interacting bodies. **[6]**

Answer. Linear momentum p = mv, the product of an object's mass and its velocity (a vector). The principle of conservation of momentum states that the total momentum of a system of interacting bodies remains constant provided no resultant **external** force acts on the system.
Consistency with Newton's third law: during an interaction, body 1 exerts force F on body 2 and, by the third law, body 2 exerts -F on body 1. These act for the same time t, so the impulses are Ft and -Ft — equal and opposite. The momentum gained by one equals the momentum lost by the other, so the total momentum is unchanged.

Q48. An object of mass 2.0 kg experiences a resultant force that varies with time as shown: the force rises linearly from 0 to 12 N over the first 4.0 s, then stays constant at 12 N for the next 3.0 s. The object starts from rest. **[7]**

1. Use the area under the force–time graph to find the impulse over the 7.0 s.
2. Hence find the final velocity.



Answer. **(a)** Impulse = area under F–t graph. Triangle (0–4 s): ½(4.0)(12) = 24 N s. Rectangle (4–7 s): (3.0)(12) = 36 N s. Total impulse = 24 + 36 = **60 N s**.

(b) Impulse = change in momentum = $m\Delta v$: 60 = 2.0 × v (starts from rest) $\Rightarrow$ v = **30 m s^{-1}**.

Q49. A raindrop falling through still air reaches a terminal velocity. **[8]**

1. Explain, in terms of the forces acting, how terminal velocity is reached.
2. Sketch and describe the acceleration–time graph from the instant the drop starts to fall.
3. State and explain what happens to the terminal velocity if the drop coalesces with others to form a larger drop.

Answer. **(a)** Initially only weight acts, so acceleration = g. As speed increases, the drag (air-resistance) force increases. The resultant force (weight - drag) decreases, so the acceleration decreases. When drag has grown equal to weight, the resultant force is zero, acceleration is zero and the drop falls at constant (terminal) velocity.

(b) Acceleration starts at g ($\approx$ 9.81 m s^{-2}) and decreases as a decaying curve towards zero (never negative).

(c) A larger drop has a greater weight-to-drag ratio (weight $\in$ volume $\in$ r^3, while drag depends on a lower power of r, roughly the cross-sectional area $\in$ r^2). A greater speed is therefore needed before drag balances weight, so the terminal velocity **increases**.

Q50. A 1500 kg car travelling at 20 m s^{-1} collides with and sticks to a stationary 900 kg car (a perfectly inelastic collision). The combined wreck then slides to rest on a road for which the constant resistive (frictional) force is 8.4×10^3 N. **[9]**

1. Find the common velocity immediately after impact.
2. Find the distance the wreck slides before stopping.
3. Find the kinetic energy lost in the collision itself.

Answer. **(a)** Momentum: 1500(20) = (1500 + 900)v $\Rightarrow$ 30000 = 2400v $\Rightarrow$ v = **12.5 m s^{-1}**.
(b) KE after impact = ½(2400)$(12.5)^2$ = 1.875×10^5 J. Work done against friction = Fd: $8.4\times10^3 \times d = 1.875\times10^5 \Rightarrow$ d = **22.3 m**.
(c) KE before impact = ½(1500)$(20)^2$ = 3.00×10^5 J. KE lost in collision = $3.00\times10^5 - 1.875\times10^5$ = **1.13×10^5 J** (transferred to heat, sound and deformation).

Q51. A particle of mass 2.0 kg moving at 4.0 m s^{-1} along the +x direction collides with a 3.0 kg particle at rest. After the collision the 2.0 kg particle moves at 2.0 m s^{-1} at 60° above the +x axis. Using conservation of momentum in two dimensions, find the velocity (magnitude and direction) of the 3.0 kg particle after the collision. **[9]**

Answer. Momentum conserved in x and y. Before: p_x = 2.0×4.0 = 8.0, p_y = 0.
2.0 kg after: v_x = 2.0 cos60° = 1.0, v_y = 2.0 sin60° = 1.732 m s^{-1}. Its momentum: (2.0, 3.46) kg m s^{-1}.
3.0 kg particle momentum = total - that of 2.0 kg: x: 8.0 - 2.0 = 6.0; y: 0 - 3.46 = -3.46 kg m s^{-1}.

Speed = $\sqrt{6.0^2 + 3.46^2}$/3.0 = $\sqrt{36 + 12.0}$/3.0 = $\sqrt{48}$/3.0 = 6.93/3.0 = **2.3 m s^{-1}**. Direction = $\tan^{-1}$(3.46/6.0) = **30° below the +x axis**.

Q52. A constant horizontal force of 50 N is applied to a 12 kg box on a rough horizontal floor. The box accelerates at 2.5 m s^{-2}. **[8]**

1. Find the frictional force acting on the box.
2. The force is then removed. Find the deceleration of the box, assuming the frictional force is unchanged, and the distance it travels before stopping if it was moving at 6.0 m s^{-1} when the force was removed.

> **Answer.** **(a)** Resultant = ma = 12 × 2.5 = 30 N. Applied 50 N, so friction = 50 - 30 = **20 N** (opposing motion).
> **(b)** With the applied force removed, only friction acts: deceleration = F/m = 20/12 = 1.67 m s^{-2}.
>
> Distance: $v^2 = u^2 - 2as$, $0 = 6.0^2 - 2(1.67)s \Rightarrow s = 36/3.33 =$ **10.8 m**.

Q53. Explain the distinction between mass and weight. A probe of mass 250 kg is taken to a planet where the gravitational field strength is 3.7 N kg^{-1}. **[7]**

1. State its mass and calculate its weight on the planet.
2. Calculate the resultant force and acceleration if a 1200 N thrust acts vertically upward on the probe at the planet's surface.

> **Answer.** Mass is the quantity of matter / the property resisting change in motion (a scalar, in kg), the same everywhere. Weight is the gravitational force on the mass, W = mg, a vector in N that depends on the local field strength.
> **(a)** Mass = **250 kg** (unchanged). Weight = mg = 250 × 3.7 = **925 N**.
> **(b)** Resultant (up positive) = 1200 - 925 = 275 N upward. Acceleration = 275/250 = **1.1 m s^{-2}** upward.

Q54. A chain of total mass 3.0 kg and length 2.0 m hangs vertically with its lower end just touching a table. It is released and falls; as it lands, the links come to rest on the table. Consider the instant when a length x of chain has already landed. **[6]**

1. Explain why the force the table exerts on the chain is greater than the weight of the chain already at rest on it.
2. Outline the two contributions to this force.

> **Answer.** **(a)** As well as supporting the weight of the stationary chain already on it, the table must also bring the moving links to rest as they arrive, i.e. it exerts a force to change their momentum (F = rate of change of momentum). This extra impulsive force makes the total force exceed the static weight of the landed portion.
> **(b)** (i) The **weight** of the length x already lying on the table, $(\rho_L x)g$ where ρ_L = mass per unit length; (ii) the **rate of change of momentum** of the links arriving each second, $(\rho_L v) \times v = \rho_L v^2$, where v is the speed of the arriving links. The total is the sum of these two terms.

Q55. State Newton's first law. A book rests on a table. Identify the forces on the book, state which pairs are Newton's-third-law pairs, and explain why the upward push of the table on the book and the weight of the book are **not** a Newton's-third-law pair even though they are equal and opposite here. **[6]**

> **Answer.** Newton's first law: an object continues at rest or at constant velocity unless acted on by a resultant external force.
> Forces on the book: its weight W (Earth pulling book down) and the normal contact force N (table pushing book up).
> Third-law pairs: (i) Earth pulls book down (W) □ book pulls Earth up with equal force; (ii) table pushes book up (N) □ book pushes table down with equal force.
> N and W are not a third-law pair because they act on the **same** body (the book), are forces of different types (contact vs gravitational), and involve only the book–Earth or book–table interactions separately. They are equal here only because the book is in equilibrium (first law), not because of the third law — if the table accelerated, N would no longer equal W.

Q56. Two gliders on a frictionless air track approach each other: glider P (0.30 kg) moves right at 0.80 m s^{-1}; glider Q (0.50 kg) moves left at 0.40 m s^{-1}. They collide and the collision is perfectly elastic. Find the velocity of each glider after the collision. **[9]**

> **Answer.** Take right as positive. Momentum: 0.30(0.80) + 0.50(-0.40) = 0.24 - 0.20 = 0.04 kg m s^{-1}. So $0.30v_P + 0.50v_Q = 0.04$. (1)
> Elastic: relative speed of approach = relative speed of separation. Approach = 0.80 - (-0.40) = 1.20 m s^{-1}. So $v_Q - v_P = 1.20$. (2)
>
> From (2) $v_Q = v_P + 1.20$. Sub into (1): $0.30v_P + 0.50(v_P + 1.20) = 0.04$ ⇒ $0.80v_P + 0.60 = 0.04$ ⇒ v_P = -0.56/0.80 = **-0.70 m s^{-1}** (P moves left at 0.70). v_Q = -0.70 + 1.20 = **+0.50 m s^{-1}** (Q moves right). Check KE: before 0.096 + 0.040 = 0.136 J; after ½(0.30)$(0.70)^2$ + ½(0.50)$(0.50)^2$ = 0.0735 + 0.0625 = 0.136 J — conserved. □

Q57. A fireworks shell of mass 2.0 kg is moving vertically upward at 15 m s^{-1} when it explodes into two fragments. One fragment of mass 0.80 kg is thrown vertically downward at 12 m s^{-1} immediately after the explosion. **[7]**

1. Find the velocity of the other fragment immediately after the explosion.
2. State and justify the total momentum immediately before and after the explosion.

> **Answer.** Take upward positive. Mass of second fragment = 2.0 - 0.80 = 1.20 kg.
>
> **(a)** Momentum: 2.0(15) = 0.80(-12) + 1.20v ⇒ 30 = -9.6 + 1.20v ⇒ 1.20v = 39.6 ⇒ v = **+33 m s^{-1}** (upward).
> **(b)** The total momentum is 30 kg m s^{-1} upward both immediately before and immediately after. During the very short explosion the internal forces are far larger than the external force (gravity), so to a good approximation there is no external impulse and momentum is conserved.

Q58. Two masses, 3.0 kg and 5.0 kg, hang from the ends of a light inextensible string passing over a frictionless pulley (an Atwood machine). The system is released from rest. **[9]**

1. By applying Newton's second law to each mass, find the acceleration of the system.
2. Find the tension in the string.
3. State the force exerted by the string on the pulley axle.

Answer. Let a be the acceleration; the 5.0 kg falls, the 3.0 kg rises.

(a) For 5.0 kg: 5.0g - T = 5.0a. For 3.0 kg: T - 3.0g = 3.0a. Adding: (5.0 - 3.0)g = 8.0a $\Rightarrow$ a = 2.0 × 9.81/8.0 = **2.45 m s^{-2}**.
(b) T = 3.0(g + a) = 3.0(9.81 + 2.45) = 3.0 × 12.26 = **36.8 N** (check: 5.0(g - a) = 5.0 × 7.36 = 36.8 N ✓).
(c) The string pulls down on the pulley on both sides, so the axle supports 2T = 2 × 36.8 = **73.6 N** (less than the total weight 78.5 N because the system is accelerating).

Q59. Sand falls vertically at a steady rate of 3.0 kg s^{-1} onto a horizontal conveyor belt that is moving at a constant 1.5 m s^{-1}. **[8]**

1. Calculate the horizontal force needed to keep the belt moving at constant speed (ignore belt friction).
2. Calculate the rate at which kinetic energy is given to the sand.
3. Explain why the power delivered by the driving force is not equal to your answer to (b), and find where the difference goes.

Answer. **(a)** The belt must accelerate the sand from 0 to 1.5 m s^{-1}: force = rate of change of momentum = (dm/dt)v = 3.0 × 1.5 = **4.5 N**.
(b) Rate of KE gain = ½(dm/dt)v^2 = ½ × 3.0 × 1.5^2 = **3.4 W**.
(c) Power delivered by the force = Fv = 4.5 × 1.5 = 6.75 W. This is **twice** the rate of KE gain. The other 3.4 W is dissipated as heat (and sound) as the sand slips on the belt before reaching belt speed — the collisions between sand grains and belt are inelastic.

Q60. A ball of mass 0.25 kg is dropped from a height of 1.8 m onto a floor and rebounds to a height of 1.0 m. The ball is in contact with the floor for 0.015 s. **[9]**

1. Find the speeds of the ball just before and just after impact.
2. Find the average resultant force on the ball during contact.
3. Hence find the average force exerted by the floor on the ball.

Answer. **(a)** Before: v = $\sqrt{2 \times 9.81 \times 1.8}$ = $\sqrt{35.3}$ = 5.94 m s^{-1} (down). After: v = $\sqrt{2 \times 9.81 \times 1.0}$ = $\sqrt{19.6}$ = 4.43 m s^{-1} (up).
(b) Take up positive. Δp = m(v_{up} - v_{down}) = 0.25(4.43 - (-5.94)) = 0.25 × 10.37 = 2.59 kg m s^{-1}. Resultant force = Δp/Δt = 2.59/0.015 = **173 N** (upward).
(c) Resultant = N - mg, so N = 173 + mg = 173 + 0.25 × 9.81 = 173 + 2.45 = **175 N** (upward). The weight is small compared with the impact force.

Q61. A radioactive nucleus of mass 220 u, initially at rest, emits an α-particle of mass 4.0 u. The α-particle leaves with a speed of 1.5×10^7 m s^{-1}. **[8]**

1. Using conservation of momentum, find the recoil speed of the daughter nucleus.
2. Find the ratio of the kinetic energy of the α-particle to that of the daughter nucleus.

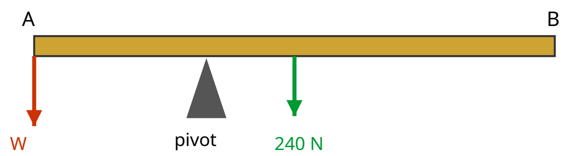
Answer. Daughter mass = 220 - 4.0 = 216 u.

(a) Momentum conserved (initially zero): $0 = 4.0v_\alpha + 216v_d \Rightarrow v_d = -(4.0/216)(1.5\times10^7)$ = **-2.8×10^5 m s^{-1}** (opposite direction).
(b) $KE = p^2/(2m)$; the two momenta are equal in magnitude, so $KE_\alpha/KE_d = m_d/m_\alpha = 216/4.0 =$ **54**. The light α-particle carries 54 times more kinetic energy than the heavy recoiling nucleus.

4 Forces, density and pressure

Q62. A uniform beam AB of weight 240 N and length 3.0 m is pivoted at a point 1.0 m from end A. A load of weight W hangs from end A, and a downward force is applied at end B to keep the beam horizontal and in equilibrium. **[7]**

1. Taking moments about the pivot, find the relationship needed for equilibrium if no force acts at B and only W balances the beam's weight.
2. Hence find the value of W that keeps the beam horizontal with no force at B.



Answer. The beam's weight (240 N) acts at its centre of gravity, the midpoint, which is 1.5 m from A, i.e. 0.5 m to the right of the pivot. W acts at A, 1.0 m to the left of the pivot.
(a) Principle of moments about the pivot: clockwise moment = anticlockwise moment. Anticlockwise (W at A): W × 1.0. Clockwise (weight at centre): 240 × 0.5.

(b) W × 1.0 = 240 × 0.5 = 120 ⇒ **W = 120 N**.

Q63. Define the **moment of a force** and the **torque of a couple**. A steering wheel of diameter 0.36 m is turned by a driver applying two equal, oppositely-directed tangential forces of 15 N at opposite ends of a diameter. **[6]**

1. Calculate the torque of the couple.
2. Explain why a couple produces rotation but no translational acceleration.

Answer. The moment of a force about a point = force × perpendicular distance from the point to the line of action of the force. The torque of a couple = one force × the perpendicular distance between the two forces.
(a) Separation = diameter = 0.36 m. Torque = 15 × 0.36 = **5.4 N m**.
(b) A couple consists of two equal and opposite forces, so their vector sum (resultant force) is zero — hence no translational acceleration (Newton's second law). However, because the forces act along different lines, they produce a net turning effect (torque), causing rotation.

Q64. A rectangular block of dimensions 0.20 m × 0.15 m × 0.10 m has a mass of 8.1 kg. **[7]**

1. Calculate its density.
2. Calculate the maximum and minimum pressures it can exert when resting on a horizontal surface on different faces.

> **Answer.** Volume = 0.20 × 0.15 × 0.10 = 3.0×10^{-3} m^3. Weight = 8.1 × 9.81 = 79.5 N.
> **(a)** Density = m/V = 8.1 / 3.0×10^{-3} = **2700 kg m^{-3}**.
> **(b)** Maximum pressure uses the smallest face: 0.15×0.10 = 0.015 m^2: p_{max} = 79.5/0.015 = **5.3×10^3 Pa**. Minimum pressure uses the largest face: 0.20×0.15 = 0.030 m^2: p_{min} = 79.5/0.030 = **2.65×10^3 Pa**.

Q65. Starting from the definitions of pressure and density, derive the expression $\Delta p = \rho g \Delta h$ for the pressure difference between two points separated by a vertical height Δh in a fluid of uniform density ρ. **[6]**

> **Answer.** Consider a horizontal area A at depth, supporting the column of fluid of height Δh above it. Volume of the column = AΔh; mass = ρAΔh. Weight of column = mg = ρAΔh g.
> This weight is supported by the extra pressure Δp acting over area A, giving a force F = Δp × A. In equilibrium these balance:
>
> $\Delta p \times A = \rho A \Delta h\, g \Rightarrow$ **$\Delta p = \rho g \Delta h$** (the area A cancels, so the result is independent of the column's cross-section).

Q66. A U-tube contains water. Oil of density 800 kg m^{-3} is poured into the left arm and floats on the water without mixing. The oil column is 12 cm long. When equilibrium is reached, the water level in the right arm is higher than the oil–water boundary in the left arm. **[7]**

1. Explain why there is a height difference.
2. Calculate the difference in the levels of the two liquid surfaces (density of water = 1000 kg m^{-3}).

> **Answer.** **(a)** At the level of the oil–water interface, the pressure from both arms must be equal (same fluid, connected). The oil is less dense, so a taller oil column is needed to produce the same pressure as the shorter, denser water column in the other arm — hence the surfaces differ in height.
> **(b)** Equate pressures at the interface level: $\rho_{oil} g h_{oil} = \rho_{water} g h_{water}$. So h_{water} = (800/1000) × 0.12 = 0.096 m = 9.6 cm of water balances the 12 cm oil column. The difference in the top surface levels = 12 - 9.6 = **2.4 cm**.

Q67. An object of volume 5.0×10^{-4} m^3 and weight 6.0 N is fully submerged and held stationary in water (density 1000 kg m^{-3}). **[8]**

1. Calculate the upthrust on the object.
2. Determine the tension in the string holding it, and state whether the string is above or below the object.
3. Explain the origin of the upthrust in terms of pressure.

Answer. **(a)** Upthrust = ρgV = 1000 × 9.81 × 5.0×10^{-4} = **4.9 N**.
(b) Weight (6.0 N down) > upthrust (4.9 N up), so the object tends to sink; the string must pull **upward** (attached above). Equilibrium: T + upthrust = weight ⇒ T = 6.0 - 4.9 = **1.1 N**.
(c) Pressure in a fluid increases with depth (Δp = ρgΔh). The pressure on the bottom face of the object is greater than on the top face, so the upward force on the bottom exceeds the downward force on the top. This net upward force is the upthrust.

Q68. A non-uniform plank of length 4.0 m rests horizontally on two supports, one at each end. The support at end A reads 320 N and the support at end B reads 480 N. **[7]**

1. Find the weight of the plank.
2. Find the distance of the centre of gravity from end A.

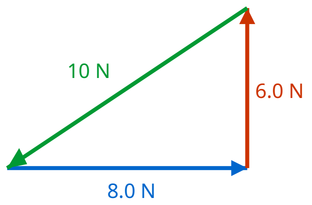
Answer. **(a)** Vertical equilibrium: weight = sum of support forces = 320 + 480 = **800 N**.
(b) Take moments about A (let centre of gravity be distance d from A). The weight (800 N) acts at d; support B (480 N) acts at 4.0 m. Moments about A: 800 × d = 480 × 4.0 ⇒ d = 1920/800 = **2.4 m from A**. (Check: it is nearer B, consistent with B carrying more load.)

Q69. State the two conditions for a rigid body to be in equilibrium. A ladder leans against a smooth vertical wall, resting on rough ground. Explain, without calculation, why the frictional force at the ground is essential for equilibrium and in which direction it acts. **[6]**

Answer. Conditions for equilibrium: (i) the resultant force in any direction is zero (no translational acceleration); (ii) the resultant torque (moment) about any point is zero (no angular acceleration). The smooth wall can only exert a horizontal (normal) force on the ladder, pushing it away from the wall. For horizontal equilibrium this must be balanced by an equal and opposite horizontal force at the base. Only friction can provide this, so friction is essential and acts **horizontally towards the wall** (preventing the base from sliding outward). Without it, there would be an unbalanced horizontal force and the ladder would slip.

Q70. Three coplanar forces act on a body in equilibrium. Forces of 6.0 N and 8.0 N act at right angles to each other. **[6]**

1. Use a vector triangle to find the magnitude and direction of the third force.
2. State the general rule that three forces in equilibrium must satisfy when drawn head-to-tail.



> **Answer.** **(a)** The third force must balance the resultant of the other two. Resultant of the 6.0 N and 8.0 N perpendicular forces = $\sqrt{6.0^2 + 8.0^2} = \sqrt{100} = 10$ N, at $\tan^{-1}(6.0/8.0) = 36.9°$ from the 8.0 N force. The third force is **10 N**, acting in exactly the **opposite direction** to this resultant (i.e. 180° from it).
> **(b)** When three forces in equilibrium are drawn head-to-tail (each starting where the previous ends), they form a **closed triangle** — the vector sum is zero.

Q71. A hydraulic press has a small piston of area 2.0×10^{-4} m² and a large piston of area 5.0×10^{-2} m². A force of 40 N is applied to the small piston. **[8]**

1. Using the fact that pressure is transmitted equally through the fluid, find the force on the large piston.
2. The small piston moves down 0.30 m. Find the distance the large piston rises, and show energy is conserved.

> **Answer.** **(a)** Pressure = F/A is the same on both pistons: $p = 40 / 2.0\times10^{-4} = 2.0\times10^{5}$ Pa. Force on large piston = $pA = 2.0\times10^{5} \times 5.0\times10^{-2}$ = **1.0×10^{4} N**.
>
> **(b)** Incompressible fluid: volume displaced is equal. $A_1d_1 = A_2d_2 \Rightarrow d_2 = (2.0\times10^{-4} \times 0.30)/(5.0\times10^{-2})$ = **1.2×10^{-3} m**. Energy check: input work = 40 × 0.30 = 12 J; output work = $1.0\times10^{4} \times 1.2\times10^{-3}$ = 12 J. Equal — energy conserved (force multiplied, distance reduced).

Q72. A uniform rod of weight 50 N and length 1.2 m is hinged at a wall at end A and held horizontal by a cable attached at end B, making an angle of 30° with the rod. **[7]**

1. By taking moments about A, find the tension in the cable.
2. Hence state one advantage of taking moments about A rather than about the centre.

> **Answer.** **(a)** The rod's weight (50 N) acts at its midpoint, 0.60 m from A. The cable tension T at B has a component perpendicular to the rod of T sin30°, acting at 1.2 m from A. Moments about A: $T \sin30° \times 1.2 = 50 \times 0.60 \Rightarrow T \times 0.5 \times 1.2 = 30 \Rightarrow 0.60T = 30 \Rightarrow T =$ **50 N**.
> **(b)** Taking moments about A **eliminates the unknown hinge force** (which acts at A and therefore has zero moment about A), so the equation contains only the tension — simplifying the solution.

Q73. A helium balloon of volume 0.40 m^3 is filled with helium of density 0.18 kg m^{-3}. The mass of the (empty) balloon fabric and its attachments is 0.050 kg. The surrounding air has density 1.2 kg m^{-3}. **[8]**

1. Calculate the upthrust on the balloon.
2. Determine the maximum additional load the balloon can just lift.

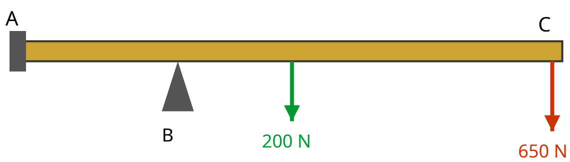
> **Answer.** **(a)** Upthrust = weight of displaced air = $\rho_{air}gV$ = 1.2 × 9.81 × 0.40 = **4.71 N**.
> **(b)** Downward weights: helium = 0.18 × 0.40 × 9.81 = 0.706 N; fabric = 0.050 × 9.81 = 0.491 N; load = mg. For the balloon to just lift: upthrust = total weight $\Rightarrow$ 4.71 = 0.706 + 0.491 + m(9.81) $\Rightarrow$ m(9.81) = 3.51 $\Rightarrow$ m = **0.358 kg** ($\approx$ 0.36 kg).

Q74. Explain what is meant by the **centre of gravity** of an object. Describe how you would find the centre of gravity of an irregular flat lamina experimentally, and explain the physics that makes the method work. **[6]**

> **Answer.** The centre of gravity is the single point at which the entire weight of the object may be taken to act.
> Method: suspend the lamina freely from a pin through a hole near its edge and let it hang in equilibrium. Hang a plumb line from the same pin and mark the vertical line on the lamina. Repeat from a different point. The centre of gravity is where the lines intersect.
> Physics: when freely suspended, the object settles so that its centre of gravity lies vertically below the pivot — otherwise the weight would exert a moment about the pivot and cause rotation. Since the centre of gravity lies on each plumb-line, their intersection locates it.

Q75. A uniform diving board of weight 200 N and length 4.0 m is bolted at one end (A) and rests on a support (B) 1.2 m from A. A diver of weight 650 N stands at the free end (C). **[9]**

1. Taking moments about the support B, find the force the bolt at A exerts on the board, stating its direction.
2. Find the force exerted by the support B on the board.



> **Answer.** The board's weight (200 N) acts at its centre, 2.0 m from A, i.e. 0.8 m to the right of B. The diver (650 N) is at C, 4.0 - 1.2 = 2.8 m to the right of B. A is 1.2 m to the left of B.
> **(a)** Moments about B (let F_A be the bolt force): the diver and board weight turn the board clockwise about B; F_A must turn it anticlockwise, so the bolt pulls the board **down**. F_A × 1.2 = 200 × 0.8 + 650 × 2.8 = 160 + 1820 = 1980 $\Rightarrow$ F_A = **1650 N downward**.
> **(b)** Vertical equilibrium: F_B = total downward forces = 200 + 650 + 1650 = **2500 N upward**. (Check moments about A: F_B×1.2 = 200×2.0 + 650×4.0 = 400 + 2600 = 3000; F_B = 2500 N □)

Q76. A simple mercury barometer consists of a vertical tube, closed at the top, with its open lower end in a dish of mercury (density 1.36×10^4 kg m^{-3}). The mercury column stands 756 mm above the surface in the dish. **[7]**

1. Calculate the atmospheric pressure.
2. State what occupies the space above the mercury in the closed tube, and explain why the height would be unchanged if a wider tube were used.

Answer. **(a)** $p = \rho g h = 1.36\times10^4 \times 9.81 \times 0.756 =$ **1.01×10^5 Pa**.
(b) Above the mercury is a near-**vacuum** (a little mercury vapour), so it exerts negligible pressure. The height depends only on the balance between atmospheric pressure and $\rho g h$, which is independent of the tube's cross-sectional area (a wider tube holds more mercury but the pressure at the base depends only on the vertical height), so h is unchanged.

Q77. A hydrometer floats vertically in a liquid. It consists of a weighted bulb and a thin uniform stem of cross-sectional area 4.0×10^{-5} m². Its total mass is 12 g. In pure water (1000 kg m^{-3}) the water line is at a certain mark on the stem. **[8]**

1. Find the total volume submerged when it floats in water.
2. When placed in a salt solution of density 1050 kg m^{-3}, find how far up the stem the liquid line moves (relative to the water mark).

Answer. Floating: upthrust = weight $\Rightarrow$ $\rho V_{sub} g = mg$ $\Rightarrow$ $V_{sub} = m/\rho$.
(a) In water: $V_{sub} = 0.012/1000 =$ **1.2×10^{-5} m³**.
(b) In salt solution: $V'_{sub} = 0.012/1050 = 1.143\times10^{-5}$ m³. Less volume is submerged, so the hydrometer rises. The change in submerged volume $= 1.2\times10^{-5} - 1.143\times10^{-5} = 5.7\times10^{-7}$ m³. This corresponds to a length of stem $\Delta L = \Delta V/A = 5.7\times10^{-7}/4.0\times10^{-5} =$ **1.4×10^{-2} m** (about 14 mm higher out of the liquid).

Q78. A uniform rod AB of weight 40 N is hinged to a wall at A and held horizontal by a light cable from B to a point on the wall directly above A, so the cable makes an angle of 40° with the rod. A load of 60 N hangs from B. **[9]**

1. By taking moments about A, find the tension in the cable.
2. Find the horizontal and vertical components of the force exerted by the hinge on the rod.

Answer. Let the rod length be L. Rod weight (40 N) acts at L/2; load (60 N) and cable act at B (distance L). Cable tension T has a component perpendicular to the rod of T sin40°.

(a) Moments about A: $T \sin40° \times L = 40 \times (L/2) + 60 \times L$. Divide by L: $T \sin40° = 20 + 60 = 80$ $\Rightarrow$ $T = 80/\sin40° = 80/0.643 =$ **124 N**.
(b) Cable pulls towards the wall and upward: horizontal component $T \cos40° = 124 \times 0.766 = 95.2$ N; vertical component $T \sin40° = 80$ N. Hinge horizontal force balances $T \cos40°$: H = **95 N (away from wall)**. Vertical: $V + T \sin40° = 40 + 60$ $\Rightarrow$ $V = 100 - 80 =$ **20 N upward**.

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5 Work, energy and power

Q79. A cyclist and bicycle of total mass 90 kg free-wheel from rest down a hill that drops a vertical height of 40 m over a slope length of 500 m. At the bottom the speed is 22 m s^{-1}. **[8]**

1. Calculate the loss in gravitational potential energy.
2. Calculate the kinetic energy gained.
3. Hence find the average resistive force acting along the slope.

Answer. **(a)** $\Delta E_P = mg\Delta h = 90 \times 9.81 \times 40 =$ **3.53×10^4 J**.
(b) $E_K = ½mv^2 = ½ \times 90 \times 22^2 = ½ \times 90 \times 484 =$ **2.18×10^4 J**.
(c) Energy dissipated by resistance = $\Delta E_P - E_K = 3.53\times10^4 - 2.18\times10^4 = 1.35\times10^4$ J. Work done against resistance = F × (slope length): $1.35\times10^4 = F \times 500 \Rightarrow F =$ **27 N**.

Q80. Derive the expression P = Fv for the power delivered by a constant force F acting on an object moving at constant velocity v in the direction of the force. A car engine delivers a useful power of 24 kW to drive a car at a steady 30 m s^{-1} on a level road. **[8]**

1. Calculate the total resistive force at this speed.
2. If the resistive force is proportional to v^2, find the useful power needed to travel at 40 m s^{-1}.

Answer. Derivation: work done in time t by force F moving the object a distance s = Fs. Power P = work/time = Fs/t = F(s/t) = **Fv** (since v = s/t).

(a) At steady speed, driving force = resistive force. $P = Fv \Rightarrow F = P/v = 24000/30 =$ **800 N**.

(b) $F \in v^2$, so at 40 m s^{-1}: $F' = 800 \times (40/30)^2 = 800 \times 1.778 = 1422$ N. $P' = F'v' = 1422 \times 40 =$ **5.7×10^4 W** (57 kW). (Note $P \in v^3$: $24 \times (40/30)^3 = 56.9$ kW.)

Q81. A pump raises water from a well 8.0 m deep and delivers it at the surface through a pipe at a speed of 3.0 m s^{-1}. The mass of water delivered per second is 15 kg. **[8]**

1. Calculate the power needed to raise the water (gain in gravitational PE per second).
2. Calculate the power needed to give the water its kinetic energy.
3. If the pump's input power is 2.0 kW, calculate its efficiency.

Answer. **(a)** $P_{PE} = (mass/s)g\Delta h = 15 \times 9.81 \times 8.0 =$ **1177 W**.
(b) $P_{KE} = ½(mass/s)v^2 = ½ \times 15 \times 3.0^2 = ½ \times 15 \times 9.0 =$ **67.5 W**.
(c) Useful output = 1177 + 67.5 = 1245 W. Efficiency = useful/input = 1245/2000 = 0.62 = **62%**.

Q82. A ball of mass 0.20 kg is dropped from a height of 2.0 m onto a hard floor and rebounds to a height of 1.4 m. **[8]**

1. Calculate the speed of the ball just before and just after the bounce.
2. Calculate the energy lost in the bounce and express it as a percentage of the initial PE.
3. State the main form into which the lost energy is transferred.

Answer. **(a)** Before: $v = \sqrt{2gh} = \sqrt{2 \times 9.81 \times 2.0} = \sqrt{39.24}$ = **6.26 m s^{-1}**. After: $v = \sqrt{2 \times 9.81 \times 1.4} = \sqrt{27.47}$ = **5.24 m s^{-1}**.
(b) Initial PE = mgh = 0.20 × 9.81 × 2.0 = 3.92 J. Rebound PE = 0.20 × 9.81 × 1.4 = 2.75 J. Energy lost = 3.92 - 2.75 = **1.18 J** = (1.18/3.92) × 100 = **30%** of the initial PE.
(c) Mainly **internal (thermal) energy** (heating of ball and floor), with some sound.

Q83. A skier of mass 70 kg starts from rest at the top of a slope and reaches a speed of 18 m s^{-1} at the bottom, having descended a vertical height of 25 m along a track of length 90 m. **[8]**

1. Show that the motion is not frictionless and find the total energy dissipated.
2. Find the average frictional force and the efficiency of the descent in converting PE to KE.

Answer. **(a)** If frictionless, $v = \sqrt{2gh} = \sqrt{2 \times 9.81 \times 25}$ = 22.1 m s^{-1}, but the actual speed is only 18 m s^{-1}, so energy is lost.
PE lost = mgh = 70 × 9.81 × 25 = 1.717×10^4 J. KE gained = ½(70)(18)^2 = 1.134×10^4 J. Energy dissipated = $1.717\times10^4 - 1.134\times10^4$ = **5.83×10^3 J**.
(b) Friction force = dissipated energy / distance = $5.83\times10^3/90$ = **65 N**. Efficiency = KE/PE = $1.134\times10^4/1.717\times10^4$ = **66%**.

Q84. State the principle of conservation of energy. A 0.050 kg arrow is fired vertically upward and leaves the bow with 30 J of kinetic energy. Air resistance does 6.0 J of work on the arrow during the ascent. **[7]**

1. Find the maximum height reached.
2. Explain why the arrow returns to the ground with less than 30 J of kinetic energy.

Answer. Conservation of energy: energy cannot be created or destroyed, only transferred from one form to another; the total energy of an isolated system is constant.
(a) At maximum height all KE has become PE plus work done against air resistance: 30 = mgh + 6.0 ⇒ mgh = 24 J ⇒ h = 24/(0.050 × 9.81) = **48.9 m**.
(b) On the way down, air resistance again does negative work on the arrow, dissipating more energy as heat. So of the 24 J of PE at the top, less than 24 J returns as KE at the ground, and the arrow arrives with less KE than the 30 J it started with.

Q85. A conveyor belt lifts gravel at a rate of 25 kg s^{-1} onto a platform 4.0 m higher than the loading point. The gravel is placed on the belt essentially at rest and leaves it moving at the belt speed of 1.5 m s^{-1}. **[7]**

1. Calculate the power needed to raise the gravel.
2. Calculate the additional power needed to give the gravel its kinetic energy.
3. State one reason the motor's actual power input must exceed the sum of these.

Answer. (a) P_{PE} = (mass/s)gΔh = 25 × 9.81 × 4.0 = **981 W**.
(b) P_{KE} = ½(mass/s)v^2 = ½ × 25 × 1.5^2 = ½ × 25 × 2.25 = **28 W**.
(c) The motor must also do work against friction in the belt/bearings and against any drag, and some energy is dissipated as heat and sound; the belt is not 100% efficient, so input power exceeds the useful 1009 W.

Q86. A 1200 kg car accelerates on a level road from 10 m s^{-1} to 25 m s^{-1} in 8.0 s. The average resistive force during this time is 500 N. **[9]**

1. Calculate the gain in kinetic energy.
2. Calculate the average driving force provided by the engine.
3. Calculate the average useful output power of the engine over this period.

Answer. (a) ΔE_K = ½m(v^2 - u^2) = ½(1200)(25^2 - 10^2) = 600(625 - 100) = 600 × 525 = **3.15×10^5 J**.
(b) Distance travelled = average speed × time = ½(10 + 25) × 8.0 = 140 m. Work by driving force = ΔE_K + work against resistance = 3.15×10^5 + 500 × 140 = 3.15×10^5 + 7.0×10^4 = 3.85×10^5 J. Driving force = 3.85×10^5/140 = **2750 N**.
(c) Useful output power = total work by driving force / time = 3.85×10^5/8.0 = **4.8×10^4 W** (48 kW).

Q87. A simple pendulum bob of mass 0.30 kg is pulled aside so that it rises a vertical height of 0.20 m above its lowest point, then released. **[8]**

1. Find the speed of the bob at the lowest point (neglecting resistance).
2. The string is 1.0 m long. Find the tension in the string at the lowest point.

Answer. (a) Energy conservation: ½mv^2 = mgh ⇒ $v = \sqrt{2gh} = \sqrt{2 \times 9.81 \times 0.20} = \sqrt{3.924}$ = **1.98 m s^{-1}**.
(b) At the lowest point the bob moves in a circle, so the resultant upward force provides the centripetal force: T - mg = mv^2/r. T = m(g + v^2/r) = 0.30(9.81 + 3.924/1.0) = 0.30 × 13.73 = **4.1 N**. (The tension exceeds the weight because it must also supply the centripetal force.)

Q88. Define **efficiency**. An electric motor rated at 500 W is used to lift a 30 kg load. In a test, it raises the load 6.0 m in 5.0 s at constant speed. [7]

1. Calculate the useful power output.
2. Calculate the efficiency.
3. Explain, in energy terms, where the 'lost' input energy goes.

> **Answer.** Efficiency = (useful energy or power output) / (total energy or power input), a ratio (often ×100%).
> **(a)** Useful work = mgh = 30 × 9.81 × 6.0 = 1766 J. Useful power = 1766/5.0 = **353 W**.
> **(b)** Efficiency = 353/500 = 0.71 = **71%**.
> **(c)** The remaining 147 W is dissipated mainly as heat — in the motor windings (resistive I^2R heating), in friction in the bearings and gears, and a little as sound. Energy is conserved: input = useful output + energy dissipated.

Q89. A spring-loaded toy of mass 0.040 kg is compressed against a spring storing 0.90 J of elastic potential energy, then released to fire vertically upward. Air resistance is negligible. [7]

1. Assuming all the elastic PE becomes kinetic energy, find the launch speed.
2. Find the maximum height reached above the launch point.

> **Answer.** **(a)** $E_{elastic} = ½mv^2 \Rightarrow 0.90 = ½(0.040)v^2 \Rightarrow v^2 = 1.80/0.040 = 45 \Rightarrow v = \sqrt{45}$ = **6.7 m s⁻¹**.
>
> **(b)** At maximum height all KE becomes PE: mgh = 0.90 $\Rightarrow$ h = 0.90/(0.040 × 9.81) = **2.3 m**. (Equivalently the 0.90 J stored is entirely converted to gravitational PE at the top.)

Q90. Explain why, for a car braking to rest, the braking distance is proportional to the square of the initial speed (assuming a constant braking force). A car travelling at 15 m s^{-1} stops in 18 m. Estimate its stopping distance from 30 m s^{-1} with the same braking force. [6]

> **Answer.** The braking force F does work Fd to remove the kinetic energy: Fd = ½mv^2. For a fixed F and m, d = (m/2F)v^2, so d $\in$ v^2 — doubling the speed quadruples the braking distance.
>
> Since d $\in$ v^2: $d_2/d_1 = (v_2/v_1)^2 = (30/15)^2 = 4$. So d_2 = 4 × 18 = **72 m**.

Q91. A ski-lift carries skiers up a 30° slope at a constant 2.5 m s^{-1}. Each chair plus skier has an average mass of 85 kg and there are 40 loaded chairs on the ascending side at any instant. Friction is negligible. [8]

1. Find the component of weight along the slope for one chair.
2. Find the total useful power the motor must supply to the ascending chairs.

> **Answer.** **(a)** Component of weight down the slope = mg sin30° = 85 × 9.81 × 0.5 = **417 N** per chair.
> **(b)** Total force to be overcome = 40 × 417 = $1.668×10^4$ N. The chairs move along the slope at 2.5 m s^{-1}, so useful power = Fv = $1.668×10^4$ × 2.5 = **$4.2×10^4$ W** (42 kW). (Equivalently, each chair rises vertically at 2.5 sin30° = 1.25 m s^{-1}, giving the same total via mg × $v_{vertical}$.)

Q92. A 0.60 kg block slides along a horizontal frictionless surface at 5.0 m s^{-1}, then encounters a rough patch 2.0 m long where the frictional force is 1.8 N, after which the surface is frictionless again and rises into a smooth ramp. **[7]**

1. Find the block's kinetic energy after crossing the rough patch.
2. Find the maximum vertical height it rises up the ramp.

Answer. **(a)** Initial KE = ½(0.60)(5.0)² = 7.5 J. Work done against friction = 1.8 × 2.0 = 3.6 J. KE after rough patch = 7.5 - 3.6 = **3.9 J**.

(b) On the smooth ramp KE converts to PE: mgh = 3.9 $\Rightarrow$ h = 3.9/(0.60 × 9.81) = **0.66 m**.

Q93. A car of mass 1100 kg climbs a straight hill inclined at 6.0° to the horizontal at a steady speed of 18 m s^{-1}. The total resistive force (friction + air) is 620 N. **[9]**

1. Find the component of the car's weight acting down the slope.
2. Find the useful output power of the engine.
3. The car then reaches a level road. Assuming the engine power and resistive force are unchanged, find the initial acceleration on the level road.

Answer. **(a)** W_{slope} = mg sin6.0° = 1100 × 9.81 × 0.1045 = **1128 N**.
(b) At steady speed the driving force = resistance + weight component = 620 + 1128 = 1748 N. P = Fv = 1748 × 18 = **$3.1×10^4$ W** (31 kW).
(c) On the level at the same instant (v = 18 m s^{-1}): driving force = P/v = 31470/18 = 1748 N. Resultant = driving - resistance = 1748 - 620 = 1128 N. a = F/m = 1128/1100 = **1.03 m s^{-2}**.

Q94. A pile driver of mass 250 kg falls freely from rest through 2.4 m onto the top of a pile of mass 150 kg, and the two move together immediately after impact. The pile is then driven 0.12 m into the ground before stopping. **[10]**

1. Find the speed of the driver just before impact.
2. Find the common speed of driver and pile immediately after the (inelastic) impact.
3. Find the average resistive force exerted by the ground on the pile.

Answer. **(a)** v = $\sqrt{2gh}$ = $\sqrt{2 × 9.81 × 2.4}$ = $\sqrt{47.1}$ = **6.86 m s^{-1}**.

(b) Momentum: 250 × 6.86 = (250 + 150)v' $\Rightarrow$ v' = 1715/400 = **4.29 m s^{-1}**.
(c) After impact KE = ½(400)(4.29)² = $3.68×10^3$ J. During the 0.12 m of penetration, the ground's resistance R and gravity both act. Work-energy: (R - (400)g) × 0.12 = KE $\Rightarrow$ R × 0.12 = 3680 + 400×9.81×0.12 = 3680 + 471 = 4151 $\Rightarrow$ R = 4151/0.12 = **$3.5×10^4$ N**.

Q95. On a fairground ride a car of total mass 400 kg starts from rest at the top of a track, descends a vertical drop of 22 m, and then travels round the inside of a vertical circular loop of radius 6.0 m. Friction is negligible. **[9]**

1. Find the speed of the car at the bottom of the drop.
2. Find the speed of the car at the top of the loop.
3. Determine whether the car maintains contact with the track at the top of the loop, justifying your answer.

Answer. **(a)** $\frac{1}{2}mv^2 = mgh \Rightarrow v = \sqrt{2 \times 9.81 \times 22} = \sqrt{431.6} = \mathbf{20.8\ m\ s^{-1}}$.

(b) The top of the loop is 2r = 12 m above the bottom. Energy: $\frac{1}{2}v_{top}^2 = \frac{1}{2}v_{bot}^2 - g(12) \Rightarrow v_{top}^2 = 431.6 - 2\times9.81\times12 = 431.6 - 235.4 = 196.2 \Rightarrow v_{top} = \mathbf{14.0\ m\ s^{-1}}$.

(c) Minimum speed to maintain contact: $mg = mv_{min}^2/r \Rightarrow v_{min} = \sqrt{gr} = \sqrt{9.81 \times 6.0} = 7.67\ m\ s^{-1}$. Since 14.0 > 7.67, the required centripetal force exceeds the weight, so the track must push inward too — the car **stays in contact** with the track.

Q96. A wind turbine has blades that sweep out a circle of radius 25 m. Air of density 1.2 kg m^{-3} moves towards the turbine at 12 m s^{-1}. **[9]**

1. Show that the mass of air passing through the swept area per second is about 2.8×10^4 kg s^{-1}.
2. Calculate the maximum theoretical power available in this moving air.
3. The turbine has an overall efficiency of 35%. Find its electrical power output.

Answer. Swept area $A = \pi r^2 = \pi(25)^2 = 1963\ m^2$.
(a) Mass per second = $\rho Av = 1.2 \times 1963 \times 12 = \mathbf{2.83\times10^4\ kg\ s^{-1}}$. ✓
(b) Power in wind = $\frac{1}{2}(mass/s)v^2 = \frac{1}{2} \times 2.83\times10^4 \times 12^2 = \frac{1}{2} \times 2.83\times10^4 \times 144 = \mathbf{2.0\times10^6\ W}$ (2.0 MW).
(c) Output = $0.35 \times 2.0\times10^6 = \mathbf{7.1\times10^5\ W}$ (0.71 MW).

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6 Deformation of solids

Q97. A steel wire of natural length 2.5 m and diameter 0.40 mm is stretched by a load of 45 N, producing an extension of 2.8 mm. **[8]**

1. Calculate the stress in the wire.
2. Calculate the strain.
3. Hence calculate the Young modulus of steel.

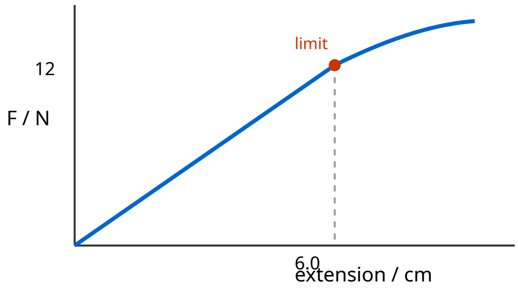
Answer. Cross-sectional area $A = \pi(d/2)^2 = \pi(0.20\times10^{-3})^2 = 1.257\times10^{-7}$ m².
(a) Stress = F/A = 45 / 1.257×10^{-7} = **3.58×10^{8} Pa**.
(b) Strain = e/L = $2.8\times10^{-3}/2.5$ = **1.12×10^{-3}** (no units).
(c) E = stress/strain = $3.58\times10^{8}/1.12\times10^{-3}$ = **3.2×10^{11} Pa**.

Q98. Describe an experiment to determine the Young modulus of a metal in the form of a long wire. State the measurements taken and the instrument used for each, how the result is obtained, and two precautions that improve accuracy. **[8]**

Answer. Clamp a long, thin wire horizontally over a pulley (or vertically from a support) with a reference marker; hang loads from the free end.
Measurements: original length L (metre rule); diameter d at several points and orientations (micrometer screw gauge), then average and use $A = \pi d^2/4$; extension e for each load (vernier scale/travelling microscope against the marker); load F = mg (known slotted masses).
Plot F (y) against e (x): gradient = EA/L, so E = gradient × L/A. (Or plot stress against strain; gradient = E.)
Precautions: use a long, thin wire so the extension is large and measurable; measure diameter in several places (wire may not be uniform) and average; keep temperature constant; ensure loading stays within the limit of proportionality (and use a comparison wire to allow for support yielding/thermal effects).

Q99. The force–extension graph for a spring is a straight line from the origin up to a load of 12 N at an extension of 6.0 cm, beyond which the line curves (the limit of proportionality is reached at 12 N). **[7]**

1. Calculate the spring constant.
2. Calculate the elastic potential energy stored at an extension of 6.0 cm.
3. State what the area under the graph beyond the limit of proportionality represents.



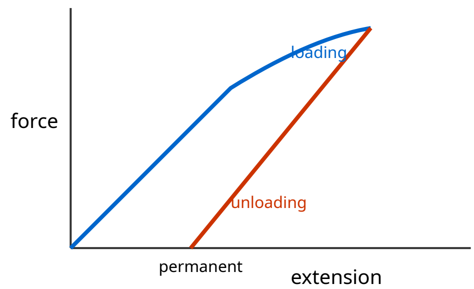
Answer. **(a)** $k = F/x = 12/0.060 =$ **200 N m^{-1}**.
(b) Within the linear region $E_P = ½Fx = ½ \times 12 \times 0.060 =$ **0.36 J** ($= ½kx^2 = ½ \times 200 \times 0.060^2 = 0.36$ J).
(c) The area under the curve beyond the limit of proportionality still represents the **work done** in stretching the spring (energy transferred to it), but this is no longer given by ½Fx and not all of it is recoverable elastic PE if the elastic limit is exceeded.

Q100. Two wires, X and Y, are made of the same material. Y has twice the diameter and half the length of X. The same load is hung from each. **[7]**

1. Compare the stress in Y with that in X.
2. Compare the extension of Y with that of X.

Answer. Same material $\Rightarrow$ same Young modulus E. Let X have diameter d, length L; then Y has diameter 2d, length L/2. Area $\in d^2$, so $A_Y = 4A_X$.
(a) Stress = F/A. Same F, but $A_Y = 4A_X$, so $\text{stress}_Y = ¼\,\text{stress}_X$ — the stress in Y is **one quarter** that in X.
(b) Extension $e = FL/(AE)$. Compared with X: $e_Y/e_X = (L_Y/L_X) \times (A_X/A_Y) = (½) \times (¼) = ⅛$. The extension of Y is **one eighth** that of X.

Q101. Distinguish between **elastic** and **plastic** deformation, and define the **elastic limit**. A metal wire is loaded beyond its elastic limit and then completely unloaded. Sketch and describe the loading and unloading force–extension lines, and identify what the area between them represents. **[7]**



Answer. Elastic deformation: the material returns to its original dimensions when the load is removed. Plastic deformation: the material does not fully return; a permanent extension remains. The elastic limit is the maximum load (or stress) beyond which the material no longer returns to its original length — deformation becomes permanent.
Loading beyond the elastic limit follows a curve; on unloading, the line comes back roughly parallel to the initial straight (elastic) portion but is displaced, meeting the extension axis at a non-zero (permanent) extension. The two lines form a loop.
The area enclosed between the loading and unloading lines represents the **energy dissipated** (as heat) in permanently deforming the material.

Q102. A vertical spring of spring constant 25 N m^{-1} hangs with a 0.15 kg mass attached at rest. **[8]**

1. Find the extension of the spring.
2. The mass is pulled down a further 4.0 cm and released. Using energy methods, find the elastic PE stored at the moment of release relative to the equilibrium position, and hence the maximum speed of the mass as it passes back through equilibrium.

Answer. **(a)** At equilibrium spring force = weight: $kx = mg \Rightarrow x = mg/k = (0.15 \times 9.81)/25 =$ **0.0589 m** (5.9 cm).
(b) Additional stretch A = 0.040 m. The extra elastic PE relative to equilibrium is $\frac{1}{2}kA^2 = \frac{1}{2} \times 25 \times 0.040^2 = 0.020$ J. As it returns to equilibrium this converts to kinetic energy (the gravitational term is already accounted for by measuring from equilibrium): $\frac{1}{2}mv^2 = 0.020 \Rightarrow v = \sqrt{2 \times 0.020/0.15} = \sqrt{0.267} =$ **0.52 m s^{-1}**.

Q103. Define **stress**, **strain** and the **Young modulus**, giving the SI unit of each. Explain why the Young modulus is a property of the material, whereas the spring constant is a property of the particular specimen. **[6]**

> **Answer.** Stress = force per unit cross-sectional area, σ = F/A (unit Pa = N m^{-2}). Strain = extension per unit original length, ε = x/L (no unit). Young modulus = stress/strain, E = σ/ε (unit Pa) — valid within the limit of proportionality.
> E is intrinsic to the material because dividing by area and by original length removes the dependence on the specimen's size and shape; a given material always has the same E. The spring constant k = F/x depends on the object's dimensions (length and cross-section) as well as the material, so two springs of the same material but different dimensions have different k.

Q104. A composite hangs from a support: a steel wire of stiffness (force constant) 800 N m^{-1} is joined end-to-end (in series) to a second wire of stiffness 1200 N m^{-1}. A load of 24 N is hung from the lower end. **[7]**

1. Explain why the two wires experience the same force.
2. Find the total extension of the combination.

> **Answer.** **(a)** The wires are in series, so the load is transmitted undiminished through both: the tension is the same (24 N) throughout, otherwise the massless junction would have a resultant force and accelerate. Hence each wire carries the full 24 N.
> **(b)** Extension of each: e = F/k. Steel: 24/800 = 0.030 m; second: 24/1200 = 0.020 m. Total extension = 0.030 + 0.020 = **0.050 m**. (Equivalently the combined stiffness is 1/k = 1/800 + 1/1200 ⇒ k = 480 N m^{-1}, giving e = 24/480 = 0.050 m.)

Q105. The graph of stress against strain for a ductile metal wire is drawn. Mark and explain the significance of (i) the region where the graph is a straight line through the origin, (ii) the point where the graph first departs from a straight line, and (iii) the region of large strain for small increases in stress. **[6]**

> **Answer.** (i) The initial straight line through the origin is the region where **stress** ∈ **strain** (Hooke's law obeyed); the gradient is the Young modulus. The material is elastic here.
> (ii) The point where the line first curves is the **limit of proportionality** — beyond it stress is no longer proportional to strain (the elastic limit lies close to or just beyond this point; beyond the elastic limit deformation becomes permanent).
> (iii) The region where the strain increases greatly for little extra stress is the region of **plastic (ductile) flow / yielding**; the wire is being permanently stretched and will not return to its original length.

Q106. A rubber cord and a metal wire are each stretched and then released. State and explain, with reference to their force–extension behaviour, one important way in which rubber differs from the metal in how it stores and returns energy. **[5]**

> **Answer.** For the metal (loaded within its elastic limit) the loading and unloading force–extension lines coincide (a straight line), so essentially all the work done in stretching is returned as elastic PE on release — little energy is lost.
> Rubber shows **hysteresis**: its loading and unloading curves differ, forming a loop, and the unloading line lies below the loading line. Less energy is returned than was put in; the area of the loop represents energy converted to heat, so the rubber warms up when repeatedly stretched and released.

Q107. A wire obeys Hooke's law up to a load of 60 N, at which its extension is 4.0 mm. **[7]**

1. Calculate the work done in stretching it to this extension.
2. An identical wire is stretched to only 3.0 mm. Calculate the elastic PE stored, and state what fraction of the answer to (a) this represents.

> **Answer.** Spring constant $k = F/x = 60/4.0\times10^{-3} = 1.5\times10^{4}$ N m^{-1}.
> **(a)** $W = ½Fx = ½ \times 60 \times 4.0\times10^{-3}$ = **0.12 J**.
> **(b)** $E_P = ½kx^2 = ½ \times 1.5\times10^{4} \times (3.0\times10^{-3})^2 = ½ \times 1.5\times10^{4} \times 9.0\times10^{-6}$ = **0.0675 J**. As a fraction of (a):
>
> 0.0675/0.12 = **0.56** (= $(3.0/4.0)^2$ = 9/16, since $E_P \in x^2$).

Q108. A load is suspended from two vertical wires attached side by side to the same support and the same load bar, so they stretch by the same amount (they are in parallel). Wire S is steel ($E = 2.0 \times 10^{11}$ Pa) and wire B is brass ($E = 1.0 \times 10^{11}$ Pa); both have the same length and the same cross-sectional area. **[8]**

1. Explain why the two wires have the same strain but different stresses.
2. Find the ratio of the tension in the steel wire to that in the brass wire.
3. If the total load is 300 N, find the tension in each wire.

> **Answer.** **(a)** They share the same support and load bar and extend by the same amount over the same original length, so strain (= extension/length) is equal. Since stress = E × strain and the two materials have different E, the stresses differ (steel carries more stress for the same strain).
> **(b)** Same area A and same strain ε: tension = stress × A = EεA. $T_S/T_B = E_S/E_B = 2.0/1.0$ = **2**.
>
> **(c)** $T_S + T_B = 300$ and $T_S = 2T_B \Rightarrow 3T_B = 300 \Rightarrow T_B$ = **100 N**, T_S = **200 N**.

Q109. A nylon climbing rope of unstretched length 50 m and cross-sectional area 8.0×10^{-5} m^2 has a Young modulus of 3.0×10^9 Pa. A climber of mass 80 kg falls and is brought to rest by the rope, which stretches elastically. **[8]**

1. Find the tension in the rope when the climber hangs at rest, and the resulting extension.
2. Find the elastic potential energy stored in the rope at this extension.

> **Answer.** **(a)** Static tension = weight = 80 × 9.81 = 785 N. Extension e = FL/(AE) = $(785 \times 50)/(8.0\times10^{-5} \times 3.0\times10^9) = 39250/2.4\times10^5$ = **0.164 m**.
> **(b)** Within the limit of proportionality E_P = ½Fe = ½ × 785 × 0.164 = **64 J**. (Equivalently the rope's stiffness k = AE/L = 4800 N m^{-1}, and $½ke^2 = ½\times4800\times0.164^2$ = 64 J.)

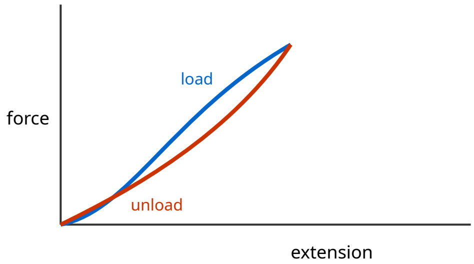
Q110. The stress–strain graph for a metal wire is a straight line from the origin to the limit of proportionality at a stress of 2.5×10^8 Pa and strain 1.25×10^{-3}, then curves upward to a maximum (the ultimate tensile stress, UTS) of 4.0×10^8 Pa before the wire breaks. **[8]**

1. Calculate the Young modulus of the metal.
2. Calculate the elastic potential energy stored per unit volume at the limit of proportionality.
3. State what the UTS represents.

> **Answer.** **(a)** E = stress/strain = $2.5\times10^8/1.25\times10^{-3}$ = **2.0×10^{11} Pa**.
> **(b)** Energy per unit volume = area under the stress–strain line (up to the limit) = ½ × stress × strain = ½ × $2.5\times10^8 \times 1.25\times10^{-3}$ = **1.6×10^5 J m^{-3}**.
> **(c)** The UTS is the **maximum stress** the material can withstand before it fails (the greatest tensile stress it can bear).

Q111. A student stretches a rubber band and a copper wire, in turn, and plots force against extension for each through a full loading-then-unloading cycle. **[8]**

1. Sketch and describe the shape of each loading–unloading curve.
2. Explain what the area enclosed by the rubber band's loop represents and its practical consequence.
3. State why the copper wire, once loaded beyond its elastic limit, does not return to its original length.



Answer. **(a)** Rubber: loading and unloading follow different S-shaped curves forming a closed loop (hysteresis); unloading lies below loading. Copper (small extensions): loading is nearly a straight line and unloading retraces it; loaded beyond the elastic limit, the unloading line is parallel to the initial line but offset, ending at a permanent extension.
(b) The area of the rubber loop is the **energy dissipated as heat** per cycle. Consequence: repeatedly stretched rubber warms up, and not all stored energy is returned (important for, e.g., car tyres and shock absorbers).
(c) Beyond the elastic limit the deformation is **plastic**: planes of atoms have slipped past one another to new permanent positions, so the wire keeps a permanent extension when unloaded.

7 Waves

Q112. Define **displacement**, **amplitude**, **frequency** and **phase difference** for a progressive wave. **[7]**
Two points on a progressive wave of wavelength 0.80 m are separated by 0.30 m along the direction of travel.

1. Calculate their phase difference in degrees and in radians.
2. State the smallest separation of two points that oscillate in antiphase.

Answer. Displacement: distance of a particle from its equilibrium position (with direction). Amplitude: the maximum displacement. Frequency: number of complete oscillations per unit time. Phase difference: the fraction of a cycle (expressed as an angle) by which one oscillation leads or lags another.
(a) One wavelength = 360° = 2π rad. Phase difference = (0.30/0.80) × 360° = **135°** = (0.30/0.80) × 2π = **2.36 rad** (= 3π/4).
(b) Antiphase means a phase difference of 180°, i.e. half a wavelength: smallest separation = λ/2 = **0.40 m**.

Q113. The trace on a cathode-ray oscilloscope (CRO) shows a sinusoidal signal. The time-base is set to 2.0 ms per division and one complete cycle occupies 3.5 divisions horizontally. The y-gain is 0.50 V per division and the trace has a peak-to-peak height of 6.0 divisions. **[6]**

1. Calculate the frequency of the signal.
2. Calculate the amplitude (peak voltage) of the signal.

Answer. **(a)** Period T = 3.5 div × 2.0 ms/div = 7.0 ms = 7.0×10^{-3} s. Frequency f = 1/T = $1/7.0\times10^{-3}$ = **143 Hz**.
(b) Peak-to-peak voltage = 6.0 div × 0.50 V/div = 3.0 V. Amplitude (peak) = half of this = **1.5 V**.

Q114. Derive the wave equation v = fλ from the definitions of speed, frequency and wavelength. A sound wave of frequency 256 Hz travels in air at 340 m s^{-1} and then passes into water where its speed becomes 1480 m s^{-1}. **[7]**

1. Find its wavelength in air and in water.
2. State, with a reason, what happens to the frequency as the wave crosses into the water.

Answer. Derivation: in one period T the wave advances exactly one wavelength λ. Speed = distance/time = λ/T. Since f = 1/T, v = λ/T = fλ.
(a) Air: λ = v/f = 340/256 = **1.33 m**. Water: λ = 1480/256 = **5.78 m**.
(b) The frequency is **unchanged** (256 Hz). Frequency is set by the source; the number of wavefronts arriving per second must equal the number leaving the boundary per second, so f is continuous across the boundary while v and λ change.

Q115. A point source emits sound uniformly in all directions with a power of 0.80 W. **[7]**

1. Calculate the intensity at a distance of 5.0 m from the source.
2. At what distance is the intensity one quarter of this value?
3. State how the amplitude of the wave at that greater distance compares with the amplitude at 5.0 m.

> **Answer.** **(a)** Intensity I = power/area = $P/(4\pi r^2) = 0.80/(4\pi \times 5.0^2) = 0.80/314 =$ **2.5×10^{-3} W m^{-2}**.
>
> **(b)** $I \in 1/r^2$. For I to fall to ¼, r^2 must increase ×4, so r increases ×2: r = **10 m**.
>
> **(c)** Intensity $\in$ amplitude^2, so if I becomes ¼, the amplitude becomes $\sqrt{¼}$ = **½** of the amplitude at 5.0 m.

Q116. Compare **transverse** and **longitudinal** waves, giving one example of each from the syllabus. **[6]** Explain, in terms of the direction of particle oscillation, why one of these wave types can be polarised and the other cannot.

> **Answer.** Transverse wave: particle oscillations are perpendicular to the direction of energy travel (e.g. electromagnetic waves / waves on a string). Longitudinal wave: particle oscillations are parallel to the direction of energy travel, producing compressions and rarefactions (e.g. sound waves). Polarisation restricts the oscillations to a single plane. Only transverse waves can be polarised, because their oscillations are perpendicular to the propagation direction and so there are many possible planes to select from. In a longitudinal wave the oscillation is along the single propagation direction, so there is no choice of plane to restrict — it cannot be polarised.

Q117. A police car siren emits sound of frequency 900 Hz. The car moves at 30 m s^{-1} along a straight road; the speed of sound is 340 m s^{-1}. **[8]**

1. Calculate the frequency heard by a stationary observer as the car approaches.
2. Calculate the frequency heard as the car recedes.
3. Explain physically why the approaching frequency is higher.

> **Answer.** Use $f_o = f_s\, v/(v \pm v_s)$, minus for approach.
> **(a)** Approaching: $f_o = 900 \times 340/(340 - 30) = 900 \times 340/310 =$ **987 Hz**.
> **(b)** Receding: $f_o = 900 \times 340/(340 + 30) = 900 \times 340/370 =$ **827 Hz**.
> **(c)** As the source approaches, each successive wavefront is emitted from a position closer to the observer, so the wavefronts are 'bunched' — the wavelength reaching the observer is shortened. Since the wave speed in air is unchanged, a shorter wavelength means a higher observed frequency.

Q118. State the approximate range of wavelengths (in free space) of the following regions of the electromagnetic spectrum and place them in order of increasing wavelength: X-rays, microwaves, visible light, infrared. State two properties that all electromagnetic waves share. **[6]**

Answer. Order of increasing wavelength: X-rays ($\approx 10^{-10}$ m) < visible (400–700 nm, i.e. $\approx$ 4–7 × 10^{-7} m) < infrared ($\approx 10^{-6}$ to 10^{-3} m) < microwaves ($\approx 10^{-3}$ to 10^{-1} m).
Shared properties (any two): all are transverse waves; all travel at the same speed $c = 3.0 \times 10^8$ m s^{-1} in free space; all can be polarised; all can be reflected, refracted, diffracted and can interfere; all transfer energy.

Q119. Unpolarised light of intensity I_0 passes through a polarising filter, then through a second (analyser) whose transmission axis is at 60° to the first. **[7]**

1. State the intensity of the light after the first filter (in terms of I_0).
2. Use Malus's law to find the intensity after the second filter.
3. State the effect on the final intensity of rotating the analyser to 90° to the first.

Answer. **(a)** An ideal polariser transmits half the incident unpolarised intensity: I_1 = **$\frac{1}{2}I_0$** (and the transmitted light is now plane-polarised).
(b) Malus: $I = I_1\cos^2\theta = \frac{1}{2}I_0 \times \cos^2 60° = \frac{1}{2}I_0 \times (0.5)^2 = \frac{1}{2}I_0 \times 0.25 =$ **$0.125\ I_0$** ($I_0/8$).
(c) At 90°, $\cos^2 90° = 0$, so the final intensity is **zero** — crossed polarisers block the light.

Q120. A wave travelling in the +x direction is described by a displacement–position graph (a 'snapshot' at one instant) and, separately, by a displacement–time graph for one particle. State what physical quantity is read directly from each graph, and explain how the two graphs differ even though both look sinusoidal. **[6]**

Answer. From the displacement–**position** graph (a snapshot of the whole wave at one instant) you read the **wavelength** λ (distance between adjacent points in phase) and the amplitude.
From the displacement–**time** graph for a single particle you read the **period** T (and hence frequency f = 1/T) and the amplitude.
They differ because the horizontal axes represent different quantities (distance vs time): the position graph shows how displacement varies across space at a frozen instant, whereas the time graph shows how one fixed particle's displacement varies as time passes. Both are sinusoidal but carry different information (λ vs T).

Q121. A loudspeaker connected to a signal generator produces a sound of constant frequency. As the output power of the generator is doubled, describe and explain the change (if any) in (i) the frequency of the sound, (ii) the wavelength, and (iii) the amplitude of the air-particle oscillations, assuming the speed of sound is constant. [6]

> **Answer.** (i) Frequency is unchanged — it is set by the signal generator, not by the power.
> (ii) Wavelength is unchanged: with $v = f\lambda$ and both v and f constant, λ is fixed.
> (iii) Intensity $\in$ power, and intensity $\in$ amplitude^2. Doubling the power doubles the intensity, so the amplitude increases by a factor $\sqrt{2} \approx 1.41$ — the air particles oscillate through larger displacements, making the sound louder.

Q122. The speed of a transverse wave on a stretched string depends on the tension and the mass per unit length. In an investigation, a wave of frequency 50 Hz on a string has a wavelength of 1.2 m. [6]

1. Calculate the wave speed.
2. The tension is then increased so the speed becomes 90 m s^{-1} while the frequency is kept at 50 Hz. Find the new wavelength.

> **Answer.** **(a)** $v = f\lambda = 50 \times 1.2 =$ **60 m s^{-1}**.
> **(b)** With f fixed at 50 Hz and v = 90 m s^{-1}: $\lambda = v/f = 90/50 =$ **1.8 m**. (The wavelength increases because the wave now travels faster at the same frequency.)

Q123. Explain what is meant by saying that a progressive wave **transfers energy** without transferring matter. Use the example of a cork floating on water as a water wave passes to support your explanation. [5]

> **Answer.** As a progressive wave travels, each particle of the medium oscillates about a fixed equilibrium position, passing on energy to the next particle, but the particles themselves are not carried along with the wave — there is no net transport of matter in the direction of travel.
> A cork on water bobs up and down (and slightly back and forth) as the wave passes but stays in roughly the same place; it is not carried across the pond. Its gain in energy (kinetic and potential as it rises) shows that energy has been delivered to it by the wave, even though the water (and the cork) has not moved along with the wave.

Q124. A beam of vertically plane-polarised light of intensity I_0 is incident on a series of three polarising filters whose transmission axes are at 0°, 45° and 90° to the vertical respectively. [7]

1. Find the intensity transmitted through all three, in terms of I_0.
2. Compare this with the intensity if the middle filter were removed, and comment.

> **Answer.** **(a)** The light is already polarised along 0°. After filter 1 (axis 0°): I_0 (unchanged). After filter 2 (45°): $I_0\cos^2 45° = 0.5\ I_0$. After filter 3 (45° from filter 2): $0.5\ I_0 \times \cos^2 45° = 0.5 \times 0.5\ I_0 =$ **$0.25\ I_0$**.
> **(b)** Without the middle filter, filters 1 and 3 are crossed (0° and 90°), so $I_0\cos^2 90° =$ **0**. Remarkably, inserting an extra filter between crossed polarisers lets light through ($0.25\ I_0$), because the middle filter rotates the plane of polarisation in two 45° steps rather than one blocked 90° step.

Q125. A microwave source and detector are used to show that microwaves are transverse and to measure their properties. Describe (i) how you would demonstrate that the microwaves are polarised, and (ii) outline how you could estimate the wavelength using a metal reflector. **[6]**

Answer. **(i) Polarisation:** place a metal grille (parallel wires) between the transmitter and receiver and rotate it about the beam axis. The detected signal varies from a maximum to (near) zero as the grille is rotated through 90°. This dependence on orientation shows the wave is plane-polarised — and hence transverse (a longitudinal wave would show no such variation).
(ii) Wavelength: direct the transmitter at a flat metal reflector so the reflected and incident waves superpose to form a stationary wave. Move the detector along the beam and locate adjacent minima (nodes); the distance between adjacent nodes is $\lambda/2$, so λ = twice that spacing.

Q126. A progressive transverse wave travels in the +x direction along a string. At time $t = 0$ the displacement–position graph is a sine curve of amplitude 8.0 mm and wavelength 0.60 m; the wave speed is 24 m s^{-1}. **[8]**

1. Calculate the frequency and period.
2. A particle is at maximum positive displacement at $t = 0$. Find the first two times after $t = 0$ when it is at zero displacement.
3. State the phase difference between two particles 0.15 m apart.

Answer. **(a)** $f = v/\lambda = 24/0.60 =$ **40 Hz**; $T = 1/f =$ **0.025 s** (25 ms).
(b) Starting at maximum, the particle first reaches zero after a quarter period and again after three-quarters: $t_1 = T/4 =$ **6.25×10^{-3} s**; $t_2 = 3T/4 =$ **1.875×10^{-2} s**.
(c) Phase difference = (0.15/0.60) × 360° = **90°** ($\pi/2$ rad).

Q127. A train sounds a horn of constant frequency 660 Hz. A stationary observer on the platform hears the frequency fall from one steady value to another as the train passes through the station without stopping. (Speed of sound = 340 m s^{-1}.) **[8]**

1. The observer measures the approaching frequency as 700 Hz. Calculate the speed of the train.
2. Calculate the frequency the observer hears as the train recedes.

Answer. **(a)** Approaching: $f_o = f_s v/(v - v_s) \Rightarrow 700 = 660 \times 340/(340 - v_s)$. $340 - v_s = 660 \times 340/700 = 320.6 \Rightarrow v_s =$ **19.4 m s^{-1}**.
(b) Receding: $f_o = 660 \times 340/(340 + 19.4) = 660 \times 340/359.4 =$ **624 Hz**.

Q128. A laser emits a parallel beam of light of power 2.0 mW and wavelength 630 nm. The beam has a circular cross-section of diameter 1.2 mm. **[7]**

1. Calculate the intensity of the beam.
2. The beam is expanded so its diameter becomes 6.0 mm with the same power. State the new intensity and the factor by which the amplitude of the light wave changes.

Answer. **(a)** Area = $\pi(0.6\times10^{-3})^2 = 1.13\times10^{-6}$ m². $I = P/A = 2.0\times10^{-3}/1.13\times10^{-6}$ = **1.77×10^3 W m^{-2}**.

(b) Diameter ×5 $\Rightarrow$ area ×25, so with the same power I becomes 1/25 of before: I' = 1770/25 = **71 W m^{-2}**. Since I $\in$ amplitude², the amplitude changes by $\sqrt{1/25}$ = **1/5** (falls to one fifth).

Q129. Vertically plane-polarised light of intensity I_0 is incident on two polarising filters. The first has its axis at angle θ to the vertical; the second (analyser) has its axis horizontal. **[8]**

1. Write an expression for the intensity transmitted through both filters in terms of I_0 and θ.
2. Show that this transmitted intensity is a maximum when θ = 45°, and find that maximum in terms of I_0.

Answer. **(a)** After filter 1: $I_1 = I_0\cos^2\theta$. Filter 2 is at (90° - θ) to filter 1, so $I = I_1\cos^2(90° - \theta) = I_0\cos^2\theta \sin^2\theta$. Using $\sin\theta\cos\theta = \frac{1}{2}\sin2\theta$: **$I = \frac{1}{4}I_0\sin^2(2\theta)$**.

(b) I is greatest when $\sin^2(2\theta) = 1$, i.e. 2θ = 90° $\Rightarrow$ θ = 45°. Then $I_{max} = \frac{1}{4}I_0 \times 1$ = **$I_0/4$**. (With no middle filter the crossed pair would transmit zero.)

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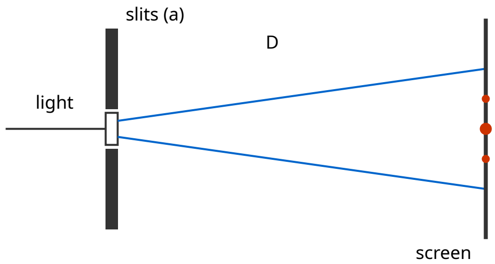
8 Superposition

Q130. State the **principle of superposition**. In a Young's double-slit experiment, coherent light of wavelength 590 nm illuminates two slits 0.45 mm apart, and fringes are observed on a screen 2.4 m away. **[7]**

1. Calculate the fringe separation.
2. Calculate the distance from the central bright fringe to the 4th-order bright fringe.



Answer. Principle of superposition: when two or more waves meet at a point, the resultant displacement is the vector sum of the individual displacements at that point.
(a) Fringe separation x = λD/a = $(590\times10^{-9} \times 2.4)/(0.45\times10^{-3}) = 1.416\times10^{-6}/4.5\times10^{-4}$ = $\mathbf{3.1\times10^{-3}}$ **m** (3.1 mm).

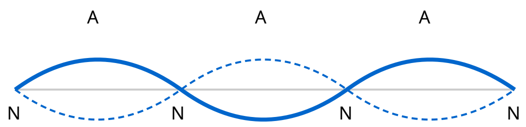
(b) Distance to the 4th bright fringe = 4x = $4 \times 3.1\times10^{-3}$ = $\mathbf{1.26\times10^{-2}}$ **m** (≈ 12.6 mm).

Q131. Explain the meaning of the terms **interference** and **coherence**. State the conditions necessary for a stable two-source interference pattern of light to be observed, and explain why two separate filament lamps of the same colour cannot produce such a pattern. **[7]**

Answer. Interference: the superposition of two (or more) waves to give a resultant of larger or smaller amplitude — constructive where they arrive in phase, destructive where in antiphase.
Coherence: two sources are coherent if they maintain a **constant phase difference** (which requires the same frequency).
Conditions for observable fringes: the sources must be coherent (constant phase difference, same frequency), have roughly equal amplitudes (for good contrast), and (for transverse waves) the same polarisation; the path difference must be within the coherence length.
Two separate lamps emit light in short, random bursts with rapidly and randomly changing phase, so their phase difference is not constant — they are incoherent. Any interference pattern changes far too quickly to be seen, averaging out to uniform illumination.

Q132. A stationary (standing) wave is set up on a string of length 1.5 m fixed at both ends, vibrating in its third harmonic (three loops). **[7]**

1. Sketch the string, marking nodes (N) and antinodes (A).
2. Determine the wavelength.
3. If the wave speed on the string is 120 m s^{-1}, find the frequency.



Answer. **(a)** Three loops between the fixed ends: nodes at both ends and at two interior points (4 nodes total), with an antinode at the centre of each loop (3 antinodes): N–A–N–A–N–A–N.

(b) Each loop is $\lambda/2$ long; three loops span the length: $3(\lambda/2) = 1.5$ m $\Rightarrow$ $\lambda = 2 \times 1.5/3 =$ **1.0 m**.
(c) $f = v/\lambda = 120/1.0 =$ **120 Hz**.

Q133. A diffraction grating has 500 lines per millimetre. It is illuminated normally by monochromatic light of wavelength 640 nm. **[8]**

1. Calculate the grating spacing d.
2. Calculate the angle of the second-order maximum.
3. Determine the highest order that can be observed.

Answer. **(a)** $d = 1/(500$ lines per mm$) = 1/(500\times10^{3}$ lines per m$) = 2.0\times10^{-6}$ m.

(b) $d \sin\theta = n\lambda$: $\sin\theta = 2 \times 640\times10^{-9}/2.0\times10^{-6} = 1.28\times10^{-6}/2.0\times10^{-6} = 0.640 \Rightarrow \theta =$ **39.8°**.

(c) Maximum order when $\sin\theta \le 1$: $n \le d/\lambda = 2.0\times10^{-6}/640\times10^{-9} = 3.13$. So the highest order is $\mathbf{n = 3}$.

Q134. Explain the formation of a stationary wave on a stretched string in terms of two progressive waves. State two ways in which a stationary wave differs from a progressive wave. **[7]**

Answer. A stationary wave forms when two progressive waves of the same frequency, wavelength and (ideally) amplitude travel in opposite directions along the string and superpose — typically an incident wave and its reflection from a fixed end. At points where the two waves are always in antiphase they cancel, giving permanent **nodes** (zero amplitude); where they are always in phase they reinforce, giving **antinodes** (maximum amplitude).
Differences (any two): (i) in a stationary wave energy is stored/not transferred along the wave, whereas a progressive wave transfers energy; (ii) amplitude varies with position (nodes to antinodes) in a stationary wave but is the same for all particles in a progressive wave; (iii) between adjacent nodes all particles oscillate in phase in a stationary wave, whereas phase changes continuously along a progressive wave.

Q135. Explain the meaning of **diffraction**. Describe, using a ripple tank, how the extent of diffraction of water waves through a gap depends on the size of the gap relative to the wavelength, and state the condition for the most pronounced diffraction. **[6]**

Answer. Diffraction is the spreading of a wave as it passes through a gap or around an obstacle/edge, into the region that would otherwise be a geometrical shadow.
In a ripple tank, straight water waves are directed at a gap in a barrier. When the gap is much wider than the wavelength, the waves pass through with only slight spreading at the edges. As the gap is narrowed towards the wavelength, the emerging waves spread out much more, becoming nearly circular. The most pronounced diffraction occurs when the **gap width is approximately equal to the wavelength**.

Q136. In a double-slit experiment the fringe separation is measured for red light and then for blue light with the same slits and screen distance. **[6]**

1. State and explain which colour gives the wider fringe spacing.
2. The screen is then moved twice as far away. State the effect on the fringe separation.
3. Instead, the slit separation is doubled. State the effect on the fringe separation.

Answer. Fringe separation $x = \lambda D/a$.

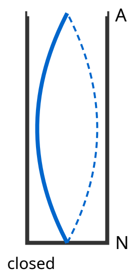
(a) Red light gives wider fringes: $x \in \lambda$, and red light has the longer wavelength than blue.

(b) $x \in D$, so doubling D **doubles** the fringe separation.

(c) $x \in 1/a$, so doubling the slit separation **halves** the fringe separation.

Q137. A tube closed at one end and open at the other resonates with sound. The shortest air column that gives resonance with a tuning fork of frequency 512 Hz is 16.2 cm long. (Speed of sound in air = 330 m s^{-1}; ignore end corrections.) **[7]**

1. Sketch the displacement pattern for this fundamental resonance, marking the node and antinode.
2. Show that the data are consistent with the speed of sound quoted.



Answer. (a) For a closed tube the closed end is a displacement **node** and the open end is an **antinode**. The fundamental has a quarter-wavelength in the tube: N at the closed end, A at the open end.

(b) Length = $\lambda/4 \Rightarrow \lambda = 4 \times 0.162 = 0.648$ m. Then $v = f\lambda = 512 \times 0.648 =$ **332 m s^{-1}**, which agrees with 330 m s^{-1} to within about 0.5% — consistent (the small excess is largely due to the neglected end correction).

Q138. Two identical loudspeakers, driven in phase by the same signal generator at a frequency of 1.7 kHz, are placed 1.5 m apart facing a listener who walks along a line parallel to the speakers. The listener notices alternate loud and quiet points. (Speed of sound = 340 m s^{-1}.) **[7]**

1. Explain the origin of the loud and quiet points.
2. State the path difference (in wavelengths) at a quiet point nearest the centre.

Answer. $\lambda = v/f = 340/1700 = 0.20$ m.
(a) Sound from the two coherent speakers superposes. Where the path difference is a whole number of wavelengths ($n\lambda$) the waves arrive in phase and interfere constructively $\Rightarrow$ loud points; where it is an odd number of half-wavelengths ($(n+½)\lambda$) they arrive in antiphase and interfere destructively $\Rightarrow$ quiet points.
(b) The first quiet point either side of the centre occurs where the path difference is **½λ** (= 0.10 m).

Q139. Describe how a diffraction grating can be used to measure the wavelength of monochromatic light. State the measurements taken and how the wavelength is calculated, and explain one advantage of using a grating rather than a double slit for this purpose. **[7]**

Answer. Shine the light normally onto a grating of known spacing d (from the lines per metre). Measure the angle θ of a diffracted maximum of order n (e.g. using a spectrometer table or by measuring the distance of the maximum from the straight-through beam and the grating–screen distance, then $\theta = \tan^{-1}(y/L)$). Apply $d \sin\theta = n\lambda$, so $\lambda = d \sin\theta/n$. Repeat for several orders and average.
Advantage over a double slit: a grating has many slits, so the maxima are much **sharper and brighter**, and the diffraction angles are large, allowing θ (and hence λ) to be measured much more precisely.

Q140. White light is passed through a diffraction grating. **[6]**

1. Explain why the central (zero-order) maximum is white but the higher orders are spread into spectra.
2. State, with a reason, which colour is diffracted through the largest angle in a given order.

Answer. **(a)** At the zero order ($n = 0$), $d \sin\theta = 0$ for all wavelengths, so $\theta = 0$ for every colour — all wavelengths are transmitted straight through together and combine to give white. For $n \geq 1$, $\sin\theta = n\lambda/d$ depends on λ, so each colour is diffracted through a different angle, spreading the light into a spectrum.

(b) Since $\sin\theta \in \lambda$, the longest wavelength is diffracted most: **red** light is diffracted through the largest angle in a given order (violet the least).

Q141. Microwaves of wavelength 2.8 cm from a transmitter are reflected back by a metal sheet, forming a stationary wave. A small probe detector is moved along the line between transmitter and sheet. **[6]**

1. Explain what the detector registers as it is moved, and why.
2. Calculate the distance the detector moves between five successive positions of minimum signal.

Answer. **(a)** The incident and reflected microwaves superpose to form a stationary wave with fixed nodes and antinodes. As the probe moves, the detected signal rises and falls: it registers minima at the nodes (waves in antiphase, cancel) and maxima at the antinodes (in phase, reinforce).
(b) Adjacent nodes are $\lambda/2$ = 1.4 cm apart. Five successive minima span four gaps: distance = 4 × 1.4 = **5.6 cm**.

Q142. In a Young's double-slit arrangement, the fringes are found to have low contrast (bright fringes not very bright, dark fringes not fully dark). Suggest two possible reasons for the poor contrast and, for each, state how it could be improved. **[6]**

Answer. Reason 1: the two slits transmit unequal amplitudes (e.g. slits of different width or uneven illumination), so destructive interference is incomplete and dark fringes are not fully dark. Improve by making the slits equal in width and illuminating them symmetrically.
Reason 2: the source is not sufficiently monochromatic (a range of wavelengths gives overlapping fringe patterns of different spacing that wash out) or the slit–source arrangement gives poor coherence. Improve by using monochromatic (e.g. laser or filtered) light and a single narrow source slit to ensure coherence. (Also reducing background/ambient light raises contrast.)

Q143. In a Young's double-slit experiment, light of unknown wavelength illuminates two slits 0.30 mm apart. On a screen 1.8 m away, the distance across 10 bright-fringe spacings is measured as 31.5 mm. **[8]**

1. Calculate the fringe separation and hence the wavelength of the light.
2. Explain why the distance across several fringes is measured rather than a single fringe spacing.
3. State and explain the effect on the pattern of replacing the light with one of longer wavelength.

Answer. **(a)** Fringe separation x = 31.5/10 = 3.15 mm = 3.15×10^{-3} m. $\lambda = ax/D = (0.30\times10^{-3} \times 3.15\times10^{-3})/1.8 = 9.45\times10^{-7}/1.8$ = **5.3×10^{-7} m** (530 nm).
(b) The fringes are close together, so measuring one spacing has a large percentage uncertainty; measuring across many spacings and dividing reduces the percentage uncertainty in x (the absolute measuring uncertainty is spread over a larger distance).

(c) $x = \lambda D/a \ ^{\in} \lambda$, so a longer wavelength gives **wider** fringe spacing (the pattern spreads out).

Q144. A diffraction grating is marked '600 lines per mm'. It is illuminated normally by light containing two wavelengths, 480 nm and 640 nm. **[9]**

1. Calculate the grating spacing.
2. Calculate the angular separation of the two wavelengths in the first-order spectrum.
3. Determine the highest order in which the 640 nm line can be seen.

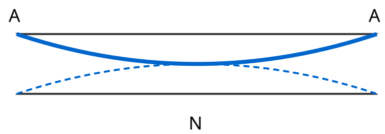
Answer. **(a)** $d = 1/(600\times10^3) =$ **1.667×10^{-6} m**.

(b) First order ($n = 1$): $\sin\theta_{480} = 480\times10^{-9}/1.667\times10^{-6} = 0.288 \Rightarrow \theta = 16.7°$. $\sin\theta_{640} = 640\times10^{-9}/1.667\times10^{-6} = 0.384 \Rightarrow \theta = 22.6°$. Angular separation = 22.6 - 16.7 = **5.9°**.

(c) Highest order for 640 nm: $n \leq d/\lambda = 1.667\times10^{-6}/640\times10^{-9} = 2.6$, so **n = 2**.

Q145. A tube of length 0.85 m is open at both ends. The speed of sound in air is 340 m s^{-1}. (Neglect end corrections.) **[8]**

1. Sketch the displacement pattern of the fundamental (first harmonic) and state the positions of the antinodes.
2. Calculate the fundamental frequency.
3. Calculate the frequency of the third harmonic.



Answer. **(a)** An open–open tube has displacement **antinodes at both open ends** and a node in the middle for the fundamental. The fundamental fits half a wavelength in the tube (A–N–A).

(b) $L = \lambda/2 \Rightarrow \lambda = 2L = 1.70$ m. $f_1 = v/\lambda = 340/1.70 =$ **200 Hz**.
(c) For an open tube all harmonics are present: $f_3 = 3f_1 = 3 \times 200 =$ **600 Hz**.

Q146. Two loudspeakers, 0.80 m apart, are driven in phase at the same frequency and face a wall 5.0 m away. A microphone moved along a line parallel to the speakers detects a series of maxima and minima. Adjacent maxima are found to be 0.66 m apart. **[8]**

1. Treating this as a two-source interference pattern, calculate the wavelength of the sound.
2. Calculate the frequency (speed of sound = 340 m s^{-1}).
3. State and explain what happens to the spacing of the maxima if the frequency is increased.

Answer. **(a)** Using $x = \lambda D/a$ with $x = 0.66$ m, $D = 5.0$ m, $a = 0.80$ m: $\lambda = xa/D = (0.66 \times 0.80)/5.0 =$ **0.106 m** ($\approx$ 0.11 m).
(b) $f = v/\lambda = 340/0.106 =$ **3.2×10^3 Hz** (3.2 kHz).

(c) Higher frequency $\Rightarrow$ shorter wavelength; since $x = \lambda D/a$, the spacing of the maxima **decreases** (the pattern gets closer together).

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9 Electricity

Q147. A copper wire of cross-sectional area 1.5×10^{-6} m² carries a current of 3.2 A. Copper has 8.5×10^{28} free electrons per cubic metre. **[7]**

1. Calculate the drift speed of the electrons.
2. Comment on the size of your answer, given that electrical signals travel near the speed of light.

Answer. **(a)** $I = Anvq \Rightarrow v = I/(Anq) = 3.2/(1.5\times10^{-6} \times 8.5\times10^{28} \times 1.60\times10^{-19})$. Denominator = $1.5\times10^{-6} \times 8.5\times10^{28} \times 1.60\times10^{-19} = 2.04\times10^{4}$. $v = 3.2/2.04\times10^{4}$ = $\mathbf{1.6\times10^{-4}\ m\ s^{-1}}$ ($\approx$ 0.16 mm s^{-1}).
(b) The drift speed is extremely small, yet a lamp lights almost instantly. This is because the electric field is established through the whole circuit at near light speed, so all the free electrons everywhere start drifting almost simultaneously — it is not necessary for an electron to travel from switch to lamp.

Q148. State what is meant by 'the charge on charge carriers is quantised'. A charge of 96 C flows past a point in a conductor in 40 s. **[6]**

1. Calculate the current.
2. Calculate the number of electrons that pass the point in this time.

Answer. Quantised charge means charge exists only in whole-number multiples of the elementary charge $e = 1.60\times10^{-19}$ C; you cannot have a fraction of e as free charge.
(a) I = Q/t = 96/40 = **2.4 A**.
(b) Number of electrons = Q/e = $96/1.60\times10^{-19}$ = $\mathbf{6.0\times10^{20}}$.

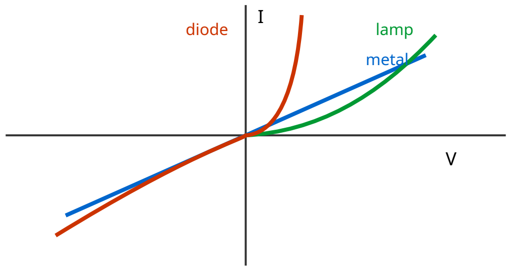
Q149. Define **potential difference** across a component. A 12 V battery drives a current of 0.50 A through a motor for 2.0 minutes, lifting a load. **[8]**

1. Calculate the charge that flows.
2. Calculate the electrical energy transferred.
3. If the motor lifts a 1.5 kg load by 40 cm in this time, find its efficiency.

Answer. Potential difference across a component = the electrical energy transferred to other forms per unit charge passing through it (V = W/Q).
(a) Q = It = 0.50 × 120 = **60 C**.
(b) W = VQ = 12 × 60 = **720 J** (= VIt).
(c) Useful output = mgh = 1.5 × 9.81 × 0.40 = 5.89 J. Efficiency = 5.89/720 = 0.0082 = **0.82%** (very low — most energy is dissipated as heat in the windings and against friction).

Q150. The I–V characteristics of a metallic conductor at constant temperature, a filament lamp, and a semiconductor diode differ markedly. **[8]**

1. Sketch all three characteristics on the same axes.
2. Explain the shape of the filament-lamp characteristic.
3. State how the resistance of the diode varies as the forward voltage increases from zero.



Answer. **(a)** Metallic conductor: a straight line through the origin (constant gradient). Filament lamp: an S-shaped curve through the origin that becomes less steep (gradient decreases) as |V| increases. Diode: negligible current for reverse and small forward voltages, then a sharp rise in current above the forward 'turn-on' voltage (≈ 0.6 V); essentially zero current when reverse-biased.
(b) As the current increases, the filament heats up; the higher temperature increases the resistance (metal ions vibrate more, impeding electron flow), so a given increase in V produces a smaller increase in I — hence the curve bends over.
(c) The diode's resistance is very high at zero/low forward voltage and **decreases** rapidly once the turn-on voltage is exceeded (it conducts freely).

Q151. State **Ohm's law**. A nichrome wire has resistance 6.0 Ω. A second nichrome wire of the same material has twice the length and half the diameter of the first. **[7]**

1. Calculate the resistance of the second wire.
2. Explain, using R = ρL/A, each factor by which the resistance changes.

Answer. Ohm's law: the current in a metallic conductor is directly proportional to the potential difference across it, provided physical conditions (especially temperature) remain constant.

(a) R = ρL/A. Doubling length ×2. Half the diameter ⇒ area ×¼ (A ∈ d^2), so 1/A ×4. Overall factor = 2 × 4 = 8. R_2 = 8 × 6.0 = **48 Ω**.

(b) L doubles ⇒ R doubles (longer path, more collisions). The diameter halving reduces the cross-sectional area to one quarter, and since R ∈ 1/A this multiplies R by 4. The two effects combine to multiply R by 8.

Q152. A resistivity experiment gives, for a wire of uniform diameter 0.38 mm and length 1.20 m, a resistance of 4.6 Ω. **[8]**

1. Calculate the resistivity of the material.
2. The diameter is measured with a micrometer to ±0.01 mm and the length to ±1 cm. State which measurement contributes most to the percentage uncertainty in the resistivity, and why.

> **Answer.** $A = \pi d^2/4 = \pi(0.38\times10^{-3})^2/4 = 1.134\times10^{-7}$ m².
> **(a)** $\rho = RA/L = (4.6 \times 1.134\times10^{-7})/1.20 = 5.216\times10^{-7}/1.20 =$ **4.3×10^{-7} Ω m**.
>
> **(b)** Percentage uncertainties: length 1/120 ≈ 0.83%; diameter 0.01/0.38 = 2.6%, but d is **squared** in the area, so it contributes 2 × 2.6% = 5.2%. The **diameter** dominates, both because its fractional uncertainty is larger and because it enters as d^2.

Q153. Define **resistance**. A 60 W, 240 V lamp and a 100 W, 240 V lamp are each operated at their rated values. **[7]**

1. Calculate the operating resistance of each lamp.
2. State, with a reason, which lamp has the thicker or shorter filament (assuming same material and operating temperature).

> **Answer.** Resistance = the ratio of the potential difference across a component to the current through it, R = V/I.
>
> **(a)** $P = V^2/R \Rightarrow R = V^2/P$. 60 W lamp: $R = 240^2/60 = 57600/60 =$ **960 Ω**. 100 W lamp: $R = 240^2/100 =$ **576 Ω**.
> **(b)** The 100 W lamp has the lower resistance. Since $R = \rho L/A$, a lower R means a shorter and/or thicker filament; the higher-power (100 W) lamp therefore has the **thicker (or shorter) filament** to allow a larger current at the same voltage.

Q154. Explain, in terms of charge carriers, why the resistance of (i) a thermistor decreases as its temperature rises and (ii) a light-dependent resistor (LDR) decreases as light intensity increases. Contrast this with the behaviour of a metal as its temperature rises. **[7]**

> **Answer. (i) Thermistor:** it is a semiconductor. As temperature rises, more electrons gain enough energy to become free charge carriers, so the number density n of carriers increases greatly. This increase outweighs the increased vibration of the lattice, so resistance falls.
> **(ii) LDR:** incident light delivers energy that frees more charge carriers, again increasing n and so decreasing the resistance as intensity increases.
> **Metal contrast:** in a metal the number of free electrons is essentially fixed; raising the temperature makes the positive ions vibrate more, scattering the electrons more often, so the resistance **increases**.

Q155. A 2.5 kW electric heater is designed for a 230 V mains supply. **[7]**

1. Calculate the current it draws and its resistance.
2. The heater is instead connected to a 115 V supply. Assuming its resistance is unchanged, calculate the power now dissipated and comment on the result.

> **Answer.** **(a)** $I = P/V = 2500/230 =$ **10.9 A**. $R = V/I = 230/10.9 =$ **21.2 Ω** (or $R = V^2/P = 230^2/2500 = 21.2$ Ω).
>
> **(b)** $P = V^2/R = 115^2/21.2 = 13225/21.2 =$ **625 W**. Halving the voltage quarters the power ($P \in V^2$), so the heater delivers only one quarter of its rated output — it would heat much less effectively.

Q156. The current in a conductor is not the same thing as the drift of individual electrons. Using $I = Anvq$, explain what happens to the drift speed when a wire of the same material narrows to half its cross-sectional area, if the current is unchanged. State what this implies about charge conservation. **[6]**

> **Answer.** For a given material n and q are fixed, and the current I is the same all along a series conductor (charge is conserved — charge cannot pile up). Since $I = Anvq$, if A halves then v must double to keep $I = Anvq$ constant. So the electrons drift **twice as fast** in the narrow section.
> This is consistent with conservation of charge: the same number of coulombs pass every cross-section per second, so where the wire is thinner the carriers simply move faster to carry the identical current.

Q157. A student has a length of resistance wire and wants a 5.0 Ω resistor. The wire has resistance 2.0 Ω per metre. **[7]**

1. Find the length needed.
2. The student instead cuts three equal lengths of this same 2.5 m wire and connects them in parallel. Find the resistance of the combination.

> **Answer.** **(a)** Length = R/(resistance per metre) = 5.0/2.0 = **2.5 m**.
> **(b)** A 2.5 m length has resistance 2.5 × 2.0 = 5.0 Ω. Cut into three equal pieces: each is 5.0/3 = 1.667 Ω. In parallel: $1/R = 3/1.667 = 1.80$, so R = **0.56 Ω**. (Equivalently three equal resistors in parallel give $R_{each}/3 = 1.667/3 = 0.56$ Ω.)

Q158. A 100 m length of overhead power cable has a total resistance of 0.40 Ω and carries a current of 250 A. **[6]**

1. Calculate the power dissipated as heat in the cable.
2. Explain, with reference to $P = I^2R$, why electrical power is transmitted at high voltage and low current.

Answer. **(a)** $P = I^2R = 250^2 \times 0.40 = 62500 \times 0.40 =$ **2.5×10^4 W** (25 kW).
(b) For a given power $P = VI$ to be delivered, using a higher voltage allows a smaller current. Since the heating loss in the cables is I^2R, reducing the current has a large effect (loss $\in I^2$): e.g. halving the current quarters the wasted power. Transmitting at high voltage and low current therefore minimises energy wasted as heat in the transmission lines.

Q159. Two wires P and Q are connected in series across a battery. P has resistance 4.0 Ω and Q has resistance 8.0 Ω. **[6]**

1. Compare the current in P with that in Q.
2. Compare the power dissipated in P with that in Q.
3. Explain your answer to (b) using $P = I^2R$.

Answer. **(a)** In series the current is the **same** through both: $I_P = I_Q$.
(b) Power in Q is twice that in P: $P_Q/P_P = 8.0/4.0 = 2$. So Q dissipates **twice** the power of P.

(c) With the same current I in both, $P = I^2R \in R$. Since Q has double the resistance of P, it dissipates double the power. (The larger resistor gets hotter in a series circuit.)

Q160. A metal wire of diameter 0.50 mm carries a current of 1.8 A. The metal has 5.9×10^{28} free electrons per cubic metre. **[8]**

1. Calculate the mean drift speed of the electrons.
2. The wire is joined to a second wire of the same material but half the diameter. Find the drift speed in the thinner wire and explain your reasoning.

Answer. $A = \pi(0.25\times10^{-3})^2 = 1.963\times10^{-7}$ m².
(a) $v = I/(Anq) = 1.8/(1.963\times10^{-7} \times 5.9\times10^{28} \times 1.60\times10^{-19})$. Denominator $= 1.853\times10^{3}$. $v = 1.8/1.853\times10^{3} =$ **9.7×10^{-4} m s^{-1}**.

(b) The current is the same (series). Half the diameter $\Rightarrow$ quarter the area. Since $I = Anvq$ with I, n, q fixed, $v \in 1/A$, so v increases ×4: $v' = 4 \times 9.7\times10^{-4} =$ **3.9×10^{-3} m s^{-1}**. Charge conservation requires the same current everywhere, so carriers move faster where the wire is thinner.

Q161. In an experiment to measure the resistivity of a metal, the resistance of a wire is measured for several lengths, giving a straight-line graph of resistance R against length L. The graph passes through the origin with a gradient of 6.8 Ω m^{-1}. The wire has a uniform diameter of 0.34 mm. **[8]**

1. Explain why a graphical method is preferable to a single measurement.
2. Use the gradient to calculate the resistivity of the metal.

> **Answer.** **(a)** Plotting many points and drawing a best-fit line averages out random errors in individual readings and reveals systematic problems (e.g. a non-zero intercept from contact/lead resistance). A line through the origin here confirms R $\in$ L and gives a more reliable gradient than one measurement.
>
> **(b)** R = ρL/A $\Rightarrow$ gradient R/L = ρ/A, so ρ = gradient × A. A = $\pi(0.17\times10^{-3})^2 = 9.08\times10^{-8}$ m^2. ρ = 6.8 × 9.08×10^{-8} = **6.2×10^{-7} Ω m**.

Q162. A semiconductor diode and a fixed 220 Ω resistor are connected in series across a variable supply. When the supply is 2.0 V, the current is 6.0 mA. **[7]**

1. Calculate the potential difference across the resistor and hence across the diode.
2. Explain why the diode does not obey Ohm's law, referring to its I–V characteristic.

> **Answer.** **(a)** V_R = IR = 6.0×10^{-3} × 220 = 1.32 V. By Kirchhoff's second law V_{diode} = 2.0 - 1.32 = **0.68 V** (a typical forward turn-on voltage).
> **(b)** Ohm's law requires current proportional to p.d. (constant resistance). The diode's I–V graph is highly non-linear: almost no current until the forward voltage reaches $\approx$ 0.6 V, then a steep rise; and negligible current when reverse-biased. The ratio V/I is not constant, so the diode is non-ohmic.

Q163. An electric shower is rated at 8.5 kW when connected to a 230 V mains supply. **[8]**

1. Calculate the current drawn and the resistance of the heating element.
2. The element is a nichrome wire of resistivity 1.1×10^{-6} Ω m and cross-sectional area 4.0×10^{-7} m^2. Calculate the required length of wire.

> **Answer.** **(a)** I = P/V = 8500/230 = **37 A**. R = V^2/P = 230^2/8500 = 52900/8500 = **6.2 Ω**.
>
> **(b)** R = ρL/A $\Rightarrow$ L = RA/ρ = $(6.2 \times 4.0\times10^{-7})/1.1\times10^{-6} = 2.49\times10^{-6}/1.1\times10^{-6}$ = **2.3 m**.

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10 D.C. circuits

Q164. A cell of e.m.f. 1.5 V and internal resistance 0.50 Ω is connected to an external resistor of 2.5 Ω. **[7]**

1. Calculate the current in the circuit.
2. Calculate the terminal potential difference.
3. Calculate the power wasted inside the cell.

Answer. **(a)** $\varepsilon = I(R + r)$: $I = \varepsilon/(R + r) = 1.5/(2.5 + 0.50) = 1.5/3.0 =$ **0.50 A**.
(b) Terminal p.d. $V = IR = 0.50 \times 2.5 =$ **1.25 V** ($= \varepsilon - Ir = 1.5 - 0.50 \times 0.50 = 1.25$ V).
(c) Power in internal resistance $= I^2r = 0.50^2 \times 0.50 =$ **0.125 W**.

Q165. State **Kirchhoff's first and second laws**, and state the conservation principle underlying each. **[6]** Explain briefly why the first law must hold at any junction in a circuit that carries a steady current.

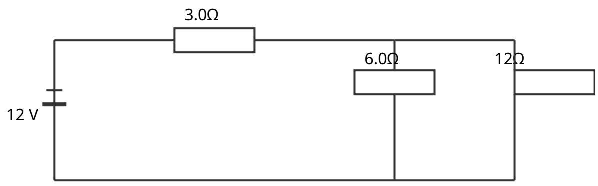
Answer. First law: the sum of the currents entering a junction equals the sum of the currents leaving it ($\Sigma I_{in} = \Sigma I_{out}$). This follows from conservation of **charge**.
Second law: around any closed loop, the sum of the e.m.f.s equals the sum of the p.d.s ($\Sigma\varepsilon = \Sigma IR$). This follows from conservation of **energy**.
The first law must hold for a steady current because charge cannot accumulate at a junction (there is no build-up): every coulomb per second arriving must leave, otherwise charge (and hence the current) would change with time, contradicting the steady state.

Q166. Derive, using Kirchhoff's laws, the formula for the combined resistance of two resistors R_1 and R_2 connected in parallel across a supply of p.d. V. **[6]**

Answer. The two resistors share the same p.d. V (both across the supply). By Kirchhoff's first law the total current splits: $I = I_1 + I_2$.
For each resistor $I_1 = V/R_1$ and $I_2 = V/R_2$. For the single equivalent resistance R: $I = V/R$.
Substituting: $V/R = V/R_1 + V/R_2$. Dividing through by V gives $\mathbf{1/R = 1/R_1 + 1/R_2}$.

Q167. In the circuit, a 12 V battery of negligible internal resistance is connected to a 3.0 Ω resistor in series with a parallel combination of a 6.0 Ω and a 12 Ω resistor. **[8]**

1. Find the total resistance.
2. Find the current drawn from the battery.
3. Find the current in the 12 Ω resistor.



> **Answer.** **(a)** Parallel pair: $1/R_p = 1/6.0 + 1/12 = 2/12 + 1/12 = 3/12 \Rightarrow R_p = 4.0\ \Omega$. Total = 3.0 + 4.0 = **7.0 Ω**.
> **(b)** I = V/R = 12/7.0 = **1.71 A**.
> **(c)** P.d. across the parallel pair = $I \times R_p$ = 1.71 × 4.0 = 6.86 V. Current in 12 Ω: I = 6.86/12 = **0.57 A**.

Q168. Distinguish between the **e.m.f.** of a source and the **potential difference** across its terminals, in terms of energy. Explain why the terminal p.d. falls below the e.m.f. when the source delivers a current, and describe one situation in which the terminal p.d. equals the e.m.f. **[6]**

> **Answer.** The e.m.f. is the electrical energy transferred (from other forms) per unit charge driven around the **complete** circuit by the source. The terminal p.d. is the electrical energy transferred to the **external** circuit per unit charge.
> When a current I flows, some energy per unit charge is dissipated inside the source across its internal resistance r (an amount Ir). Hence $V = \varepsilon - Ir < \varepsilon$.
> The terminal p.d. equals the e.m.f. when no current flows (open circuit, I = 0), so there is no 'lost volts' Ir — for example when the terminal p.d. is measured by a voltmeter of (ideally) infinite resistance.

Q169. A potential divider consists of two fixed resistors, R_1 = 4.0 kΩ (top) and R_2 = 6.0 kΩ (bottom), across a 9.0 V supply, with the output taken across R_2. **[8]**

1. Calculate the output voltage on no load.
2. A voltmeter of resistance 6.0 kΩ is now connected across the output. Calculate the new output reading, and comment.

> **Answer.** **(a)** $V_{out} = V R_2/(R_1 + R_2)$ = 9.0 × 6.0/(4.0 + 6.0) = 9.0 × 0.60 = **5.4 V**.
> **(b)** The 6.0 kΩ voltmeter is in parallel with R_2 (6.0 kΩ): combined = (6.0 × 6.0)/(6.0 + 6.0) = 3.0 kΩ. V_{out} = 9.0 × 3.0/(4.0 + 3.0) = 9.0 × 3.0/7.0 = **3.9 V**.
>
> Comment: the reading has dropped substantially (5.4 → 3.9 V) — the voltmeter 'loads' the divider by drawing current. A high-resistance voltmeter would disturb the circuit far less.

Q170. Explain the principle of a **potentiometer** used as a means of comparing two potential differences, and describe the role of the galvanometer in the null method. State one advantage of this method over using a voltmeter. **[7]**

Answer. A potentiometer is a uniform resistance wire carrying a steady current from a driver cell, so the p.d. along it is proportional to length. A source to be measured is connected (via a galvanometer) between one end and a sliding contact. The contact is moved until the galvanometer reads **zero** (the balance/null point): then the p.d. of the source equals the p.d. across that length of wire, so p.d. $\in$ balance length. Comparing two sources' balance lengths gives the ratio of their p.d.s ($V_1/V_2 = L_1/L_2$).
The galvanometer detects the balance condition: at null it carries no current, indicating the two p.d.s are equal and opposite.
Advantage: at balance the source drives **no current**, so no current flows through its internal resistance — the true e.m.f./p.d. is measured without the 'lost volts' error that a current-drawing voltmeter would introduce.

Q171. Two identical cells, each of e.m.f. 1.5 V and internal resistance 0.30 Ω, are connected to a 1.4 Ω external resistor. **[8]**

1. Find the current if the cells are connected in series.
2. Find the current if the cells are connected in parallel.

Answer. **(a)** Series: total e.m.f. = 1.5 + 1.5 = 3.0 V; total internal resistance = 0.30 + 0.30 = 0.60 Ω. I = 3.0/(1.4 + 0.60) = 3.0/2.0 = **1.5 A**.
(b) Parallel: e.m.f. stays 1.5 V; internal resistances in parallel = 0.30/2 = 0.15 Ω. I = 1.5/(1.4 + 0.15) = 1.5/1.55 = **0.97 A**. (Series gives more current here because the doubled e.m.f. outweighs the increased internal resistance for this relatively small external resistor.)

Q172. A thermistor is used in a potential divider to make a temperature-sensing circuit that switches on a warning when the temperature becomes too high. The thermistor (negative temperature coefficient) is placed in series with a fixed resistor across a 6.0 V supply. **[7]**

1. State how the thermistor's resistance changes as temperature rises.
2. Explain how you would connect the output so that the output voltage **rises** as the temperature rises, and justify your choice.

Answer. **(a)** As temperature rises, the thermistor's resistance **decreases**.
(b) Take the output across the **fixed resistor** (i.e. put the thermistor in the 'top' position). Then $V_{out} = V \times R_{fixed}/(R_{thermistor} + R_{fixed})$. As temperature rises, $R_{thermistor}$ falls, so the denominator falls and the fraction increases — V_{out} rises. This rising voltage can be fed to a switching circuit to trigger the warning at high temperature.

Q173. In the circuit shown, a 6.0 V battery of internal resistance 0.50 Ω supplies two resistors: R_1 = 10 Ω in parallel with R_2 = 15 Ω. Use Kirchhoff's laws to find the terminal p.d. and the current in each resistor. **[8]**

> **Answer.** Parallel combination: $1/R_p = 1/10 + 1/15 = 3/30 + 2/30 = 5/30 \Rightarrow R_p = 6.0\ \Omega$.
>
> Total current from battery (loop, Kirchhoff II): $\varepsilon = I(R_p + r)$: $6.0 = I(6.0 + 0.50) \Rightarrow I = 6.0/6.5 = 0.923$ A.
> Terminal p.d. = $\varepsilon - Ir = 6.0 - 0.923 \times 0.50$ = **5.54 V** (= p.d. across the parallel pair).
> Currents (Kirchhoff I): $I_1 = 5.54/10$ = **0.55 A**; $I_2 = 5.54/15$ = **0.37 A**. Check: 0.55 + 0.37 = 0.92 A = total.
> ✓

Q174. A battery of e.m.f. ε and internal resistance r delivers current to a variable external resistance R. Explain how the terminal p.d. and the power delivered to R vary as R is decreased from a very large value towards zero, and state the condition for maximum power transfer to R. **[7]**

> **Answer.** As R decreases, the current $I = \varepsilon/(R + r)$ increases. The terminal p.d. $V = \varepsilon - Ir$ therefore **decreases**: from ε (open circuit, $R \to \infty$) down towards 0 (short circuit, $R \to 0$).
>
> The power in R, $P = I^2R = \varepsilon^2R/(R + r)^2$, is zero at both extremes ($R \to 0$ gives $R \to 0$; $R \to \infty$ gives $I \to 0$), so it rises to a maximum in between. Maximum power is delivered to R when **R = r** (the external resistance matches the internal resistance); then the transfer efficiency is 50%.

Q175. A student connects a 12 V battery (internal resistance 1.0 Ω) to a network: a 4.0 Ω resistor in series with two 6.0 Ω resistors that are in parallel with each other. **[9]**

1. Find the current drawn from the battery.
2. Find the power dissipated in the 4.0 Ω resistor.
3. Find the total power delivered by the battery and the fraction wasted internally.

> **Answer.** Two 6.0 Ω in parallel = 3.0 Ω. External total = 4.0 + 3.0 = 7.0 Ω. Including internal: 7.0 + 1.0 = 8.0 Ω.
> **(a)** $I = \varepsilon/(R + r) = 12/8.0$ = **1.5 A**.
> **(b)** $P_{4\Omega} = I^2R = 1.5^2 \times 4.0 = 2.25 \times 4.0$ = **9.0 W**.
> **(c)** Total power from battery = $\varepsilon I = 12 \times 1.5 = 18$ W. Wasted internally = $I^2r = 1.5^2 \times 1.0 = 2.25$ W.
> Fraction wasted = 2.25/18 = **0.125 (12.5%)**.

Q176. Describe how a light-dependent resistor (LDR) can be used in a potential divider to switch on a lamp automatically when it gets dark. Draw/describe the arrangement and explain the operation. **[6]**

> **Answer.** Connect the LDR in series with a fixed resistor across a supply, taking the output across the **LDR** to a switching circuit (transistor/relay).
> In the light the LDR has low resistance, so only a small fraction of the supply voltage appears across it — V_{out} is low and the lamp stays off. As it gets dark the LDR's resistance rises sharply, so a larger share of the supply voltage now appears across the LDR — V_{out} rises. When V_{out} exceeds the switching threshold, the circuit turns the lamp on. (Swapping the LDR and fixed resistor would instead switch the lamp on in bright light.)

Q177. Two resistors of 200 Ω and 300 Ω are available. A 5.0 V supply of negligible internal resistance is connected across them. **[7]**

1. Find the total power dissipated when they are in series and when they are in parallel.
2. State which arrangement dissipates more power and explain why using $P = V^2/R$.

> **Answer.** **(a)** Series: R = 200 + 300 = 500 Ω; $P = V^2/R = 5.0^2/500 = 25/500 =$ **0.050 W**. Parallel: 1/R = 1/200 + 1/300 = 3/600 + 2/600 = 5/600 ⇒ R = 120 Ω; P = 25/120 = **0.208 W**.
> **(b)** The **parallel** arrangement dissipates more power. Both resistors have the full 5.0 V across them in parallel, and with $P = V^2/R$ a smaller total resistance (120 Ω vs 500 Ω) at the same voltage means greater power. (In series the shared voltage and larger total resistance reduce the power.)

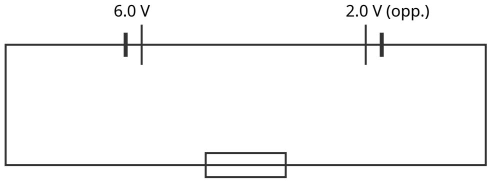
Q178. A battery is connected in turn to two different external resistors. With a 4.0 Ω resistor the current is 0.60 A; with a 9.0 Ω resistor the current is 0.30 A. **[9]**

1. Set up two equations using ε = I(R + r) and solve them to find the e.m.f. and internal resistance of the battery.
2. Calculate the terminal p.d. in each case and comment.

> **Answer.** **(a)** ε = I(R + r): Case 1: ε = 0.60(4.0 + r) = 2.40 + 0.60r. Case 2: ε = 0.30(9.0 + r) = 2.70 + 0.30r.
> Set equal: 2.40 + 0.60r = 2.70 + 0.30r ⇒ 0.30r = 0.30 ⇒ r = **1.0 Ω**. Then ε = 2.40 + 0.60 = **3.0 V**.
> **(b)** Case 1: V = IR = 0.60 × 4.0 = 2.4 V. Case 2: V = 0.30 × 9.0 = 2.7 V. The terminal p.d. is higher when the external resistance is larger (smaller current ⇒ smaller 'lost volts' Ir), approaching ε as R increases.

Q179. In the circuit, two cells are connected in the same loop with two resistors. Cell 1 has e.m.f. 6.0 V **[8]** and internal resistance 0.50 Ω; cell 2 has e.m.f. 2.0 V and internal resistance 0.30 Ω and is connected so that it opposes cell 1. They drive current through an external 3.2 Ω resistor.

1. Using Kirchhoff's second law, find the current in the circuit.
2. Find the potential difference across the 3.2 Ω resistor.



Answer. **(a)** The cells oppose, so the net e.m.f. = 6.0 - 2.0 = 4.0 V. Total resistance = 0.50 + 0.30 + 3.2 = 4.0 Ω. Kirchhoff II: $\Sigma\varepsilon = \Sigma IR \Rightarrow 4.0 = I \times 4.0 \Rightarrow I =$ **1.0 A** (driven by the stronger cell).
(b) V = IR = 1.0 × 3.2 = **3.2 V**.

Q180. A potential divider is made from a thermistor and a 2.2 kΩ fixed resistor in series across a 12 V **[8]** supply, with the output taken across the fixed resistor. At 20°C the thermistor has resistance 3.0 kΩ; at 60°C it has resistance 0.80 kΩ.

1. Calculate the output voltage at each temperature.
2. State and explain how this circuit could be used to switch on a cooling fan when the temperature rises.

Answer. $V_{out} = V \times R_{fixed}/(R_{th} + R_{fixed})$.
(a) At 20°C: V_{out} = 12 × 2.2/(3.0 + 2.2) = 12 × 2.2/5.2 = **5.1 V**. At 60°C: V_{out} = 12 × 2.2/(0.80 + 2.2) = 12 × 2.2/3.0 = **8.8 V**.
(b) As temperature rises the thermistor's resistance falls, so V_{out} (across the fixed resistor) rises. Feeding V_{out} to a switching circuit (transistor/relay or comparator) that turns on the fan once V_{out} exceeds a set threshold means the fan switches on automatically at high temperature.

Q181. A uniform potentiometer wire AB is 100.0 cm long and carries a steady current from a driver cell. A standard cell of e.m.f. 1.018 V is balanced at 63.6 cm from A. An unknown cell is then balanced at 48.2 cm from A. **[8]**

1. Explain why no current flows through the cell being tested at the balance point.
2. Calculate the e.m.f. of the unknown cell.
3. State one advantage of this method for measuring e.m.f.

Answer. **(a)** At balance the p.d. across the length of potentiometer wire exactly equals and opposes the cell's e.m.f., so there is no net driving p.d. around that loop and the galvanometer reads zero — no current flows through the tested cell.

(b) E.m.f. $\in$ balance length: $\varepsilon_x/\varepsilon_{std} = L_x/L_{std}$ $\Rightarrow$ $\varepsilon_x = 1.018 \times 48.2/63.6 =$ **0.771 V**.
(c) Because no current is drawn from the cell at balance, there is no 'lost volts' across its internal resistance, so the **true e.m.f.** is measured (unlike a voltmeter, which draws current).

11 Particle physics

Q182. In the α-particle scattering experiment, a beam of α-particles is directed at a thin gold foil. **[7]**
Most pass almost straight through, a small fraction are deflected through large angles, and a very few are reflected back.

1. State what each of these three observations tells us about the atom.
2. Explain why a thin foil and a vacuum are used.

Answer. **(a)** Most pass straight through $^{\Rightarrow}$ the atom is mostly empty space. A small fraction deflected through large angles $^{\Rightarrow}$ there is a concentrated region of positive charge (the α-particles are positive and repelled). A very few rebounding almost backwards $^{\Rightarrow}$ this positive charge (and almost all the mass) is concentrated in a very small, dense **nucleus** that can turn a fast α-particle around.
(b) A thin foil ensures α-particles pass through with (mostly) a single scattering event, so deflections are due to individual nuclei; a vacuum prevents the α-particles from being absorbed or scattered by air molecules before reaching the foil/detector.

Q183. A nuclide is written as $^{A}{}_{Z}X$. **[7]**

1. State what A and Z represent.
2. The nuclide $^{238}{}_{92}U$ decays by α-emission to thorium (Th). Write the balanced decay equation and state the values of A and Z for the thorium nuclide.
3. Explain what is conserved in the decay.

Answer. **(a)** A = nucleon (mass) number = number of protons + neutrons; Z = proton (atomic) number = number of protons (which also equals the nuclear charge in units of e).

(b) $^{238}{}_{92}U$ $^{\rightarrow}$ $^{234}{}_{90}Th$ + $^{4}{}_{2}\alpha$ (i.e. $^{4}{}_{2}He$). Thorium: **A = 234, Z = 90**.
(c) Both **nucleon number** (238 = 234 + 4) and **charge**/proton number (92 = 90 + 2) are conserved (as are mass–energy and momentum).

Q184. Describe the composition, relative mass and relative charge of α-, β^-- and γ-radiations. Explain why α-particles from a given decay have discrete energies whereas β-particles have a continuous range of energies. **[8]**

> **Answer.** α: a helium nucleus (2 protons + 2 neutrons); relative mass 4 (u), relative charge +2e. β^-: a fast electron; relative mass $\approx$ 1/1840 u, charge -e. γ: a high-energy photon (electromagnetic); zero mass, zero charge.
> In α-decay the energy released is shared between just two bodies (the α-particle and the recoiling nucleus), so conservation of momentum and energy fixes the α energy at discrete value(s) — a line spectrum. In β-decay a third particle, the (anti)neutrino, is also emitted; the fixed decay energy is shared among **three** bodies in varying proportions, so the β-particle can take any energy up to a maximum — giving a continuous spectrum.

Q185. State what is meant by an **antiparticle**. Write the nuclear equation for the β^+ decay of a nucleus ${}^{A}_{Z}X$, naming all particles produced, and state the change (if any) in A and Z. **[7]**

> **Answer.** An antiparticle has the same mass as its corresponding particle but the opposite charge (and other opposite quantum properties); e.g. the positron is the antiparticle of the electron.
>
> β^+ decay: a proton changes to a neutron, emitting a positron and an electron neutrino: ${}^{A}_{Z}X \rightarrow {}^{A}_{Z-1}Y + {}^{0}_{+1}e + \nu_e$ (positron e^+ and neutrino ν).
> Nucleon number A is **unchanged**; proton number Z **decreases by 1** (charge conserved: Z = (Z-1) + 1).

Q186. Quarks are fundamental particles. **[7]**

1. State the charges (in units of e) of the up, down and strange quarks.
2. Show that the quark composition of a proton (uud) and a neutron (udd) gives the correct charges.
3. State the composition of a baryon and of a meson in terms of quarks.

> **Answer.** **(a)** up: +⅔e; down: -⅓e; strange: -⅓e. (Each corresponding antiquark has the opposite charge.)
> **(b)** Proton uud: (+⅔) + (+⅔) + (-⅓) = +1e. □ Neutron udd: (+⅔) + (-⅓) + (-⅓) = 0. □
> **(c)** A baryon is composed of **three quarks**; a meson is composed of **one quark and one antiquark**.

Q187. Describe, in terms of quarks, the changes that occur during (i) β^- decay and (ii) β^+ decay. State which fundamental particles (leptons) are emitted in each. **[7]**

> **Answer.** **(i) β^- decay:** a neutron becomes a proton, i.e. a **down quark changes to an up quark** (udd $\rightarrow$ uud). An electron and an electron **antineutrino** ($\bar{\nu}_e$) are emitted.
>
> **(ii) β^+ decay:** a proton becomes a neutron, i.e. an **up quark changes to a down quark** (uud $\rightarrow$ udd). A positron and an electron **neutrino** (ν_e) are emitted.
> Electrons, positrons and neutrinos are all leptons (fundamental particles).

Q188. The unified atomic mass unit (u) is used as a unit of mass in nuclear physics. **[6]**

1. State the approximate value of 1 u in kilograms.
2. A carbon-12 nucleus has 6 protons and 6 neutrons. Using masses of 1.007 u (proton) and 1.009 u (neutron), calculate the total mass of the separate nucleons in u, and comment on why this exceeds 12 u.

Answer. (a) 1 u $\approx$ 1.66×10^{-27} kg.
(b) Total = 6(1.007) + 6(1.009) = 6.042 + 6.054 = **12.096 u**. This is greater than the 12 u mass of the assembled carbon-12 nucleus: when the nucleons bind together, energy (the binding energy) is released, and by mass–energy equivalence this appears as a loss of mass (the 'mass defect'). So the bound nucleus is lighter than its separated parts.

Q189. Classify each of the following as a lepton, a baryon or a meson, giving a brief reason: (i) a proton, (ii) an electron, (iii) a particle consisting of an up quark and a down antiquark, (iv) a neutrino. **[6]**

Answer. (i) Proton — a **baryon**: it is made of three quarks (uud).
(ii) Electron — a **lepton**: it is a fundamental particle, not made of quarks.
(iii) up + anti-down — a **meson**: one quark plus one antiquark (this is a π^+).
(iv) Neutrino — a **lepton**: a fundamental particle (emitted in β-decay).

Q190. A nucleus ${}^{14}_{6}C$ undergoes β^- decay to nitrogen (N). **[6]**

1. Write the full decay equation, including the antineutrino.
2. State how the nucleon number and proton number of the daughter compare with the parent, and confirm charge is conserved.

Answer. (a) ${}^{14}_{6}C \rightarrow {}^{14}_{7}N + {}^{0}_{-1}e + \bar{\nu}_e$ (an electron and an electron antineutrino).

(b) Nucleon number is unchanged (14 → 14); proton number increases by 1 (6 → 7) as a neutron converts to a proton. Charge: parent +6e = daughter (+7e) + electron (-e) + antineutrino (0) = +6e. ✓ Charge is conserved.

Q191. A nucleus of ${}^{228}_{90}Th$ decays by emitting an α-particle to form radium (Ra), which then decays by emitting a β^- particle to form actinium (Ac). **[8]**

1. Write the equation for the α-decay, giving the nucleon and proton numbers of the radium nuclide.
2. Write the equation for the β^--decay, including the antineutrino, giving the nucleon and proton numbers of the actinium nuclide.

Answer. (a) ${}^{228}_{90}Th \rightarrow {}^{224}_{88}Ra + {}^{4}_{2}\alpha$. Radium: **A = 224, Z = 88** (A: 228 = 224 + 4; Z: 90 = 88 + 2). ✓

(b) ${}^{224}_{88}Ra \rightarrow {}^{224}_{89}Ac + {}^{0}_{-1}e + \bar{\nu}_e$. Actinium: **A = 224, Z = 89** (nucleon number unchanged; a neutron becomes a proton so Z rises by 1). Charge: 88 = 89 + (-1) ✓

Q192. A Σ^+ baryon has a charge of +1e and the quark composition uus; a K^- meson has a charge of -1e. **[7]**

1. Verify that the composition uus gives the correct charge for the Σ^+.
2. The K^- is composed of a strange quark and an up antiquark. Show that this gives a charge of -1e, using the fact that an antiquark has the opposite charge to its quark.

Answer. Quark charges: u = +⅔e, s = -⅓e; antiquarks have opposite sign so ū (anti-up) = -⅔e.
(a) uus: (+⅔) + (+⅔) + (-⅓) = +1e. ▯ Correct for Σ^+.
(b) K^- = s + ū (strange quark + up antiquark): (-⅓) + (-⅔) = -1e. ▯ A meson (one quark + one antiquark) with charge -1e.

Q193. In a nuclear process, a proton within a nucleus is converted into a neutron. **[7]**

1. Name this type of decay and state the two leptons emitted.
2. Describe the change at the quark level.
3. Explain why the emitted positron has a range of kinetic energies rather than a single value.

Answer. **(a)** This is **β^+ decay**. The particles emitted are a **positron** (e^+) and an **electron neutrino** (ν_e) — both leptons.
(b) A proton (uud) becomes a neutron (udd): an **up quark changes into a down quark**.
(c) The fixed decay energy is shared among three bodies — the daughter nucleus, the positron and the neutrino. Because the neutrino carries away a variable share, the positron can take any energy from zero up to a maximum, giving a continuous energy spectrum.

Q194. The masses of some particles are: proton 1.00728 u, neutron 1.00867 u, helium-4 nucleus 4.00151 u. (1 u = 1.66×10^{-27} kg; c = 3.00×10^{8} m s^{-1}.) **[7]**

1. Calculate the mass defect when two protons and two neutrons combine to form a helium-4 nucleus.
2. Explain the physical meaning of this mass defect.

Answer. **(a)** Mass of separate nucleons = 2(1.00728) + 2(1.00867) = 2.01456 + 2.01734 = 4.03190 u.
Mass defect Δm = 4.03190 - 4.00151 = **0.03039 u** (= $0.03039 \times 1.66\times10^{-27} = 5.04\times10^{-29}$ kg).
(b) The assembled nucleus has less mass than its separate parts. When the nucleons bind together, energy (the binding energy) is released; by mass–energy equivalence ($E = \Delta mc^2$) this energy corresponds to the 'lost' mass. To pull the nucleus apart again, the same energy would have to be supplied.

SECTION B — Multiple choice answers [Q1–Q122]

1 Physical quantities and units

1. The power P dissipated in a resistor is $P = kI^aR^b$, where I is current and R is resistance. For the equation to be homogeneous, a and b are:

- **A** a = 1, b = 1
- **B** **a = 2, b = 1**
- **C** a = 2, b = 2
- **D** a = 1, b = 2

Correct: B. Power = kg m^2 s^{-3}; R has units kg m^2 s^{-3} A^{-2}. For the amperes to cancel, $A^{a-2b} = A^0 \Rightarrow$ a = 2b; matching gives b = 1, a = 2. This is $P = I^2R$.

---

2. A quantity is calculated from $Y = p^2q / \sqrt{r}$. The percentage uncertainties are p: 2%, q: 3%, r: 8%. The percentage uncertainty in Y is:

- **A** 9%
- **B** **11%**
- **C** 13%
- **D** 17%

Correct: B. Add percentage uncertainties, each × its power: $p^2 \rightarrow$ 2(2%) = 4%; q $\rightarrow$ 3%; $\sqrt{r} = r^{1/2} \rightarrow$ ½(8%) = 4%. Total = 4 + 3 + 4 = 11%.

---

3. Which of the following is a set of three vector quantities only?

- **A** mass, velocity, force
- **B** **displacement, acceleration, momentum**
- **C** work, power, energy
- **D** velocity, speed, weight

Correct: B. Vectors have magnitude and direction. Displacement, acceleration and momentum are all vectors. The others contain scalars (mass, work, power, energy, speed).

---

4. The Young modulus has SI base units:

- **A** kg m s^{-2}
- **B** **kg m^{-1} s^{-2}**
- **C** kg m^2 s^{-2}
- **D** kg m^{-2} s^{-1}

> **Correct: B.** Young modulus = stress/strain = pressure (strain is dimensionless). Pressure = force/area = (kg m s^{-2})/m^2 = kg m^{-1} s^{-2}.

---

5. A micrometer has a zero error of -0.03 mm (it reads -0.03 mm when closed). A wire measured with it gives a reading of 0.42 mm. The true diameter is:

- **A** 0.39 mm
- **B** 0.42 mm
- **C** **0.45 mm**
- **D** 0.48 mm

> **Correct: C.** A reading of -0.03 when closed means all readings are 0.03 mm too small, so the correction is to add 0.03: true = 0.42 + 0.03 = 0.45 mm. (True = reading - zero error = 0.42 - (-0.03).)

---

6. Which estimate is closest to the order of magnitude of the kinetic energy of a 60 kg sprinter running at 10 m s^{-1}?

- **A** 3×10^2 J
- **B** **3×10^3 J**
- **C** 3×10^4 J
- **D** 3×10^5 J

> **Correct: B.** $E_K = ½mv^2 = ½ \times 60 \times 100 = 3000$ J = 3×10^3 J.

---

7. Two forces of 7.0 N and 4.0 N act at a point. Which value could NOT be the magnitude of their resultant?

- **A** 3.0 N
- **B** 7.0 N
- **C** 11.0 N
- **D** **2.0 N**

> **Correct: D.** The resultant of two forces lies between |7-4| = 3 N and 7+4 = 11 N inclusive. A value of 2.0 N is below the minimum of 3.0 N, so it is impossible.

---

8. A student states a result as $(4.0 \pm 0.2) \times 10^3$ and another quantity as $(2.00 \pm 0.05) \times 10^2$. The percentage uncertainty in their product is closest to:

- **A** 2.5%
- **B** 5.0%
- **C** **7.5%**
- **D** 10%

Correct: C. First: 0.2/4.0 = 5.0%. Second: 0.05/2.00 = 2.5%. Product ⇒ add: 5.0 + 2.5 = 7.5%.

---

9. Which pair of quantities has the same SI base units?

- **A** energy and power
- **B** **pressure and stress**
- **C** momentum and force
- **D** density and pressure

Correct: B. Pressure and stress are both force/area (kg m^{-1} s^{-2}). The other pairs differ (e.g. momentum kg m s^{-1} vs force kg m s^{-2}).

---

10. The equation for the speed of a wave on a string is $v = \sqrt{T/\mu}$, where T is tension. For homogeneity, the base units of μ must be:

- **A** **kg m^{-1}**
- **B** kg m
- **C** kg s^{-1}
- **D** kg m^{-2}

Correct: A. $v^2 = T/\mu \Rightarrow \mu = T/v^2 = (\text{kg m s}^{-2})/(\text{m s}^{-1})^2 = (\text{kg m s}^{-2})/(\text{m}^2\ \text{s}^{-2}) = \text{kg m}^{-1}$. (μ is mass per unit length.)

---

11. The kinetic energy of a body is $E = ½mv^2$. If the mass is measured to ±2% and the speed to ±3%, the percentage uncertainty in E is:

- **A** 5%
- **B** 6%
- **C** **8%**
- **D** 11%

Correct: C. v is squared, so it contributes 2 × 3% = 6%; m contributes 2%. Total = 2 + 6 = 8%.

---

12. A rectangular plate has sides (50.0 ± 0.5) mm and (20.0 ± 0.5) mm. The percentage uncertainty in its area is closest to:

- **A** 1.0%
- **B** 2.0%
- **C** **3.5%**
- **D** 5.0%

> **Correct: C.** %unc in each side: 0.5/50.0 = 1.0% and 0.5/20.0 = 2.5%. Area is a product ⇒ add: 1.0 + 2.5 = 3.5%.

2 Kinematics

13. A ball is thrown vertically up and returns to the thrower. Taking up as positive, which graph best describes its acceleration against time during the whole flight (ignoring air resistance)?

- **A** a positive constant then negative constant
- **B** **a constant negative value throughout**
- **C** zero at the top, negative elsewhere
- **D** a straight line decreasing through zero

> **Correct: B.** Throughout the flight the only force is gravity, so a = -g (constant negative) at all times, including at the highest point where the velocity (not the acceleration) is zero.

---

14. A car accelerates uniformly from rest. In the first second it travels distance d. The distance it travels in the third second is:

- **A** 3d
- **B** 5d
- **C** **7d**
- **D** 9d

> **Correct: C.** For uniform acceleration from rest, distance in the n-th second $\in$ (2n-1). 1st second: 1 unit = d. 3rd second: 2(3)-1 = 5 units = 5d. (Total in 3 s = 9 units; in 2 s = 4 units; difference = 5.)

---

15. A projectile is launched at 30° to the horizontal with speed u. At the highest point of its path, its speed is:

- **A** 0
- **B** u sin30°
- **C** **u cos30°**
- **D** u

> **Correct: C.** At the top the vertical component is zero; only the horizontal component u cos30° remains, so the speed there is u cos30°.

---

16. The displacement–time graph of an object is a curve that becomes steeper with time. This represents:

- **A** constant velocity
- **B** **increasing velocity (acceleration)**
- **C** decreasing velocity
- **D** the object at rest

> **Correct: B.** The gradient of a displacement–time graph is the velocity. A curve getting steeper means an increasing gradient, i.e. increasing velocity — the object is accelerating.

---

17. Two stones are dropped from a cliff, the second 1.0 s after the first. As they fall (no air resistance), the vertical distance between them:

- **A** stays constant
- **B** **increases**
- **C** decreases
- **D** first increases then decreases

> **Correct: B.** Both accelerate at g, so their velocities differ by a constant g×1.0 s = 9.81 m s^{-1}. Since the first is always moving faster, the gap between them increases with time.

---

18. A ball falls from rest and passes a window 1.5 m tall in 0.20 s. Approximately how far above the top of the window was it released? (g = 9.8 m s^{-2})

- **A** 1.5 m
- **B** 2.0 m
- **C** **2.5 m**
- **D** 3.0 m

> **Correct: C.** Average speed past window = 1.5/0.20 = 7.5 m s^{-1} = speed at mid-window. Speed at top ≈ 7.5 - 9.8×0.10 = 6.52 m s^{-1}. Fall distance to top = $v^2/2g$ = $6.52^2/19.6$ = 2.17 ≈ 2.5 m (nearest option).

---

19. An object moves with constant velocity in the x-direction and constant acceleration in the y-direction (starting with zero y-velocity). Its path is:

- **A** a straight line
- **B** a circle
- **C** **a parabola**
- **D** a hyperbola

> **Correct: C.** $x \in t$ and $y \in t^2$, so $y \in x^2$ — a parabola. This is projectile-type motion.

---

20. A car travels 40 m at 20 m s^{-1} then 40 m at 40 m s^{-1}. Its average speed over the 80 m is:

- **A** 24 m s^{-1}
- **B** **26.7 m s^{-1}**
- **C** 30 m s^{-1}
- **D** 33.3 m s^{-1}

> **Correct: B.** Times: 40/20 = 2.0 s and 40/40 = 1.0 s; total 3.0 s for 80 m. Average = 80/3.0 = 26.7 m s^{-1} (not the arithmetic mean 30).

---

21. A velocity–time graph is a straight line from (0 s, 8 m s^{-1}) to (4 s, 0 m s^{-1}), then continues as a straight line to (6 s, -6 m s^{-1}). The total distance travelled (not displacement) in 6 s is:

- **A** 10 m
- **B** 16 m
- **C** **22 m**
- **D** 6 m

> **Correct: C.** Distance = total area magnitude. 0–4 s: ½(4)(8) = 16 m forward. 4–6 s: ½(2)(6) = 6 m (backward, but distance counts magnitude). Total distance = 16 + 6 = 22 m.

---

22. A stone thrown horizontally from a height h lands a horizontal distance R away. If it were thrown horizontally at twice the speed from the same height, the horizontal distance would be:

- **A** R
- **B** $\sqrt{2}$ R
- **C** **2R**
- **D** 4R

> **Correct: C.** Time to fall depends only on h (same), t = $\sqrt{2h/g}$. Horizontal distance = speed × t, so doubling the speed doubles R.

---

23. A ball is projected horizontally at 15 m s^{-1} from a height of 20 m. The speed with which it strikes the ground is about (g = 9.8 m s^{-2}):

- **A** 15 m s^{-1}
- **B** 20 m s^{-1}
- **C** **24 m s^{-1}**
- **D** 35 m s^{-1}

> **Correct: C.** $v_y = \sqrt{2gh} = \sqrt{2 \times 9.8 \times 20}$ = 19.8 m s^{-1}. Speed = $\sqrt{15^2 + 19.8^2} = \sqrt{225 + 392} = \sqrt{617}$ = 24.8 ≈ 24 m s^{-1}.

---

24. The area between a velocity–time graph and the time axis represents displacement. For a graph that goes positive then equally negative, the area gives zero. This means:

- **A** the object did not move
- **B** the total distance is zero
- **C** **the object returned to its start point**
- **D** the speed was always zero

Correct: C. Zero net area means zero displacement — the object returned to its starting point, though it did travel (distance is non-zero).

3 Dynamics

25. A 2.0 kg body moving at 3.0 m s^{-1} east experiences a constant force and, 4.0 s later, moves at 3.0 m s^{-1} west. The magnitude of the average force is:

A 0 N

B 1.5 N

C **3.0 N**

D 6.0 N

Correct: C. Change in momentum = $m\Delta v$ = 2.0(-3.0 - 3.0) = -12 kg m s^{-1} (magnitude 12). Force = $\Delta p/\Delta t$ = 12/4.0 = 3.0 N.

---

26. A ball of mass m hits the ground vertically at speed v and rebounds at speed v. The magnitude of the impulse on the ball is:

A 0

B mv

C **2mv**

D mv^2

Correct: C. Taking up as positive: impulse = Δp = m(v) - m(-v) = 2mv. The momentum reverses, so the change is twice the initial magnitude.

---

27. Two trolleys, masses m and 2m, moving towards each other at the same speed v, collide and stick together. Their common velocity has magnitude:

A 0

B **v/3**

C v/2

D 2v/3

Correct: B. Take the 2m direction as positive: total momentum = 2m(v) + m(-v) = mv. Combined mass 3m: velocity = mv/3m = v/3 (in the direction of the heavier trolley).

---

28. A resultant force–time graph for an object is a triangle: rising from 0 to F_0 over time τ then falling to 0 over the next τ. The change in momentum is:

A **$F_0\tau$**

B $\frac{1}{2}F_0\tau$

C $2F_0\tau$

D $F_0\tau/4$

Correct: A. Impulse = area under F–t graph = area of triangle = ½ × base × height = ½ × (2τ) × F_0 = $F_0\tau$.

---

29. An object falls through air and reaches terminal velocity. At terminal velocity:

- **A** the drag force is zero
- **B** the weight is zero
- **C** **the resultant force is zero**
- **D** the acceleration equals g

Correct: C. At terminal velocity the (upward) drag has grown equal to the (downward) weight, so the resultant force — and hence the acceleration — is zero; the object moves at constant velocity.

---

30. A 0.20 kg ball moving at 5.0 m s^{-1} is struck and moves off at 5.0 m s^{-1} at 90° to its original direction. The magnitude of the change in momentum is:

- **A** 0
- **B** 1.0 kg m s^{-1}
- **C** **1.41 kg m s^{-1}**
- **D** 2.0 kg m s^{-1}

Correct: C. Momentum vectors before and after are perpendicular, each of magnitude 0.20×5.0 = 1.0. $\Delta p = \sqrt{1.0^2 + 1.0^2} = \sqrt{2}$ = 1.41 kg m s^{-1}.

---

31. In an elastic collision between two equal masses, one initially at rest, the moving mass:

- **A** **stops and the other moves off with its velocity**
- **B** continues with the same velocity
- **C** reverses direction
- **D** and the target move off together

Correct: A. For equal masses in a 1-D elastic collision with one at rest, the velocities are exchanged: the incoming mass stops and the target moves off with the incoming velocity (as with Newton's cradle).

---

32. A rocket ejects 50 kg of gas per second at 800 m s^{-1} relative to the rocket. The thrust is:

- **A** 1.6×10^1 N
- **B** **4.0×10^4 N**
- **C** 4.0×10^3 N
- **D** 6.4×10^5 N

Correct: B. Thrust = rate of change of momentum of exhaust = (dm/dt)v = 50 × 800 = 4.0×10^4 N.

---

33. A body of constant mass has a momentum–time graph that is a straight line of positive gradient. This tells us that:

- **A** the velocity is constant
- **B** **the resultant force is constant and non-zero**
- **C** the body is in equilibrium
- **D** the acceleration is zero

> **Correct: B.** The gradient of a momentum–time graph is the resultant force ($F = dp/dt$). A constant positive gradient means a constant non-zero force, hence constant acceleration.

---

34. A person of weight W stands in a lift accelerating downward at a. The reading on scales beneath them is:

- **A** $W(1 + a/g)$
- **B** **$W(1 - a/g)$**
- **C** W
- **D** Wa/g

> **Correct: B.** $N = m(g - a) = mg(1 - a/g) = W(1 - a/g)$. Downward acceleration reduces the apparent weight.

---

35. Two objects interact. Which statement is always true, whatever the type of collision?

- **A** kinetic energy is conserved
- **B** **total momentum is conserved**
- **C** the objects have equal speeds afterwards
- **D** the total kinetic energy increases

> **Correct: B.** Momentum is conserved in all collisions (no external resultant force). Kinetic energy is conserved only in elastic collisions; it is not generally conserved.

---

36. A trolley of mass 1.0 kg moving at 4.0 m s^{-1} collides with a stationary 3.0 kg trolley and they move off together. The fraction of the original kinetic energy that is lost is:

- **A** 1/4
- **B** 1/2
- **C** **3/4**
- **D** 0

> **Correct: C.** Common velocity = 1.0(4.0)/4.0 = 1.0 m s^{-1}. KE before = ½(1.0)(16) = 8.0 J; after = ½(4.0)$(1.0)^2$ = 2.0 J. Lost = 6.0 J = 3/4 of 8.0 J.

---

37. A 1200 kg car changes velocity from 15 m s^{-1} east to 15 m s^{-1} north in 6.0 s. The magnitude of the average resultant force is closest to:

A 0 N

B 2500 N

C **4200 N**

D 6000 N

Correct: C. $\Delta v = \sqrt{15^2 + 15^2} = 21.2$ m s^{-1}. $F = m\Delta v/\Delta t = 1200 \times 21.2/6.0 = 4240 \approx 4200$ N.

---

38. Water leaves a hose of area 5.0×10^{-4} m^2 at 20 m s^{-1} and hits a wall, stopping. (Density 1000 kg m^{-3}.) The force on the wall is:

A 50 N

B 100 N

C **200 N**

D 400 N

Correct: C. Mass/s = $\rho Av = 1000 \times 5.0\times10^{-4} \times 20 = 10$ kg s^{-1}. Force = (mass/s)Δv = $10 \times 20 = 200$ N.

---

4 Forces, density and pressure

39. A uniform metre rule is pivoted at its centre. A 2.0 N weight hangs at the 20 cm mark. To balance it, a 4.0 N weight must hang at the:

- **A** **65 cm mark**
- **B** 70 cm mark
- **C** 80 cm mark
- **D** 90 cm mark

> **Correct: A.** The 2.0 N weight is 30 cm from the pivot (50-20): moment = 2.0×0.30 = 0.60 N m. For balance 4.0 × d = 0.60 ⇒ d = 0.15 m = 15 cm from centre, i.e. the 65 cm mark.

---

40. The pressure at a depth of 20 m in seawater (density 1030 kg m^{-3}) due to the water alone is approximately:

- **A** 2.0×10^4 Pa
- **B** 1.0×10^5 Pa
- **C** **2.0×10^5 Pa**
- **D** 2.0×10^6 Pa

> **Correct: C.** $\Delta p = \rho g \Delta h = 1030 \times 9.81 \times 20 \approx 2.02 \times 10^5$ Pa.

---

41. A couple consists of two forces of 6.0 N separated by 0.25 m. Its torque is:

- **A** 0.75 N m
- **B** **1.5 N m**
- **C** 3.0 N m
- **D** 0 N m

> **Correct: B.** Torque of a couple = one force × separation = 6.0 × 0.25 = 1.5 N m. (The two forces are equal and opposite, so the resultant force is zero, but the torque is not.)

---

42. A block floats in water with 80% of its volume submerged. Its density is:

- **A** 200 kg m^{-3}
- **B** **800 kg m^{-3}**
- **C** 1000 kg m^{-3}
- **D** 1250 kg m^{-3}

> **Correct: B.** Floating: weight = upthrust ⇒ $\rho_{block} V g = \rho_{water}(0.80V)g$ ⇒ $\rho_{block} = 0.80 \times 1000 = 800$ kg m^{-3}.

---

43. A ladder rests against a smooth wall on rough ground. The frictional force at the ground acts:

- **A** vertically up
- **B** vertically down
- **C** **horizontally towards the wall**
- **D** horizontally away from the wall

> **Correct: C.** The smooth wall pushes the ladder horizontally away from it; for horizontal equilibrium friction at the base must act horizontally towards the wall.

---

44. An object of volume V and density ρ_o is fully immersed in a fluid of density ρ_f (with $\rho_o > \rho_f$). The apparent weight (true weight minus upthrust) is:

- **A** $\rho_f Vg$
- **B** **$(\rho_o - \rho_f)Vg$**
- **C** $(\rho_o + \rho_f)Vg$
- **D** $\rho_o Vg$

> **Correct: B.** True weight = $\rho_o Vg$; upthrust = $\rho_f Vg$. Apparent weight = $(\rho_o - \rho_f)Vg$.

---

45. Three forces act on a body in equilibrium. When drawn head-to-tail they must:

- **A** all be parallel
- **B** **form a closed triangle**
- **C** form a right angle
- **D** have equal magnitudes

> **Correct: B.** For equilibrium the vector sum is zero, so three forces drawn head-to-tail form a closed triangle.

---

46. A non-uniform beam of weight 200 N rests on two supports 3.0 m apart, one at each end. The left support reads 120 N and the right support reads 80 N. The distance of the centre of gravity from the left support is:

- **A** 1.0 m
- **B** **1.2 m**
- **C** 1.5 m
- **D** 1.8 m

> **Correct: B.** Taking moments about the left support, the weight (200 N acting at distance d) is balanced by the right reaction (80 N at 3.0 m): $200 \times d = 80 \times 3.0 \Rightarrow d = 240/200 = 1.2$ m. The centre of gravity lies nearer the more heavily loaded left support, as expected.

---

47. A uniform beam of weight 60 N and length 2.0 m rests on a single pivot 0.50 m from the left end. The downward force needed at the left end to balance it is:

- **A** 30 N
- **B** **60 N**
- **C** 90 N
- **D** 120 N

Correct: B. Weight acts at centre, 0.50 m right of pivot: moment = 60 × 0.50 = 30 N m. Force F at 0.50 m left of pivot: F × 0.50 = 30 ⇒ F = 60 N.

---

48. A cube of side 0.10 m and density 600 kg m^{-3} floats in water. The depth of the cube below the surface is:

- **A** 0.04 m
- **B** **0.06 m**
- **C** 0.08 m
- **D** 0.10 m

Correct: B. Fraction submerged = ρ_{cube}/ρ_{water} = 600/1000 = 0.60. Depth = 0.60 × 0.10 = 0.06 m.

---

5 Work, energy and power

49. A 2.0 kg object is raised 5.0 m and also given a speed of 4.0 m s^{-1}. The total energy supplied (no losses) is:

A 16 J

B 98 J

C **114 J**

D 196 J

Correct: C. ΔPE = mgh = 2.0×9.81×5.0 = 98.1 J; ΔKE = ½(2.0)$(4.0)^2$ = 16 J. Total = 98.1 + 16 = 114 J.

---

50. A car of power output P travels at constant speed v against a resistive force F. Which is correct?

A **P = Fv**

B P = F/v

C P = Fv^2

D P = ½Fv

Correct: A. At constant speed the driving force equals F; power = force × velocity = Fv.

---

51. The braking distance of a car on a level road is 20 m at 15 m s^{-1}. With the same braking force, the braking distance at 30 m s^{-1} is:

A 40 m

B 60 m

C **80 m**

D 160 m

Correct: C. Fd = ½mv^2 ⇒ d ∈ v^2. Doubling v quadruples d: 4 × 20 = 80 m.

---

52. A pump raises 300 kg of water per minute to a height of 12 m. The minimum power required is approximately:

A 59 W

B **590 W**

C 3500 W

D 35 W

Correct: B. Mass/s = 300/60 = 5.0 kg s^{-1}. P = (m/t)gh = 5.0 × 9.81 × 12 ≈ 589 W.

---

53. A 60 kg skier descends a 20 m vertical drop and arrives with KE of 9.0 kJ. The energy dissipated by friction is approximately:

- **A** **2.8 kJ**
- **B** 9.0 kJ
- **C** 11.8 kJ
- **D** 20.8 kJ

> **Correct: A.** PE lost = mgh = 60×9.81×20 = 11.77 kJ. Dissipated = 11.77 - 9.0 = 2.8 kJ.

---

54. An object is projected vertically up with kinetic energy E. At the height where its speed has halved, its kinetic energy is:

- **A** E/2
- **B** **E/4**
- **C** 3E/4
- **D** E

> **Correct: B.** KE $\in$ v^2. If v halves, KE becomes $(½)^2$ = ¼ of the original, i.e. E/4.

---

55. A machine has an efficiency of 40%. To deliver 800 J of useful output, the energy input required is:

- **A** 320 J
- **B** 1120 J
- **C** **2000 J**
- **D** 3200 J

> **Correct: C.** Efficiency = useful/input $\Rightarrow$ input = useful/efficiency = 800/0.40 = 2000 J.

---

56. A constant force of 20 N pulls a box 5.0 m across a floor; the force acts at 60° to the horizontal (direction of motion). The work done by the force is:

- **A** **50 J**
- **B** 87 J
- **C** 100 J
- **D** 173 J

> **Correct: A.** W = Fs cosθ = 20 × 5.0 × cos60° = 100 × 0.5 = 50 J.

---

57. Water flows over a waterfall of height 50 m at 200 kg s^{-1}. If all the PE were converted to electrical energy, the maximum power output would be about:

- **A** 9.8 kW
- **B** **98 kW**
- **C** 980 kW
- **D** 9.8 MW

> **Correct: B.** P = (m/t)gh = 200 × 9.81 × 50 = 9.81 × 10^4 W ≈ 98 kW.

---

58. A 500 kg lift is raised at a constant 3.0 m s^{-1}. The useful power of the motor (ignoring friction) is:

- **A** 1.5 kW
- **B** 4.9 kW
- **C** 9.8 kW
- **D** **15 kW**

> **Correct: D.** At constant speed the motor force = weight = 500 × 9.81 = 4905 N. P = Fv = 4905 × 3.0 = 14.7 kW ≈ 15 kW.

---

59. A ball of mass 0.20 kg is dropped from 5.0 m and rebounds to 3.2 m. The energy dissipated in the bounce is:

- **A** 1.8 J
- **B** **3.5 J**
- **C** 6.3 J
- **D** 9.8 J

> **Correct: B.** Energy lost = mgΔh = 0.20 × 9.81 × (5.0 - 3.2) = 0.20 × 9.81 × 1.8 = 3.5 J.

---

6 Deformation of solids

60. A wire extends 2.0 mm under a load. A second wire of the same material and length but twice the diameter carries the same load. Its extension is:

- **A** **0.5 mm**
- **B** 1.0 mm
- **C** 2.0 mm
- **D** 4.0 mm

Correct: A. $e = FL/(AE)$. Twice the diameter ⇒ 4× area ⇒ extension × 1/4: 2.0/4 = 0.5 mm.

---

61. The area under a force–extension graph (within the limit of proportionality) represents:

- **A** the spring constant
- **B** the stress
- **C** **the elastic potential energy stored**
- **D** the Young modulus

Correct: C. Work done stretching = area under F–x graph = elastic PE stored (½Fx within the limit of proportionality).

---

62. Two springs each of stiffness k are joined end-to-end (in series). The combined stiffness is:

- **A** 2k
- **B** k
- **C** **k/2**
- **D** k/4

Correct: C. In series $1/k_{total} = 1/k + 1/k = 2/k$ ⇒ $k_{total} = k/2$ (softer, extends more for a given force).

---

63. A material is stretched beyond its elastic limit and then unloaded. On the force–extension graph, this shows as:

- **A** the line returning exactly along the loading line
- **B** **a permanent extension when the force reaches zero**
- **C** an increased spring constant
- **D** no extension at all

Correct: B. Beyond the elastic limit deformation is partly plastic, so on unloading a permanent (residual) extension remains when the force returns to zero.

---

64. A wire of Young modulus E, length L and cross-sectional area A is stretched by force F. The extension is:

- **A** **FL/(AE)**
- **B** FA/(LE)
- **C** EAL/F
- **D** FE/(AL)

Correct: A. E = (F/A)/(e/L) ⇒ e = FL/(AE).

---

65. A steel wire and a copper wire have the same dimensions and carry the same load within their limits of proportionality. Steel has the larger Young modulus. Compared with the copper wire, the steel wire has:

- **A** greater stress and greater strain
- **B** **the same stress but smaller strain**
- **C** smaller stress and greater strain
- **D** the same stress and the same strain

Correct: B. Same load and dimensions ⇒ same stress (F/A). Since E = stress/strain and steel has larger E, the steel wire has the smaller strain (and hence smaller extension).

---

66. A spring stores 0.80 J of elastic PE at an extension of 40 mm (within its limit of proportionality). At an extension of 20 mm it stores:

- **A** **0.20 J**
- **B** 0.40 J
- **C** 0.60 J
- **D** 0.80 J

Correct: A. $E_P = ½kx^2 \propto x^2$. Halving x ⇒ PE × 1/4: 0.80/4 = 0.20 J.

---

67. Two identical springs each of stiffness k are connected in parallel and support a load. The combined stiffness is:

- **A** k/2
- **B** k
- **C** **2k**
- **D** 4k

Correct: C. In parallel the stiffnesses add: k_{total} = k + k = 2k (stiffer, extends less for a given load).

---

68. A wire stretches 1.5 mm under a load. An identical wire of the same material and diameter but half the length stretches, under the same load, by:

- **A** 0.375 mm
- **B** **0.75 mm**
- **C** 1.5 mm
- **D** 3.0 mm

Correct: B. e = FL/(AE) ∈ L. Half the length ⇒ half the extension: 1.5/2 = 0.75 mm.

7 Waves

69. A wave has speed 340 m s^{-1} and frequency 170 Hz. Two points 1.5 m apart along the direction of travel have a phase difference of:

- **A** 90°
- **B** 180°
- **C** **270°**
- **D** 360°

Correct: C. $\lambda = v/f = 340/170 = 2.0$ m. Phase difference = $(1.5/2.0) \times 360° = 270°$.

---

70. On a CRO the time-base is 5.0 ms/div and one cycle spans 4.0 divisions. The frequency is:

- **A** 25 Hz
- **B** **50 Hz**
- **C** 100 Hz
- **D** 200 Hz

Correct: B. Period = 4.0 × 5.0 ms = 20 ms = 0.020 s. f = 1/0.020 = 50 Hz.

---

71. The intensity of a wave is quadrupled. Its amplitude changes by a factor of:

- **A** $\sqrt{2}$
- **B** **2**
- **C** 4
- **D** 16

Correct: B. $I \in A^2 \Rightarrow A \in \sqrt{I}$. Quadrupling I multiplies A by $\sqrt{4} = 2$.

---

72. A source of sound moves towards a stationary observer at speed v_S; the speed of sound is v. The observed frequency is:

- **A** $f_S(v - v_S)/v$
- **B** $f_Sv/(v + v_S)$
- **C** **$f_Sv/(v - v_S)$**
- **D** $f_S(v + v_S)/v$

Correct: C. For an approaching source use the minus sign in the denominator: $f_O = f_Sv/(v - v_S)$, giving a higher observed frequency.

---

73. Which electromagnetic radiation has the shortest wavelength in free space?

- **A** microwaves
- **B** infrared
- **C** ultraviolet
- **D** **gamma rays**

> **Correct: D.** Gamma rays have the shortest wavelength (< 10^{-12} m), shorter than UV, infrared and microwaves.

---

74. Plane-polarised light of intensity I_0 passes through a polariser whose axis is at 30° to the plane of polarisation. The transmitted intensity is:

- **A** 0.25 I_0
- **B** 0.50 I_0
- **C** **0.75 I_0**
- **D** 0.87 I_0

> **Correct: C.** Malus: $I = I_0 \cos^2 30° = I_0(0.866)^2 = 0.75\ I_0$.

---

75. A sound wave passes from air into water, where its speed is greater. Which quantity is unchanged?

- **A** speed
- **B** wavelength
- **C** **frequency**
- **D** intensity

> **Correct: C.** Frequency is fixed by the source and is continuous across the boundary. Speed and wavelength both change ($v = f\lambda$).

---

76. Which statement about a longitudinal wave is correct?

- **A** it can be polarised
- **B** particles oscillate perpendicular to travel
- **C** **it consists of compressions and rarefactions**
- **D** it cannot transfer energy

> **Correct: C.** Longitudinal waves have particle oscillations parallel to the travel direction, forming compressions and rarefactions; they cannot be polarised but do transfer energy.

---

77. Two points on a transverse progressive wave are exactly half a wavelength apart. Their motions are:

- **A** in phase
- **B** **in antiphase (180° out of phase)**
- **C** 90° out of phase
- **D** always both at maximum displacement

Correct: B. A separation of $\lambda/2$ corresponds to a phase difference of 180° — the points are in antiphase (moving in opposite directions).

---

78. Unpolarised light of intensity I_0 passes through one ideal polarising filter. The transmitted intensity is:

- **A** I_0
- **B** 0.71 I_0
- **C** **0.50 I_0**
- **D** 0.25 I_0

Correct: C. An ideal polariser transmits half the incident unpolarised intensity: 0.50 I_0 (and the output is plane-polarised).

---

79. A wave on a string has the equation form of a sine wave with period 0.040 s and wavelength 0.50 m. Its speed is:

- **A** 2.0 m s^{-1}
- **B** **12.5 m s^{-1}**
- **C** 20 m s^{-1}
- **D** 0.020 m s^{-1}

Correct: B. $f = 1/T = 25$ Hz. $v = f\lambda = 25 \times 0.50 = 12.5$ m s^{-1}.

---

80. Plane-polarised light passes through a polariser. The intensity is reduced to 50% of its incident value when the angle between the polarisation plane and the axis is:

- **A** 30°
- **B** **45°**
- **C** 60°
- **D** 90°

Correct: B. Malus: $I/I_0 = \cos^2\theta = 0.50 \Rightarrow \cos\theta = 0.707 \Rightarrow \theta = 45°$.

---

8 Superposition

81. In a double-slit experiment, the fringe separation is x. If the slit separation is halved and the screen distance is doubled, the new fringe separation is:

- **A** x
- **B** 2x
- **C** **4x**
- **D** x/4

Correct: C. $x = \lambda D/a$. Halving a multiplies by 2; doubling D multiplies by 2; combined ×4.

---

82. A diffraction grating of 300 lines per mm is used with light of wavelength 500 nm. The number of diffracted orders visible on each side of the centre is:

- **A** 4
- **B** 5
- **C** **6**
- **D** 7

Correct: C. $d = 1/(300\times10^{3}) = 3.33\times10^{-6}$ m. Max order: $n = d/\lambda = 3.33\times10^{-6}/500\times10^{-9} = 6.67$, so n = 6 (six orders each side).

---

83. Adjacent nodes on a stationary wave are 0.30 m apart. The wavelength of the waves forming it is:

- **A** 0.15 m
- **B** 0.30 m
- **C** **0.60 m**
- **D** 1.2 m

Correct: C. Adjacent nodes are $\lambda/2$ apart, so $\lambda = 2 \times 0.30 = 0.60$ m.

---

84. For a stable two-source interference pattern, the two sources must be:

- **A** of different frequency
- **B** **coherent (constant phase difference)**
- **C** incoherent
- **D** of perpendicular polarisation

Correct: B. Observable, stable fringes require coherent sources — a constant phase difference (hence the same frequency).

---

85. A stationary wave differs from a progressive wave in that a stationary wave:

- **A** transfers energy along its length
- **B** has all points oscillating with the same amplitude
- **C** **does not transfer energy along its length**
- **D** has no nodes

> **Correct: C.** A stationary wave stores energy and does not transfer it along the wave; amplitude varies from zero at nodes to a maximum at antinodes.

---

86. White light passes through a diffraction grating. In the first-order spectrum, the colour diffracted through the largest angle is:

- **A** violet
- **B** green
- **C** **red**
- **D** all diffracted equally

> **Correct: C.** $d \sin\theta = n\lambda \Rightarrow \sin\theta \in \lambda$. Red has the longest wavelength, so it is diffracted through the largest angle.

---

87. Two coherent sources produce interference. At a point the path difference is 2.5λ. This point is:

- **A** a bright fringe (constructive)
- **B** **a dark fringe (destructive)**
- **C** of intermediate intensity
- **D** outside the pattern

> **Correct: B.** A path difference of an odd number of half-wavelengths ($2.5\lambda = 5 \times \lambda/2$) gives antiphase arrival — destructive interference (dark).

---

88. The most pronounced diffraction of a wave through a gap occurs when the gap width is:

- **A** much larger than the wavelength
- **B** **approximately equal to the wavelength**
- **C** much smaller than the wavelength
- **D** independent of the wavelength

> **Correct: B.** Diffraction is greatest when the gap width is comparable to the wavelength.

---

89. In a double-slit experiment the fringe spacing is 2.0 mm. If the whole apparatus is immersed in water (refractive index 1.33), which reduces the wavelength, the fringe spacing becomes about:

- **A** **1.5 mm**
- **B** 2.0 mm
- **C** 2.7 mm
- **D** 3.3 mm

> **Correct: A.** λ reduces by a factor 1.33 in water, and $x \in \lambda$, so $x = 2.0/1.33 = 1.5$ mm.

---

90. Light of wavelength 500 nm is incident normally on a grating with slit spacing 2.0×10^{-6} m. The angle of the first-order maximum is:

- **A** 7.2°
- **B** **14.5°**
- **C** 30°
- **D** 45°

> **Correct: B.** $\sin\theta = n\lambda/d = 500\times10^{-9}/2.0\times10^{-6} = 0.25 \Rightarrow \theta = 14.5°$.

---

9 Electricity

91. A current of 5.0 A flows for 4.0 minutes. The number of electrons passing a point is ($e = 1.6 \times 10^{-19}$ C):

- **A** **7.5×10^{21}**
- **B** 1.9×10^{21}
- **C** 7.5×10^{19}
- **D** 1.2×10^{3}

> **Correct: A.** $Q = It = 5.0 \times 240 = 1200$ C. $N = Q/e = 1200/1.6\times10^{-19} = 7.5 \times 10^{21}$.

---

92. For a wire carrying current I with n charge carriers per unit volume, each of charge q, and cross-sectional area A, the drift speed is:

- **A** **I/(nAq)**
- **B** nAq/I
- **C** InAq
- **D** IA/(nq)

> **Correct: A.** $I = nAvq \Rightarrow v = I/(nAq)$.

---

93. A wire is stretched to twice its length (volume constant). Its resistance becomes:

- **A** half
- **B** double
- **C** **four times**
- **D** unchanged

> **Correct: C.** Constant volume: doubling L halves A. $R = \rho L/A \rightarrow \rho(2L)/(A/2) = 4\rho L/A = 4R$.

---

94. The I–V graph of a filament lamp is a curve that bends towards the V-axis as V increases. This is because:

- **A** the filament cools with current
- **B** the resistance decreases with current
- **C** **the resistance increases as temperature rises**
- **D** it obeys Ohm's law

> **Correct: C.** Higher current heats the filament; increasing temperature increases resistance, so I rises less steeply with V (the curve bends towards the V-axis). The lamp does not obey Ohm's law.

---

95. Two resistors 3.0 Ω and 6.0 Ω are in parallel. The combined resistance is:

- **A** 9.0 Ω
- **B** 4.5 Ω
- **C** **2.0 Ω**
- **D** 0.5 Ω

> **Correct: C.** 1/R = 1/3.0 + 1/6.0 = 2/6 + 1/6 = 3/6 ⇒ R = 2.0 Ω.

---

96. The resistance of a thermistor (NTC) as temperature rises, and of an LDR as light intensity rises, respectively:

- **A** increases; increases
- **B** **decreases; decreases**
- **C** increases; decreases
- **D** decreases; increases

> **Correct: B.** An NTC thermistor's resistance decreases as temperature rises; an LDR's resistance decreases as light intensity rises (both because more charge carriers are freed).

---

97. A 12 V, 36 W lamp operates at its rating. Its resistance is:

- **A** 0.33 Ω
- **B** 3.0 Ω
- **C** **4.0 Ω**
- **D** 48 Ω

> **Correct: C.** $R = V^2/P = 12^2/36 = 144/36 = 4.0$ Ω.

---

98. Power P is transmitted at voltage V through cables of resistance R. The power lost as heat in the cables is:

- **A** **P^2R/V^2**
- **B** PV/R
- **C** V^2/R
- **D** PR/V

> **Correct: A.** Line current I = P/V. Loss = $I^2R = (P/V)^2R = P^2R/V^2$ — hence high-V, low-I transmission minimises loss.

---

99. Two identical resistors are connected first in series, then in parallel, across the same supply. The ratio (power in parallel : power in series) is:

- **A** 1 : 1
- **B** 2 : 1
- **C** **4 : 1**
- **D** 1 : 4

> **Correct: C.** Let each be R. Series total 2R, parallel total R/2. $P = V^2/R_{tot}$. Ratio = $(V^2/(R/2))/(V^2/2R)$ = (2/R)/(1/2R) = 4. So 4 : 1.

---

100. A charge of 4.0 mC flows through a lamp in 2.0 ms. The average current is:

- **A** 0.5 A
- **B** **2.0 A**
- **C** 8.0 A
- **D** 8.0 mA

> **Correct: B.** $I = Q/t = 4.0\times10^{-3}/2.0\times10^{-3} = 2.0$ A.

---

101. Two resistors, 6.0 Ω and 3.0 Ω, are connected in parallel and a total current of 3.0 A flows into the combination. The current in the 6.0 Ω resistor is:

- **A** 0.5 A
- **B** **1.0 A**
- **C** 1.5 A
- **D** 2.0 A

> **Correct: B.** Current divides inversely with resistance. $I_6/I_{total} = R_3/(R_3+R_6) = 3.0/9.0 = 1/3$. $I_6 = 3.0/3 = 1.0$ A.

---

10 D.C. circuits

102. A cell of e.m.f. 1.5 V has internal resistance 0.50 Ω. When connected to a 1.0 Ω resistor, the terminal p.d. is:

- **A** 0.50 V
- **B** **1.0 V**
- **C** 1.25 V
- **D** 1.5 V

Correct: B. $I = 1.5/(1.0 + 0.50) = 1.0$ A. Terminal p.d. = $IR = 1.0 \times 1.0 = 1.0$ V (= $\varepsilon - Ir = 1.5 - 0.5$).

---

103. Kirchhoff's first law (current) is a consequence of the conservation of:

- **A** energy
- **B** **charge**
- **C** momentum
- **D** mass

Correct: B. The junction (current) law expresses conservation of charge — charge cannot accumulate at a point in a steady current.

---

104. In a potential divider, two equal resistors are across a 10 V supply. The output across one resistor is 5.0 V. If a load equal to R is connected across that resistor, the output becomes:

- **A** 5.0 V
- **B** **3.3 V**
- **C** 2.5 V
- **D** 6.7 V

Correct: B. The loaded resistor becomes R/2 (R in parallel with R). $V_{out} = 10 \times (R/2)/(R + R/2) = 10 \times (1/2)/(3/2) = 10/3 = 3.3$ V.

---

105. A battery delivers maximum power to an external resistor R when:

- **A** R is very large
- **B** R = 0
- **C** **R equals the internal resistance r**
- **D** R is very small

Correct: C. Maximum power transfer to the load occurs when R = r (matched resistances); efficiency is then 50%.

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106. The e.m.f. of a source is best defined as the energy transferred per unit charge:

- **A** dissipated in the internal resistance
- **B** **in driving charge around the complete circuit**
- **C** in the external circuit only
- **D** when no charge flows

> **Correct: B.** E.m.f. = total energy transferred (from other forms to electrical) per unit charge driven around the whole circuit, including the internal resistance.

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107. Three 6.0 Ω resistors are connected in parallel. The combined resistance is:

- **A** 18 Ω
- **B** 6.0 Ω
- **C** 3.0 Ω
- **D** **2.0 Ω**

> **Correct: D.** Equal resistors in parallel: R/n = 6.0/3 = 2.0 Ω.

---

108. A potentiometer is balanced (galvanometer reads zero) at length 45.0 cm for one cell and 60.0 cm for a second. The ratio of their e.m.f.s (first : second) is:

- **A** **3 : 4**
- **B** 4 : 3
- **C** 1 : 1
- **D** 9 : 16

> **Correct: A.** At balance no current flows, so e.m.f. $^{\in}$ balance length: 45.0 : 60.0 = 3 : 4.

---

109. As the current drawn from a real battery increases, its terminal potential difference:

- **A** increases
- **B** **decreases**
- **C** stays equal to the e.m.f.
- **D** becomes zero immediately

> **Correct: B.** $V = \varepsilon - Ir$; as I increases the 'lost volts' Ir increases, so the terminal p.d. decreases.

---

110. In the circuit, a 6.0 V battery (negligible internal resistance) is connected to a 2.0 Ω resistor in series with a parallel pair of 4.0 Ω resistors. The current from the battery is:

- **A** 1.0 A
- **B** **1.5 A**
- **C** 2.0 A
- **D** 3.0 A

Correct: B. Parallel pair = 2.0 Ω. Total = 2.0 + 2.0 = 4.0 Ω. I = 6.0/4.0 = 1.5 A.

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111. A cell of e.m.f. 1.5 V is short-circuited (external resistance ≈ 0) and the current is 6.0 A. The internal resistance is:

- **A** **0.25 Ω**
- **B** 0.40 Ω
- **C** 4.0 Ω
- **D** 9.0 Ω

Correct: A. Short circuit: $\varepsilon = Ir \Rightarrow r = 1.5/6.0 = 0.25\ \Omega$.

---

112. In a potential divider with a fixed resistor and an LDR (output across the LDR), moving the setup into brighter light causes the output voltage to:

- **A** increase
- **B** **decrease**
- **C** stay the same
- **D** become zero

Correct: B. Brighter light lowers the LDR resistance, so a smaller share of the supply voltage appears across the LDR — the output taken across the LDR decreases.

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11 Particle physics

113. The nuclide $^{238}_{92}U$ emits an α-particle. The resulting nuclide has:

A **A = 234, Z = 90**

B A = 236, Z = 90

C A = 234, Z = 92

D A = 238, Z = 90

Correct: A. α-emission removes 2 protons and 2 neutrons: A decreases by 4 (238→234), Z decreases by 2 (92→90).

---

114. During β^- decay, within the nucleus:

A a proton becomes a neutron

B **a neutron becomes a proton**

C an up quark becomes a down quark

D a neutron becomes a positron

Correct: B. β^- decay: a neutron → proton (a down quark changes to an up quark), emitting an electron and an antineutrino.

---

115. The charge on a neutron (udd) confirms which quark charges?

A u = +⅓, d = -⅔

B **u = +⅔, d = -⅓**

C u = -⅔, d = +⅓

D u = +1, d = 0

Correct: B. udd: (+⅔) + (-⅓) + (-⅓) = 0. Consistent with u = +⅔e, d = -⅓e.

---

116. A meson consists of:

A three quarks

B two quarks

C **one quark and one antiquark**

D a lepton and a quark

Correct: C. A meson is a hadron made of one quark and one antiquark; a baryon is made of three quarks.

---

117. β-particles are emitted with a continuous range of energies because:

- **A** the nucleus recoils
- **B** **a neutrino/antineutrino shares the energy**
- **C** γ-rays are also emitted
- **D** the electrons collide with atoms

Correct: B. The decay energy is shared among three bodies (daughter nucleus, β-particle and (anti)neutrino), so the β-particle can take any energy up to a maximum — a continuous spectrum.

---

118. Which particle is a lepton?

- **A** proton
- **B** neutron
- **C** **neutrino**
- **D** pion (meson)

Correct: C. The neutrino is a fundamental lepton. Protons and neutrons are baryons; a pion is a meson — all hadrons made of quarks.

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119. In the α-scattering experiment, the small fraction of α-particles deflected through large angles indicates that the atom's positive charge is:

- **A** spread uniformly through the atom
- **B** **concentrated in a tiny nucleus**
- **C** carried by the electrons
- **D** zero

Correct: B. Large-angle deflection of the positive α-particles shows a concentrated positive charge (and mass) in a very small nucleus — most of the atom being empty space.

---

120. Which quantities are always conserved in a nuclear decay equation?

- **A** **nucleon number and charge**
- **B** mass and kinetic energy
- **C** proton number and neutron number separately
- **D** number of α-particles

Correct: A. Nucleon number and charge (proton number balance) are conserved. Neutron and proton numbers can individually change (e.g. in β-decay), and rest mass is not conserved (mass–energy is).

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121. A nuclide $^{A}_{Z}X$ emits a β^+ particle. The resulting nuclide is:

- **A** $^{A}_{Z+1}Y$
- **B** $^{A}_{Z-1}Y$
- **C** $^{A-4}_{Z-2}Y$
- **D** $^{A-1}_{Z}Y$

Correct: B. β^+ decay: a proton becomes a neutron, so Z decreases by 1 while A is unchanged: $^{A}_{Z-1}Y$.

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122. Which combination of quarks could form a baryon of charge +2e (such as the Δ^{++})?

- **A** uud
- **B** uuu
- **C** $u\bar{u}$
- **D** udd

Correct: B. Charge of uuu = 3 × (+⅔e) = +2e; three quarks ⇒ a baryon. uud = +1e, udd = 0, and $u\bar{u}$ is a meson.
